SURVIVING THE FUTURE

Crisis, Climate, and Human Resilience

A Data-Driven, Academically Grounded Guide to Living Through the Crises Caused by Global Warming

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PREFACE: WHY FUTURE GLOBAL WARMING IS CERTAIN

Global warming is not a hypothesis; it is an observed, measured, and physically explained reality. The certainty of future warming rests on three pillars: (1) the physics of the greenhouse effect, (2) the observed accumulation of greenhouse gases in the atmosphere, and (3) the inertia of the climate system. Even if all emissions stopped today, committed warming from past emissions would continue for decades. This book is written to be debated in academic spaces: every major claim is tied to peer-reviewed literature, data, and expert statements.

The Three Pillars of Certainty

- **Physics:** Greenhouse gases absorb outgoing infrared radiation. This is established physics, tested for over a century (Arrhenius, 1896; modern radiative transfer models).
- **Observation:** Atmospheric CO2 has risen from ~280 ppm (pre-industrial) to ~420 ppm (2024). Global mean temperature has risen ~1.1-1.2 C. These are direct measurements.
- **Inertia:** The ocean stores heat; ice sheets respond slowly. Even with zero emissions, warming continues for decades (committed warming).

Data underpinning the discussion is summarized as follows:

Table P.1 Key global climate indicators (2024)

Indicator	Value	Source
Global mean temperature rise since 1850-1900	~1.1-1.2 C	IPCC (2023)
CO2 concentration (2024)	~420 ppm	NOAA (2024)
Sea level rise since 1900	~20-25 cm	IPCC (2023)
Arctic sea ice decline	~13% per decade	NASA
People affected by climate disasters (annual)	~100 million+	IFRC
Economic damage from climate disasters (annual)	~$200-300 billion	World Bank

The expert consensus underpinning this book is summarized as follows.

Dr. James Hansen, former NASA GISS Director (2013 op-ed/2016 paper): 'Sea level rise of several meters could be expected with 2°C warming — 2°C would have consequences that can be described as disastrous.'

Prof. Katharine Hayhoe, Texas Tech Chief Scientist (2014, post-IPCC AR5): 'Climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.'

Dr. Michael Mann, Penn State climatologist (2004/2021): 'There is about as much scientific consensus about human-caused climate change as there is about gravity — multiple independent lines of evidence converge.'

Dr. Friederike Otto, Imperial College / World Weather Attribution co-founder (2021): 'The Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.'

Prof. Johan Rockström, Potsdam Institute & Stockholm Resilience Centre (2009 Nature; 2023 update): 'We define planetary boundaries within which humanity can operate safely — six of nine boundaries are now transgressed, well outside the safe operating space.'

Preface (Extended)

This extended preface provides additional context for the reader. The book is organized to move from foundational science to practical action, ensuring that every reader can find relevant material. The structure reflects the complexity of the climate crisis and the need for multi-level responses.

Additional context: The climate crisis demands sustained attention and coordinated action across all sectors of society. This book aims to equip readers with the knowledge and tools needed to contribute meaningfully to that effort.

CHAPTER 1

INTRODUCTION: THE WARMING WORLD

Theme: the global crisis of warming and why survival begins with understanding it

In the summer of 2022, a monsoon system carried moisture from the Arabian Sea across Pakistan for eight consecutive weeks. By the end of August, one-third of the country was under water, thirty-three million people had been affected, and the national highway network had been cut in more than a hundred places. The flood was not a single event but a sequence: heat that dried the soil, then rain that could not infiltrate it, then reservoir releases that arrived after the rivers had already filled. Six months later, the same country was negotiating with the International Monetary Fund while its farmers reported that two million hectares of cropland would not be planted that season.

A year later, in 2023, Canada burned. More than eighteen million hectares of forest burned, an area larger than England, and the smoke crossed the Atlantic and turned the skies of New York orange. In September of the same year, two dams above the Libyan city of Derna failed after a storm that had been forecast days in advance; the water killed more than eleven thousand people in a city that had been warned but could not act. In October 2024, a year's worth of rain fell on the Spanish province of Valencia in a few hours and killed more than two hundred people in towns that had never been mapped as floodplains.

These events are usually described as natural disasters. They are not. Each of them required a physical trigger — rain, wind, heat — but the scale of the loss was determined by conditions that people created: where houses were built, how rivers were managed, whether warnings reached the people who needed them, whether the soil had been left compacted and bare, whether the population was old, poor, or living in informal settlements. The trigger is weather. The disaster is a failure of preparation, and preparation is something human beings can change.

This book is an attempt to describe that failure and the ways out of it, in both directions: backward, into the record of what has already happened, and forward, into the range of futures that the physical system has made possible. It is not a book of prophecy. It is a book about the conditions under which communities continue to function, and the conditions under which they do not.

1.1 The Decade That Changed the Baseline

The 2020s began with a global pandemic and ended their first half with the four hottest years ever measured. The World Meteorological Organization confirmed that 2024 was the first calendar year to average more than 1.5°C above the 1850–1900 baseline, at about 1.55°C, with 2023 close behind at 1.45°C. The two years were lifted by a strong El Niño, but the underlying trend was not: the previous ten years were the ten warmest in the instrumental record, and every one of them exceeded the average of the decade before it.

The warming has not been evenly distributed in time or space. The Arctic has warmed three to four times faster than the global mean; the Mediterranean and the Middle East have warmed roughly twice as fast; the North Atlantic has cooled in some patches even as the ocean as a whole has gained heat, because melting ice and shifting winds have altered circulation. A global average of 1.5°C therefore describes a world in which some regions have already experienced 2.5°C or 3°C of local warming, and in which the statistics of extreme heat have shifted far more than the mean.

The decade also changed the character of the evidence. Attribution studies that once took a year now take a month. Satellite records that began in the 1970s now span half a century. Insurance losses, crop models, hospital admissions and migration data have accumulated to the point where the effect of climate on society can be measured rather than inferred. The result is a body of evidence that is no longer primarily about the future: it is about what has already happened, to identifiable people, in identifiable places, at a cost that can be counted.

1.2 The Anatomy of a Global Crisis

A global crisis is not simply a large disaster. It is a set of stresses that arrive together, reinforce one another, and exceed the capacity of institutions to respond. Climate change qualifies because it acts on every sector at once: it raises food prices by reducing yields, raises insurance costs by increasing losses, raises migration by making some places uninhabitable, and raises the cost of borrowing by increasing uncertainty. Each effect is modest in isolation. Together they consume the fiscal and administrative space that governments need to cope with the next shock.

The reinforcement is the part that is usually missing from public discussion. A drought reduces the harvest; the reduced harvest raises food prices; higher food prices increase poverty; poverty increases vulnerability to the next flood; the flood destroys the assets that the household would have used to recover. The sequence can turn a single bad season into a decade of lost development, and it explains why the same hazard produces a few dozen deaths in one country and thousands in another.

This is also why the language of 'natural disaster' is misleading in a second sense. It suggests a boundary between the hazard and the society exposed to it, when in reality the boundary is the thing that policy controls. A levee, a building code, a crop insurance scheme, a heat action plan and a functioning weather service are all ways of moving that boundary. The chapters that follow examine each of them, but the logic is the same throughout: the physical hazard sets the pressure, and the social system determines whether the pressure becomes a catastrophe.

1.3 From Local Hazard to Systemic Failure

Most climate impacts begin as local events, but their consequences travel through markets, supply chains and migration routes. The 2011 floods in Thailand are the standard example: a sequence of monsoon rains inundated industrial estates north of Bangkok, global hard-drive production fell by about a quarter, and computer prices rose around the world. The flood was national; the shortage was global. The same mechanism now operates through semiconductor production in Taiwan, wheat exports

from the Black Sea, and container shipping through the Red Sea.

Systems fail when a shock exceeds redundancy. A power grid with spare capacity survives a heatwave; a grid without it cascades. A port with alternative routes survives a blockage; a port that is the only route does not. The 2021 Texas winter storm, the 2022 European gas crisis, and the 2023 Panama Canal drought all demonstrated the same principle in different sectors: the failure was not caused by the weather alone but by the absence of slack that would have absorbed it. Climate change is, among other things, a systematic reduction in slack.

For households, the translation is direct. A family that depends on one income, one water source, one road and one food market is exposed to every shock that touches any of them. Diversification — of income, of water storage, of food sources, of social contacts — is the household equivalent of redundancy in a grid. It is also the reason preparedness is not a single purchase but a set of overlapping capabilities, each of which buys time when another fails.

1.4 The Human Cost: What the Record Shows

The human cost of climate change is usually reported through disasters, and the disaster statistics are stark enough. The Centre for Research on the Epidemiology of Disasters recorded more than 7,000 major disasters between 2000 and 2019, affecting 4.2 billion people and killing 1.2 million. Weather and climate events accounted for the large majority of them. In the 2020s the pattern has continued: the 2022 Pakistan floods affected thirty-three million people; the 2023 Turkey–Syria earthquakes — geological rather than climatic, but a reminder of compound vulnerability — killed more than fifty thousand; the 2024 Valencia floods killed more than two hundred in a high-income country with a modern warning system.

But disasters capture only the visible part of the cost. The World Health Organization attributes about 150,000 deaths a year to climate change through heat, disease, malnutrition and displacement, a figure that most epidemiologists consider conservative because it excludes the long tail of chronic illness and lost livelihood. Heat alone is estimated to cause more than 500,000 deaths a year globally, with the European summer of 2022 accounting for about 61,700 of them. Air pollution, much of it from the same fuels that drive warming, kills about seven million people a year.

The economic record is equally uneven. The World Bank estimates that climate-related losses in developing countries already exceed US$200 billion a year, and that the figure understates the true cost because it counts damaged assets rather than lost development. A child who is malnourished during a drought does not recover the cognitive development that was lost; a farmer who sells livestock to survive a season does not rebuild the herd at the same rate. These effects are rarely counted in disaster statistics, but they are the mechanism by which a temporary shock becomes a permanent reduction in welfare.

1.5 Why the Worst Outcomes Are Not Yet Fixed

The physical system has inertia, and that inertia is usually presented as bad news: the warming already committed will continue for decades, sea level will rise for centuries, and some changes are irreversible. All of that is true. But the same inertia means that the future is not yet determined, because the emissions that would produce the worst outcomes have not yet happened. The difference between 1.8°C and 4.4°C at the end of the century is not a difference in physics; it is a difference in cumulative carbon, and cumulative carbon is still being decided.

The remaining carbon budget makes the arithmetic explicit. To have a fifty per cent chance of holding warming to 1.5°C, the world can emit roughly 250 billion tonnes of carbon dioxide more — about six years at current rates. For 2°C the budget is roughly 1,150 billion tonnes. These numbers are uncertain at the margins, but their order of magnitude is not, and it implies that every year of delay consumes a substantial share of the remaining budget. It also implies that the choice is not between action and no action but between action now and much larger, more expensive action later.

There is a second reason for qualified optimism. The cost of the technologies that reduce emissions has fallen faster than almost anyone predicted. Solar photovoltaic costs have fallen by roughly ninety per cent since 2010, onshore wind by about seventy per cent, and lithium-ion battery packs by about eighty-five per cent. In 2023 the world added more renewable generating capacity than fossil capacity for the first time, and the International Energy Agency now expects renewables to account for the majority of new power capacity through 2030. The obstacle is no longer primarily engineering; it is the speed at which grids, permits and political coalitions can change.

1.6 What 'Surviving' Actually Means

The word survival suggests a single moment: the storm passes, the flood recedes, the family is alive. In practice, survival is a process that extends long after the water leaves. A household that survives the flood but loses its documents, its savings and its access to credit may not survive the next two years. A city that survives the heatwave but loses its power grid for a week may lose thousands of people to hospitals that cannot run. The relevant question is not whether a system survives the event but whether it returns to a level of functioning from which it can absorb the next one.

This book therefore treats survival at three levels. Personal survival is about water, food, shelter, health and the knowledge to use them. Community survival is about mutual aid, local institutions and the physical infrastructure that connects households to services. Societal survival is about the fiscal capacity, the legitimacy and the administrative competence that allow a state to keep functioning when several crises arrive at once. The levels are nested: a household cannot be secure in a failed state, and a state cannot be stable when its households are desperate.

The nesting also explains why purely individual preparation is insufficient. A family that stores water and food is better off than one that does not, but it still depends on a functioning water utility, a supply chain, a hospital and a road. Preparedness that ignores the collective is a bet that everyone else will continue to provide the services on which the prepared household depends. That bet may pay off in a short emergency; it fails in a prolonged crisis, which is precisely the scenario that climate change makes

more likely.

1.7 Knowledge, Preparedness and Adaptive Capacity

Adaptive capacity is the ability of a system to adjust to change while remaining functional. It is not the same as wealth, though wealth helps; it is the combination of information, resources, institutions and social trust that allows a community to anticipate, absorb and recover. Bangladesh has far less wealth than the United States, but its cyclone preparedness is widely regarded as superior, because it has invested in shelters, volunteers, early warnings and drills that reach the village level. Adaptive capacity is therefore a choice about where to invest, not simply a by-product of income.

Knowledge is the cheapest component and the one most often neglected. A household that knows which road floods first, which neighbour has a generator, and which clinic stays open can make better decisions in the first six hours than one that has to discover those facts in the middle of the emergency. A city that knows which neighbourhoods have the oldest residents and the least tree cover can target cooling measures before the heatwave arrives. The information is rarely secret; it is simply not assembled, not practised, and not shared.

The chapters that follow treat knowledge as infrastructure. That means writing down the plan, testing it, updating it, and making sure the people who need it can find it in a language and format they can use. It also means distinguishing between knowledge that is stable — the physics of heat, the geography of a floodplain — and knowledge that changes, such as the location of a shelter or the phone number of a clinic. A plan that does not distinguish between the two will either become obsolete or become too vague to act on.

1.8 The Argument of This Book

The argument can be stated in four propositions. First, the climate has already changed enough to produce measurable harm, and the harm will increase because of warming that is already committed. Second, the harm is not distributed evenly: it falls hardest on people with the fewest resources, in places with the weakest institutions, and on hazards — heat, drought, flood — that are least insured. Third, the capacity to reduce harm exists and is largely a matter of preparation, governance and investment rather than of inventing new technology. Fourth, the difference between a manageable future and a catastrophic one is still a political choice, made year by year, in decisions about energy, land use and finance.

The book is organised around those propositions. The first part establishes the physical and observational basis: how warming works, how it is measured, and why the record is trustworthy. The second part examines the main channels through which harm arrives — heat, water, food, ecosystems, health, economies, migration and conflict. The third part turns to response: infrastructure, education, information, household preparedness, emergency health, social capital, technology and governance. The fourth part presents case studies and closes with action plans that a household, a community or a city can use.

Each chapter follows the same discipline. It begins with field conditions — what has actually happened, where, and to whom. It then brings in the research literature to explain why those conditions produced those outcomes, and it distinguishes between what is well established, what is contested, and what is not yet known. It ends with a set of future scenarios built from historical data, present observations and explicit assumptions. The aim is not to persuade the reader of a conclusion but to give the reader the materials to reach one.

1.9 How the Chapters Fit Together

The first six chapters are diagnostic. Chapters 1 to 3 establish the physical science and the observation systems; chapters 4 to 6 examine heat, water and flooding as the three hazards that account for most direct losses. Chapters 7 to 12 follow the consequences through food, ecosystems, health, economies, migration and conflict. These chapters are ordered by the sequence in which a shock typically propagates: heat and water arrive first, food and health follow within weeks, economies and migration follow within months, and conflict, where it occurs, emerges from the interaction of all of them.

Chapters 13 to 21 are about response. They cover infrastructure, education, data and misinformation, household preparedness, emergency health, social capital, renewable energy, psychology and governance. The order is deliberate: physical infrastructure is the most visible response and the slowest to build; education and information are the cheapest and the most durable; social capital is the resource that determines whether the other responses reach the people who need them. Governance appears last because it determines whether any of the others can be sustained.

Chapters 22 to 24 are applied. Two chapters of case studies examine what happened in specific places — heatwaves, fires, floods, droughts, food crises and security emergencies — and what distinguished the responses that worked from those that did not. The final chapter provides action plans at three scales: the household, the community and the city. Readers who need immediate guidance can begin there and return to the analytical chapters later; readers who want to understand the reasoning can read in order.

1.10 Reading in an Emergency

A book is a poor instrument in an emergency, and this one is designed with that limitation in mind. If a hazard is imminent, the priority is to act on the guidance of local authorities, not to consult a chapter. If the hazard has passed and the household is safe, the chapter on household preparedness provides a checklist for the next seventy-two hours and the next three months. If the reader is a planner or a teacher, the chapters on infrastructure, education and governance are the relevant starting points.

The book is also designed to be updated. Climate science, disaster statistics and technology costs change faster than books can be reprinted, and a reader who relies on a number without checking its date will eventually be misled. Each chapter therefore identifies the kind of evidence that would change its conclusions, and the final chapter lists the data sources that are updated most frequently. The habit of checking the date of

a claim is itself a survival skill, because misinformation often consists of true statements that were true once.

Finally, the book is written for readers who are not specialists. Technical terms are explained when they first appear; numbers are given with their units and their dates; and uncertainty is stated rather than concealed. The aim is not to make the reader a climate scientist but to make the reader capable of following an argument, detecting an exaggeration, and deciding what to do. That capacity — not the memorisation of facts — is what allows a community to adapt when the facts change.

1.11 Case Study: The 2022 Pakistan Floods

The floods began in June 2022 with monsoon rain that was heavy but not unprecedented. What made them catastrophic was the sequence. A spring heatwave had melted snow and glaciers early and baked the soil into a crust; when the rain arrived, it ran off instead of infiltrating. The Indus and its tributaries rose together, and the dams and barrages that had been built to control them released water downstream to protect their own structures. By late August, the government had declared a national emergency, and satellite imagery showed a lake stretching across Sindh.

The numbers are a case study in how climate impacts are counted. About 33 million people were affected, 1,700 died, and 2.2 million houses were damaged or destroyed. The Post-Disaster Needs Assessment estimated damage at US$14.9 billion and economic losses at US$15.2 billion, together about 4.8 per cent of GDP. Agriculture lost most of a season: two million hectares of crops were destroyed, and the cotton harvest fell by about a third. The damage was concentrated in Sindh and Balochistan, the two poorest provinces, which is a pattern repeated in disaster after disaster.

Attribution research found that climate change made the extreme rainfall more likely, though the magnitude of the effect varied between studies and depended on the definition of the event (Otto et al., 2023). But attribution explains only the trigger. The vulnerability had been built over decades: irrigation systems that had silted up, embankments that had not been maintained, housing in floodplains, and a population whose livelihoods depended on a single crop season. The flood was a climate event; the disaster was a development event.

The response revealed both the strengths and the limits of international aid. Donor pledges eventually exceeded US$10 billion, but they arrived over months, while the most urgent needs were in the first weeks. The Pakistani military, which has the country's largest logistics capacity, led the relief effort; provincial governments coordinated shelters; and local charities, particularly religious foundations, distributed food faster than the formal system. The pattern — rapid local response, slow international finance — is typical, and it is why the chapters on governance and social capital treat local institutions as part of the infrastructure.

Recovery has been uneven. By 2024, many households had rebuilt with the same materials in the same places, because they had no alternative and because reconstruction assistance covered only a fraction of the cost. The lesson is not that Pakistan should have prevented the rain. It is that recovery which restores the

pre-existing vulnerability is not recovery; it is the preparation of the next disaster. The same lesson appears in the case studies of every chapter that follows.

For readers outside Pakistan, the case has three implications. It shows that a climate hazard can erase years of development in a single season. It shows that vulnerability is produced by decisions about land, water and housing that are made long before the hazard arrives. And it shows that the difference between a disaster and a catastrophe is often the speed and reach of the response — which is a matter of institutions, not of meteorology.

1.12 Future Scenarios and Predictions: What the Next Two Decades Hold

The historical record provides the starting point. Since 1970, global surface temperature has risen by about 0.18°C per decade, sea level by about 3.4 millimetres per year, and the frequency of extreme heat has increased by a factor of roughly three in many regions. The number of weather-related disasters reported to EM-DAT has risen from about 100 a year in the 1970s to more than 300 a year in the 2010s, though part of that increase reflects better reporting. The losses, adjusted for inflation, have grown by about 5 to 7 per cent a year, faster than exposure alone would predict.

The present position is that the world is on a trajectory consistent with about 2.7°C of warming by 2100 if current policies are maintained, according to the International Energy Agency's assessment of announced pledges and implemented measures. That is below the 4.4°C of the highest scenario but well above the 1.5°C target. The near-term trajectory is largely locked in by emissions already released and by infrastructure already built: even if every country met its net-zero pledge, the next two decades would still warm by about 0.4°C above the current level.

The scenarios below are therefore not predictions of what will happen but conditional projections of what could happen under stated assumptions. They differ mainly in the speed of decarbonisation and in the response of slow variables such as ice sheets and the overturning circulation. Each is anchored to observed data: the 2024 concentration of 424 parts per million, the 1.55°C annual anomaly, the 4.5 millimetres per year of sea-level rise, and the current rate of energy transition.

Table 1.1 Three futures for the next two decades

Scenario	Assumption	2035	2050	Signature impacts
Accelerated transition	Emissions peak before 2025, halve by 2035, net zero by 2060	~1.7°C	~1.8°C	Heat extremes stabilise; sea-level rise ~0.3 m by 2050; coral reefs still decline
Current policies	Implemented measures only, slow tightening after 2030	~1.9°C	~2.1°C	Severe heatwaves annual in South Asia; crop losses 5–10%; 200 million displaced by 2050
Delayed transition	Emissions plateau until 2040, rapid catch-up afterwards	~2.1°C	~2.4°C	Arctic summers ice-free; reef collapse; 400 million in coastal flood zones

The difference between the first and third rows is about half a degree by 2050 and nearly a degree by 2100, but the difference in consequences is much larger than the temperature numbers suggest. Half a degree is the difference between a heatwave that kills hundreds and one that kills tens of thousands; between a harvest that fails once a decade and one that fails every other year; between a reef that bleaches and recovers and one that dissolves. The reason is that impacts are non-linear: they depend on thresholds, and thresholds are crossed at particular temperatures rather than spread evenly across the range.

What would change these projections? A faster collapse in the cost of storage would make the first row more likely; a sustained failure of international cooperation would make the third more likely. A large release of methane from permafrost or a faster loss of Antarctic ice would push all three rows upward. The one thing that would not change them is better forecasting: the projections already incorporate the best available science, and the uncertainty that remains is dominated by choices, not by ignorance.

CHAPTER 2

THE PHYSICAL SCIENCE OF GLOBAL WARMING

Theme: the physical science of global warming and its history

In March 1958, a thirty-year-old geochemist named Charles David Keeling hauled a gas analyser up the northern slope of Mauna Loa, a volcano in Hawai'i, and began measuring the carbon dioxide content of clean Pacific air. The first readings came in at about 315 parts per million. Keeling had chosen the site for the same reason meteorologists choose mountain tops: the air there has travelled thousands of kilometres over open ocean and carries no local smoke, no city traffic, no factory exhaust. What it carries is the background composition of the atmosphere. Within two years the measurements showed something no laboratory experiment could show — the background itself was rising, year after year.

That single instrument turned a nineteenth-century physics argument into an accounting problem. Fourier, Foote, Tyndall and Arrhenius had established that carbon dioxide absorbs infrared radiation and that more of it should warm the surface. Keeling established the other half of the ledger: how fast humans were adding it. By 2024 the same series stood at roughly 424 parts per million, about fifty per cent above the pre-industrial value, with the increase still accelerating (NOAA Global Monitoring Laboratory, 2025). No other number in this book is measured so directly, over so long a period, by so few instruments.

The physics matters for the rest of the book because it sets the boundary conditions for everything else. A heatwave in Karachi, a harvest failure in the Brazilian Cerrado, an insurance withdrawal in Florida and a flood in Emilia-Romagna are different events with different proximate causes, but they share a common denominator: an atmosphere holding more energy and more water vapour than it did a century ago. This chapter establishes that denominator. It explains the mechanism that warms the surface, the gases that drive it, the evidence that the warming is real, the accounting that attributes it, and the range of futures consistent with what is already in the pipeline.

It also shows where the certainty stops. The direction of change is not in serious dispute; the magnitude of the response to a given concentration still is, within limits that matter enormously for policy. The difference between a climate that warms by 1.8°C and one that warms by 4.4°C by 2100 is not a matter of scientific taste. It is the difference between a world in which adaptation is expensive and a world in which parts of it are impossible, and it turns on assumptions about emissions that no physical model can settle on its own.

2.1 The Greenhouse Effect

The argument began with a puzzle about temperature. In 1824 Joseph Fourier calculated that a body at the Earth's distance from the Sun should be colder than the Earth actually is, and proposed that the atmosphere slows the escape of heat — a blanket, in his image, though he did not know what the blanket was made of. Thirty-two years later, Eunice Foote ran a simple experiment with glass cylinders in sunlight and found that a cylinder filled with carbon dioxide heated more than one filled with

ordinary air, writing that an atmosphere of that gas 'would give to our earth a high temperature'. In 1859 John Tyndall built an apparatus sensitive enough to measure which gases absorb infrared radiation and found that water vapour was the strongest absorber, with carbon dioxide and ozone close behind (Tyndall, 1861).

The mechanism those experiments revealed is straightforward. Sunlight arrives mostly as visible light, to which the atmosphere is largely transparent. The surface absorbs it and re-emits the energy as infrared radiation, and it is at those wavelengths that greenhouse gases intercept the outgoing beam. They re-radiate in all directions, including downward, so the effective altitude at which the planet finally loses heat to space rises. Because the air at higher altitude is colder, it radiates less efficiently, and the surface must warm until the outgoing energy again balances the incoming sunlight. In 1896 Svante Arrhenius carried out the first quantitative calculation of this balance, estimating that a doubling of carbon dioxide would raise global temperature by roughly five degrees Celsius — an estimate strikingly close to modern values, reached with pencil, paper and a great deal of patience.

The most common objection to this account is that carbon dioxide is a trace gas, four hundredths of one per cent of the atmosphere, and that something so scarce cannot govern the temperature of a planet. The objection misunderstands how absorption works. A gas does not need to be abundant to be opaque at particular wavelengths; it needs only to be present in sufficient quantity to absorb the radiation that tries to pass. At the centre of the carbon dioxide bands the atmosphere is already nearly opaque, so added molecules matter most at the edges of those bands, where absorption is incomplete. That is why the warming effect grows roughly with the logarithm of concentration rather than in direct proportion: each doubling adds about the same increment of forcing as the one before, regardless of how high the starting point already is.

2.2 Greenhouse Gases and Their Sources

Four gases do almost all of the work. Carbon dioxide is the largest single contributor and the one that accumulates; methane is far more potent per molecule but short-lived; nitrous oxide is potent and long-lived in roughly equal measure; and a family of fluorinated industrial gases, present in minute quantities, traps heat thousands of times more effectively than CO2. In 2024 the global average concentration of carbon dioxide reached about 424 parts per million, methane about 1,934 parts per billion and nitrous oxide about 336 parts per billion — values that have no precedent in the ice-core record of the last 800,000 years (NOAA, 2025; IPCC, 2021).

The sources are known with unusual precision because they are counted twice: once from the bottom up, by adding up fuel sales, land-use change and industrial processes, and once from the top down, by measuring the changing composition of the atmosphere and the ocean. The two accounts now agree to within a few per cent. Fossil fuel combustion and industry released about 37.4 billion tonnes of CO2 in 2024, land-use change perhaps four billion more (Friedlingstein et al., 2024). Energy use accounts for roughly three-quarters of total greenhouse gas emissions; agriculture, forestry and land use for about a fifth; the rest comes from industry and waste (IPCC,

2022c).

The difference in lifetimes between the gases is the most policy-relevant fact in this section. A molecule of methane survives in the atmosphere for about a decade before it is oxidised, so cutting methane emissions reduces warming within years rather than centuries. Carbon dioxide is different: perhaps a third of what is emitted today will still be in the atmosphere in a thousand years. That asymmetry explains why climate strategy pursues two tracks at once — rapid methane cuts to slow the rate of warming in the near term, and deep CO2 cuts to determine where the warming finally stops. Confusing the two tracks is a common source of bad policy, because a methane success can look like progress on the problem while the cumulative CO2 ledger continues to grow.

2.3 Evidence of Warming

The evidence for warming does not rest on a single thermometer or a single group. Surface stations, weather balloons, satellites, ocean buoys, glacier surveys, tide gauges, boreholes, the timing of spring, the poleward movement of fish stocks and the retreat of mountain glaciers all point the same way. Each system has its own errors and its own blind spots; what makes the conclusion robust is that the errors are different and the signal is common. When independent records built from different instruments by different groups agree, the probability that all of them are wrong in the same direction falls sharply (IPCC, 2021, Chapter 2).

The last two years illustrate both the strength and the subtlety of the evidence. The World Meteorological Organization estimated 2023 at about 1.45°C above the 1850–1900 average and 2024 at about 1.55°C, the warmest year in the instrumental record (WMO, 2025). Neither figure should be read as 'the climate has reached 1.5°C'; a single year bounces above and below the trend because of El Niño, volcanic aerosols, solar variability and the weather of the ocean. The trend itself, averaged over a decade, is currently rising at roughly 0.2°C per decade — about twenty times faster than the average rate of change since the last ice age (IPCC, 2021).

A related dispute concerns the satellites. Microwave sounding units have measured the temperature of the lower troposphere since 1979, and early analyses appeared to show less warming than the surface record. The discrepancy was traced to the slow drift of satellite orbits, which changed the time of day at which measurements were taken, and to the decay of the instruments themselves. Once those effects were corrected, the satellite record agreed with the surface stations within their respective uncertainties, and both agreed with model simulations driven by observed forcings (IPCC, 2021). The episode is worth remembering because it illustrates a general rule: a disagreement between two measurement systems is usually a clue about the instruments, not a refutation of the phenomenon they are both trying to measure.

2.4 Temperature Records and Trends

A global temperature record is not a single measurement but a statistical construction, and it is worth knowing how it is built because the construction is where most public argument about it goes wrong. Stations move, instruments change, cities

grow around once-rural sites, and ships once measured sea temperature by dipping buckets and now measure it at the engine intake. Each of these changes introduces a bias of a few tenths of a degree, and each bias has to be estimated and removed. Four independent groups — HadCRUT in Britain, NASA GISS in the United States, Berkeley Earth and NOAA — do this work with different methods, and their global series agree to within about 0.05°C (Morice et al., 2021; Lenssen et al., 2019; Rohde and Hausfather, 2020).

What the corrected record shows is a rise of about 1.2°C since the late nineteenth century, unevenly distributed: land has warmed faster than ocean, the Arctic three to four times faster than the global average, and nights faster than days. The record also shows that the warming has not been smooth. There are decades of little change, sharp El Niño years, and pauses that lasted a decade; all of them sit on an upward slope that has steepened since the 1970s. The apparent contradiction between short pauses and long trends is the single most exploited feature of the record in public debate, and it is resolved by the same rule that governs financial markets: look at the trend, not the last quarter.

The exceptional warmth of 2023 and 2024 has a partial explanation that is worth separating from the trend. An El Niño event transferred heat from the tropical Pacific to the atmosphere; the eruption of Hunga Tonga in 2022 injected water vapour, itself a greenhouse gas, into the stratosphere; a sharp reduction in sulphur emissions from shipping removed a reflective aerosol that had been masking part of the warming; and the solar cycle was near its maximum. Each of these contributed a fraction of a degree. Taken together they can explain why the two years sat above the trend line, but not why the trend line itself is where it is. That distinction — between the noise on the signal and the signal — is the difference between a weather story and a climate one.

2.5 Industrial Revolution and Emissions

For most of human history the carbon content of the atmosphere was remarkably stable, varying by only a few parts per million between ice ages and warm periods. That stability ended with coal. The concentration had been about 280 parts per million for thousands of years; by 1900 it had reached roughly 296, and the curve began to bend upward sharply after 1950, when oil, gas and industrial agriculture were added to coal. Historians of the environment call this inflection the Great Acceleration: in a single generation after the Second World War, population, fertiliser use, water consumption, vehicle ownership and carbon emissions all rose together, faster than at any previous point in human history (Steffen et al., 2015).

The arithmetic of that rise is usually summarised by an identity rather than a story. Emissions equal population multiplied by income per person, multiplied by the energy needed for each unit of income, multiplied by the carbon emitted for each unit of energy. Improvements in the last two terms have been real and substantial; they have simply been outpaced by growth in the first two. The same identity explains why emissions are so unequally distributed. The richest tenth of the world's population accounts for roughly half of consumption-based emissions, while the poorest half accounts for about twelve per cent (Chancel, 2022). That distribution is not a moral

footnote; it is the reason historical responsibility remains the most contested question in climate negotiations.

The geography of emissions has shifted faster than the geography of responsibility. China is now the largest annual emitter, responsible for roughly thirty per cent of current fossil CO2, followed by the United States at about eleven, the European Union at about seven and India at about seven (Friedlingstein et al., 2024). Measured cumulatively since 1850, the order is different: the United States and Europe together account for roughly forty per cent of historical emissions, China for about fourteen and India for about four. Per person, the gap is wider still: an average American emits several times more than an average Indian or African. Which accounting is used determines which obligations follow, and that is why the choice of metric is itself a political act.

2.6 IPCC Assessments

The Intergovernmental Panel on Climate Change was created in 1988 by two United Nations bodies, the World Meteorological Organization and the UN Environment Programme, in response to a simple political problem: governments needed an authoritative account of a scientific literature that none of them could read in full. Its method is unusual. Hundreds of scientists volunteer to assess published research, they produce a technical report running to thousands of pages, and then governments negotiate the summary for policymakers line by line. The process is slow, conservative and occasionally painful to read, but it has one great virtue: no single government, industry or scientist can determine its conclusions.

The sixth assessment cycle, published between 2021 and 2023, is the most important single source in this book. Its first working group concluded that it is 'unequivocal that human influence has warmed the atmosphere, ocean and land', and that the best estimate of the warming already experienced is about 1.1°C above 1850–1900 (IPCC, 2021). The second group found that roughly 3.3 to 3.6 billion people live in contexts that are highly vulnerable to climate change (IPCC, 2022b). The third found that the policies in place at the time of writing would produce warming of about 2.7°C by the end of the century (IPCC, 2022c). Those three sentences, taken together, are the empirical spine of the argument that follows.

Reading an IPCC summary requires attention to its grammar. The panel uses a calibrated vocabulary in which 'likely' means a probability between 66 and 100 per cent, 'very likely' between 90 and 100, and 'virtually certain' above 99. It also distinguishes between an assessed range, which reflects the judgement of the author team across all available evidence, and the spread of model results, which reflects only the models. Statements such as 'warming of 1.5°C will be reached in the early 2030s' are assessments, not model outputs, and they carry the authority and the caution of that distinction. Readers who quote the panel without that grammar tend to overstate either its certainty or its doubt.

2.7 Ocean Heat and Sea Level

The ocean has absorbed about ninety per cent of the excess heat trapped by greenhouse gases, and roughly a quarter of the carbon dioxide. That absorption is why the surface has not warmed faster, and it is also why the consequences arrive late. Water has an enormous heat capacity, so the ocean integrates warming over decades and then releases it slowly. Ocean heat content has set a record every year for the past decade; the increase in the upper two kilometres between 2022 and 2023 alone was larger than the total energy used by humanity in a year (Cheng et al., 2024). Marine heatwaves have roughly doubled in frequency since the 1980s, and the fourth global coral bleaching event was declared in 2024.

Sea level is the slowest and most consequential of these responses. Global mean sea level rose about 21 centimetres between 1900 and 2018, and the rate has accelerated from roughly 1.3 millimetres a year in the early twentieth century to about 4.5 millimetres a year today (IPCC, 2021; NASA, 2024). Roughly forty per cent of that rise comes from thermal expansion, a quarter from glaciers and a quarter from the ice sheets, with the remainder from changes in land water storage. Because the processes that drive it respond slowly, sea level will continue to rise for centuries even if emissions fall to zero. The practical question is therefore not whether to adapt, but how much and how soon — and the answer differs by an order of magnitude between a low-emission and a high-emission trajectory.

The coast is where the global average becomes a local emergency. Global mean sea level is an abstraction; what floods a street is relative sea level, which combines the global rise with local subsidence, sediment compaction and changing storminess. Jakarta has been sinking at rates of several centimetres a year in some districts, so its relative sea level rises far faster than the global average; Semarang and parts of Manila face similar combinations. About 1.1 billion people live in low-elevation coastal zones and about 900 million of them in Asia (Neumann et al., 2015). For those populations, the difference between 0.4 and 0.8 metres of rise by 2100 is not a scientific interval; it is the difference between a managed retreat and an unmanaged one.

2.8 Ice Loss and Cryosphere

The cryosphere is where the climate system keeps its slowest and least reversible commitments. Arctic sea ice has declined by roughly 12 to 13 per cent per decade in September since satellite records began in 1979, and the remaining summer ice is thinner and younger than it was a generation ago (Notz and Stroeve, 2016). The Greenland ice sheet has been losing mass at about 260 billion tonnes a year, Antarctica at roughly 150 billion tonnes, and the world's glaciers at about 270 billion tonnes a year outside the ice sheets (Shepherd et al., 2020; IMBIE, 2018; Zemp et al., 2019). Each of those numbers is small relative to the total ice stored on the planet and large relative to anything human infrastructure has had to absorb before.

Three consequences follow. The first is a feedback: white ice reflects sunlight that dark ocean absorbs, so lost ice accelerates local warming. The second is freshwater forcing: meltwater changes the density of the North Atlantic and can slow the

circulation that carries heat northward, with consequences for European weather that remain actively debated (Caesar et al., 2021; Ditlevsen and Ditlevsen, 2023). The third is commitment: ice sheets respond to temperature over centuries, so the sea level implied by today's warming will continue to arrive long after the emissions that caused it have stopped. This is why the difference between 1.5°C and 2°C matters far more for coasts than the difference between two decades of weather.

The Arctic case also illustrates how definitions carry consequences. 'Ice-free' does not mean an ocean without ice; it means less than one million square kilometres of ice at the September minimum, a threshold chosen because it is roughly the area at which the remaining ice becomes fragmented and mobile. Under all plausible emission trajectories the Arctic is expected to reach that state at least once before 2050, and under high emissions it becomes the normal summer condition (Notz and Stroeve, 2016). The consequences extend beyond the region: darker water absorbs more sunlight, the jet stream responds to the changed temperature gradient, and new shipping routes open across a sea that was once a barrier.

2.9 Carbon Cycle and Feedbacks

Of every tonne of carbon dioxide released by human activity, roughly half stays in the atmosphere. The other half is taken up in approximately equal measure by the ocean and by land ecosystems, and that service is the reason atmospheric concentrations have risen more slowly than emissions. The service is not guaranteed. Measurements suggest that the fraction absorbed by land and ocean sinks has been declining as a share of emissions, and that the land sink in particular is becoming more variable as droughts and fires reduce uptake (Friedlingstein et al., 2024). A weakening sink means that the same emissions produce a larger atmospheric increase — a feedback that is already visible in the data.

The ocean also absorbs carbon dioxide chemically, and the resulting change in seawater chemistry is direct and measurable: surface pH has fallen by about 0.1 units since the pre-industrial period, equivalent to a roughly thirty per cent increase in hydrogen ion concentration (IPCC, 2021). Organisms that build shells from calcium carbonate are sensitive to that change, which is why acidification is treated in this book as a parallel crisis rather than a footnote to warming. The same arithmetic that makes the ocean a buffer makes it a slow victim: it has absorbed the heat and the carbon, and it will pay for both for centuries.

The carbon budget is the bridge between physics and policy, and its arithmetic is brutally simple. Every tonne of CO2 emitted adds roughly the same amount of warming, regardless of when it is emitted; the total that can be emitted while holding warming below a given threshold is therefore a finite quantity. The best estimate for a fifty per cent chance of staying below 1.5°C was about 500 billion tonnes of CO2 from 2020, and continued emissions have since reduced it to roughly 200 billion — about five years at current rates (IPCC, 2021; Friedlingstein et al., 2024). The budget does not care about pledges, announcements or targets; it responds only to the tonnage that actually leaves the ground.

2.10 Attribution of Warming

Attribution is the discipline of asking not whether the climate has changed but why. Its central tool is the fingerprint: climate models are run with all known forcings, then with human influences removed, and the resulting spatial patterns are compared with observations. The pattern of greenhouse warming — tropospheric warming with stratospheric cooling, more warming over land than ocean, more at high northern latitudes — is distinctive, and no combination of solar and volcanic variability reproduces it (Hegerl et al., 1996; IPCC, 2021). This is why the sixth assessment could describe human influence as unequivocal rather than merely likely.

A younger branch of the field applies the same logic to individual events. Event attribution compares the probability or intensity of a heatwave, flood or drought in a world with and without human influence, using large ensembles of model simulations constrained by observations. The method has produced conclusions that are now widely cited: the 2021 Pacific Northwest heat dome was judged virtually impossible without climate change, and the 2023 Derna flood in Libya was made far more likely by warming sea surface temperatures. The method has limits — it says little about vulnerability, and single-event results carry wide uncertainty — but it has changed the terms of the compensation debate, because it allows a specific loss to be connected to a specific cause.

Attribution also has a geography of evidence. Events in wealthy countries tend to be studied quickly because they are insured, instrumented and politically salient; events in poorer countries are studied later, if at all, and often with fewer observations. The result is an asymmetry in what can be said, not in what is happening. Researchers call this the attribution gap, and it matters for compensation: a country that cannot demonstrate attribution finds it harder to claim loss and damage, even when the physical case is strong. Closing the gap is therefore a matter of both science and justice.

2.11 Case Study: The Keeling Curve

The most consequential measurement in climate science began as a side project. Charles David Keeling, a chemist by training, had developed a technique for measuring carbon dioxide with far greater precision than his contemporaries, and in 1956 he took a set of flasks to the Olympic Peninsula in Washington State and to the mountains of California to test it. The results surprised him: background concentrations were higher than the accepted value, and they varied with the season in a way nobody had documented. When he presented the finding, Roger Revelle and Hans Suess at the Scripps Institution of Oceanography invited him to continue the work as part of the International Geophysical Year.

The location he chose — Mauna Loa Observatory, 3,400 metres above the Pacific — gave the record its authority. Air arriving there has spent days or weeks over open ocean, so it samples the global background rather than a local plume. Keeling insisted on calibrating his instrument against a set of reference gases, on continuous operation rather than occasional sampling, and on publishing the raw data alongside the analysis. Those choices, which seemed pedantic at the time, are the reason the curve has

survived decades of scrutiny without revision of its central finding.

The data themselves tell a story in two layers. The first is a saw-tooth: carbon dioxide rises through the northern winter and falls through the northern summer as the forests of Siberia, Canada and Europe take up carbon during the growing season. That annual pulse is the breathing of the biosphere, and its amplitude has grown as the land has warmed. The second is the trend beneath it: a steady climb from about 315 parts per million in 1958 to roughly 424 in 2024, with no year below the previous decade's average since the record began.

The curve is also a record of what one instrument cannot see. It measures the atmosphere at one latitude and one altitude; it says nothing directly about the ocean, which absorbs most of the added heat, or about the land, which absorbs a quarter of the carbon. Those measurements came later, from ships, buoys, aircraft and satellites, and they are consistent with the mountain-top series. The Keeling record functions as a keystone rather than a complete arch: remove it and the rest still stands, but the structure becomes much harder to date and to defend.

The project was nearly cancelled twice. In the 1960s the funding agency cut the budget on the grounds that the work was routine monitoring rather than research, and Keeling had to fight to keep the instruments running; a decade later similar pressures returned. Each time the record was preserved by a small group of scientists who understood that a time series is only valuable if it is unbroken. The episode is instructive for anyone designing a monitoring system today: continuity is a scientific asset that is easy to destroy and impossible to reconstruct.

What the curve did politically was to make the abstract concrete. Before Mauna Loa, the greenhouse effect was a calculation; after it, it was a graph that a parliamentarian could hold up. The same function is served today by the satellite record of sea level and by the global temperature series, and the lesson generalises beyond climate: a long, clean, public measurement is one of the most powerful instruments a society can build, because it converts a contested claim into a shared reference point.

There is a final, quieter lesson in the project's design. Keeling published his data in full, including the seasonal wiggles that his colleagues found distracting, and he insisted that the calibration gases be exchanged between laboratories so that independent groups could compare results. That practice — open data, shared standards, documented methods — became the template for every subsequent global observing system. It is also the reason the curve has survived political attack: a record that anyone can check is much harder to dismiss than a conclusion that must be trusted.

2.12 Future Scenarios and Predictions: The Physical Science

Any statement about the physical future of the climate is conditional, and it is worth being explicit about the condition. The atmosphere responds to the total accumulated carbon dioxide, not to the annual rate at which it is emitted. That single fact determines the shape of every projection in this chapter: two trajectories that emit the same cumulative total produce roughly the same peak warming, whether they reach that total quickly or slowly. What differs is the damage along the way, the options available for adaptation, and the risk of crossing thresholds that cannot be reversed on

human timescales.

The historical record supplies the baseline. Concentrations were stable at about 280 parts per million for six thousand years before 1750; they reached 315 in 1958, about 370 in 2000 and roughly 424 in 2024, with cumulative emissions since 1850 now close to 2,500 billion tonnes of CO2 (Friedlingstein et al., 2024). Global mean temperature has risen about 1.2°C over the same period and is currently rising at roughly 0.2°C per decade. Sea level, which lagged the temperature rise for most of the twentieth century, is now accelerating.

The present position can be stated with unusual precision. For a fifty per cent chance of holding warming below 1.5°C, the remaining carbon budget is roughly 200 billion tonnes of CO2 — about five years of current emissions. For 2°C the budget is larger but still finite, and the policies in place in 2025 imply warming of about 2.7°C by 2100 rather than the 1.5°C target governments have endorsed (IPCC, 2022c; Friedlingstein et al., 2024). Three trajectories bracket the plausible range, and each rests on a different assumption about policy, technology and behaviour.

The first assumes rapid decarbonisation: emissions fall by about half by 2030 and reach net zero around mid-century, with carbon removal compensating for residual sources. Warming peaks near 1.8°C and declines slowly thereafter. The second assumes that current policies continue and are strengthened gradually, producing net zero late in the century; warming reaches about 2.7°C. The third assumes that decarbonisation stalls and that coal, oil and gas remain dominant; warming reaches roughly 4.4°C, with the largest increases over land and at high northern latitudes (IPCC, 2021, Table 4.5).

A useful thought experiment clarifies what is already committed. If every emission stopped tomorrow, the atmosphere would stop accumulating CO2 within a few years, but global temperature would stabilise rather than fall, and sea level would continue to rise for centuries as the ocean and the ice sheets finished responding to the warming that has already occurred. The commitment is not a reason for fatalism; it is a reason to distinguish between the warming that cannot now be avoided and the warming that can still be prevented. The first determines how much adaptation is necessary, the second determines whether adaptation remains possible.

Regionally, the three trajectories diverge far more than the global averages suggest. Under the high trajectory, the Mediterranean and the Middle East approach conditions in which summer heat becomes a chronic hazard rather than an episodic one; South Asia faces combinations of heat and humidity that exceed the physiological limits of outdoor work for part of the year; and the Arctic warms by several degrees more than the global mean. Under the low trajectory those outcomes are attenuated but not eliminated. The difference between the scenarios is therefore not a difference in global average temperature; it is a difference in which regions remain habitable in the way their inhabitants currently understand the term.

Table 2.1 Three physical trajectories to 2100 (relative to 1850–1900)

Trajectory	Policy assumption	2050	2100	Observable markers
Rapid decarbonisation (SSP1-2.6)	Emissions halve by 2030, net zero around 2060	~1.6°C	~1.8°C (1.3–2.4)	Atmospheric CO2 peaks below 450 ppm; sea-level rise near 0.4 m
Current policies strengthened (SSP2-4.5)	Slow tightening, net zero late century	~2.0°C	~2.7°C (2.1–3.5)	CO2 near 500 ppm by 2060; Arctic summers ice-free before 2050
Stalled decarbonisation (SSP5-8.5)	Fossil dominance continues to 2100	~2.6°C	~4.4°C (3.3–5.7)	CO2 above 700 ppm; sea-level rise above 0.8 m and accelerating

What would change these numbers is worth stating as plainly as the numbers themselves. A larger climate sensitivity than the central estimate, a faster loss of the land carbon sink, or an earlier collapse of Arctic summer sea ice would all push the outcomes toward the upper end. A faster decline in methane, a stronger aerosol effect from air-quality policy, or large-scale carbon removal would push them down. None of these possibilities can be resolved by argument; they will be resolved by measurement, and the measurements that matter are the ones this chapter has described — concentration, temperature, ocean heat, ice mass and sea level.

For the reader, the practical implication is a shift in attention. The question is no longer whether the physics is settled; it is which of three trajectories is being followed, and how quickly the answer becomes visible. That question is answered not by climate models but by the energy system, which is the subject of the chapters that follow.

CHAPTER 3

CLIMATE SYSTEMS, FEEDBACKS AND MONITORING

Theme: climate systems, feedback mechanisms and monitoring

A climate is not a temperature. It is a machine with several moving parts — an atmosphere, an ocean, a cryosphere, a biosphere and a land surface — coupled so tightly that a change in one is felt in all the others, usually after a delay. The atmosphere responds in days, the ocean in decades, the ice sheets in centuries. This difference in response time is the single most important fact for anyone trying to understand why the consequences of emissions arrive long after the emissions themselves, and why some of those consequences cannot be reversed on any timetable that matters to the people experiencing them.

The couplings also explain why the climate does not warm smoothly. Heat stored in the ocean can be released years later; a shift in wind patterns can cool a continent for a decade while the planet as a whole warms; a volcanic eruption can suppress global temperature for two or three years. These internal variations are not noise in the sense of meaningless — they are the weather of the climate system, and they determine whether a particular harvest fails or survives, whether a particular city floods this year or next.

Feedback is the mechanism that turns a small push into a large response, or a large push into a small one. Warming increases atmospheric water vapour, which is itself a greenhouse gas, which produces more warming; that is a positive feedback, and it roughly doubles the effect of carbon dioxide alone. Warming melts reflective ice, exposing dark ocean that absorbs more sunlight; another positive feedback. But warming also increases the radiation the planet emits to space, which is a negative feedback and the reason the system has not run away. The balance between these effects, not the strength of any one of them, determines how sensitive the climate is.

This chapter is about that balance, about the thresholds where it can change abruptly, and about the instruments that allow us to watch it happen. It closes with the least glamorous and most consequential subject in climate science: data quality, and the question of what a measurement can and cannot tell you.

3.1 The Climate System

The Earth receives about 340 watts of solar energy for every square metre of its surface, averaged over day and night and over the seasons. Roughly thirty per cent of that is reflected back to space by clouds, ice and bright surfaces — the albedo — and the remainder is absorbed by the atmosphere, the ocean and the land. Because the planet is neither heating nor cooling catastrophically on human timescales, the energy leaving must roughly equal the energy arriving. The greenhouse effect works by slowing the departure, not by adding energy; the system compensates by warming until the outgoing radiation again balances the incoming sunlight (Trenberth, Fasullo and Kiehl, 2009).

What makes the system complicated is that its components have different memories. The atmosphere forgets a temperature anomaly within weeks; the surface ocean remembers it for months to years; the deep ocean for centuries; an ice sheet for

millennia. A pulse of heat injected into the atmosphere is therefore partitioned: some escapes to space immediately, some is absorbed by the ocean and returned decades later, and some is consumed melting ice. The same is true of carbon dioxide, which is exchanged continuously among the atmosphere, the biosphere and the ocean, with a small fraction remaining airborne for thousands of years.

The practical consequence is that the climate responds to a change in forcing with a lag that depends on the component being observed. Surface temperature responds within a few years, sea level over centuries, and ice sheets over millennia. Policymakers who expect a rapid visible response to emission cuts are therefore looking at the wrong variable; the response is real but it arrives first in the ocean and only later at the coast.

3.2 Atmosphere and Ocean Circulation

The atmosphere moves heat from the tropics toward the poles through three great cells in each hemisphere: the Hadley cell, which lifts moist air at the equator and brings it down over the subtropics; the Ferrel cell, which drives the mid-latitude westerlies; and the polar cell. The boundaries between these cells are where the jet streams run, and the position of the jets determines whether a winter storm reaches Lisbon or stalls over the North Sea. As the tropics warm faster than the poles, the temperature gradient weakens and the jets tend to shift poleward, though the effect is uneven and remains an active research question (IPCC, 2021, Chapter 4).

The ocean moves heat in parallel, and more slowly. The Atlantic Meridional Overturning Circulation carries warm surface water northward, cools it in the Nordic seas, and returns it southward as cold, dense deep water; the Gulf Stream is its most visible surface expression. The circulation transports roughly eighteen million cubic metres of water per second — about a hundred times the flow of the Amazon — and delivers heat to Europe that helps keep its winters milder than those of comparable latitudes in Canada. It is driven by differences in temperature and salinity, which is why added freshwater from Greenland's melting ice is watched so closely.

Superimposed on these steady circulations are oscillations: the El Niño–Southern Oscillation, which shifts heat across the tropical Pacific every two to seven years and alters rainfall from Peru to Indonesia; the North Atlantic Oscillation, which governs European winter weather; the Indian Ocean Dipole; the Southern Annular Mode. These patterns do not create the long-term trend, but they determine which years are hot, which are wet, and where the damage falls.

3.3 Positive and Negative Feedback Loops

The largest feedback is also the least visible. Warmer air holds more water vapour — about seven per cent more for each degree of warming, following the Clausius–Clapeyron relation — and water vapour is a powerful greenhouse gas. The increase in water vapour roughly doubles the warming that carbon dioxide would produce on its own, which is why the climate sensitivity to a doubling of CO2 is about three degrees rather than one and a half. This is not a speculative mechanism; it is measured, and it is the reason the same physics that gives a modest direct effect gives a large total one (IPCC, 2021, Chapter 7).

Ice and snow provide a second positive feedback. White surfaces reflect sunlight; dark ocean and dark land absorb it. When ice retreats, the newly exposed surface absorbs more energy, which warms the region, which melts more ice. The feedback is strongest in the Arctic, where it contributes to the three- to fourfold amplification of warming relative to the global mean. A third comes from the carbon cycle: warming increases the respiration of soil microbes and the frequency of fire, releasing carbon that had been stored, which produces further warming.

Against these stand the negative feedbacks that keep the system stable. The planet radiates more energy as it warms, following the Stefan–Boltzmann law; the lapse rate — the rate at which temperature falls with altitude — adjusts in a way that damps the response in the tropics; and increased weathering of silicate rocks removes carbon dioxide from the atmosphere over geological time. Clouds are the great uncertainty: low clouds reflect sunlight and cool, high clouds trap heat and warm, and how their balance changes with warming remains the largest single source of spread among climate models.

3.4 Tipping Points and Abrupt Change

Some parts of the climate system do not respond gradually. They behave like a see-saw held just past its balance point: a small additional push produces a large, rapid and often irreversible change. Researchers call these tipping elements, and a 2022 assessment identified sixteen of them, including the Greenland and West Antarctic ice sheets, the Atlantic overturning circulation, the Amazon rainforest, boreal permafrost and warm-water coral reefs (Armstrong McKay et al., 2022). The assessment found that several would be at risk at warming levels between 1.5 and 2°C, which is to say within the range that current policies are likely to produce.

The concept is easy to misunderstand. A tipping point is not a cliff edge at a precise temperature; it is a region of the parameter space in which the system loses stability, and the exact location is uncertain because the systems are complex and poorly observed. Nor does crossing a threshold always produce an instantaneous change: the Greenland ice sheet may take centuries to disappear once committed, and the Amazon may degrade over decades through fire and drought. What changes at the threshold is not the speed of the change but its reversibility.

The policy implication is asymmetric. If the system were linear, overshooting a target and returning later would be an acceptable strategy. If it contains thresholds, overshoot becomes a gamble with a permanent loss on the other side. That asymmetry, rather than any single temperature number, is the strongest argument for treating 1.5°C as a limit to be approached with caution rather than a target to be traded against other priorities.

3.5 Temperature and Satellite Monitoring

The global temperature record is assembled from about thirty thousand land stations, several thousand ships and buoys, and a handful of satellite instruments, all processed by four independent groups using different methods. Satellites have measured the temperature of the lower troposphere since 1979, first with microwave sounders and

later with infrared interferometers; they provide global coverage but suffer from orbital drift, calibration changes and the difficulty of separating the signal of the lower atmosphere from that of the surface. Reanalysis products, which combine observations with a weather model, fill the gaps and provide consistent fields of temperature, wind and humidity over the whole planet (Hersbach et al., 2020).

Each system has a characteristic error. Surface stations are precise but unevenly distributed, with dense coverage in Europe and North America and sparse coverage in parts of Africa and South America. Satellites are uniform but indirect, measuring radiance rather than temperature and requiring a radiative transfer model to convert one into the other. Reanalyses are complete but inherit the biases of their models. The skill of the system comes from the disagreement: when independent methods converge, the result is robust; when they diverge, the divergence points to a problem worth investigating.

The architecture of that system is itself a policy achievement. The Global Climate Observing System, established in 1992, sets standards for instruments, data exchange and continuity, and the Paris Agreement's transparency framework depends on it. When a country reports its emissions or its progress, it relies on measurements that are only credible because they are comparable across borders. Monitoring, in other words, is not a technical appendix to climate policy; it is the infrastructure that makes policy verifiable.

3.6 Ocean and Ice Monitoring

The ocean was, until recently, the least observed part of the climate system — a gap that mattered because it stores more than ninety per cent of the excess heat. That changed with Argo, an international programme that began deploying autonomous profiling floats in 2000 and now maintains roughly four thousand of them. Each float sinks to two thousand metres, measures temperature and salinity as it rises, transmits its data by satellite and repeats the cycle every ten days. The result is a three-dimensional picture of the ocean's heat content, updated continuously, available to anyone with an internet connection.

Ice is observed by a different combination of instruments. Satellite altimetry, which has measured sea surface height since 1992, also measures the height of ice sheets; the GRACE mission, launched in 2002, weighs them by detecting tiny changes in the Earth's gravity field. Radar and laser altimeters measure thinning, and microwave sensors map the extent of sea ice every day. Each method has blind spots — altimetry struggles at the steep margins of ice sheets, gravity missions have coarse spatial resolution — but together they have produced a consistent picture of accelerating loss (Shepherd et al., 2020).

The data from these systems are open, which is unusual for expensive infrastructure and essential for credibility. A government that disputes a sea-level projection can download the altimetry, the tide-gauge record and the gravity data and check the arithmetic itself. That openness does not settle political disputes, but it narrows the space in which they can be conducted by assertion rather than evidence.

3.7 Extreme Event Attribution

Attribution began with a question about a single heatwave. After the European summer of 2003 killed tens of thousands of people, a team led by Peter Stott asked whether the probability of such an event had changed because of human influence, and answered by comparing model simulations with and without greenhouse gases (Stott, Stone and Allen, 2004). The answer — that the risk had roughly doubled — established a method that has since been applied to hundreds of events, from the Russian heatwave of 2010 to the floods of 2022 in Pakistan.

The method has become faster and more standardised. The World Weather Attribution group, founded in 2014, now produces rapid analyses within weeks of an event, using peer-reviewed protocols and large ensembles of model runs. Its findings are stated as changes in probability or intensity rather than as certainties: the 2021 Pacific Northwest heat dome was judged to be virtually impossible without climate change; the 2022 Pakistan floods were made more likely by warming, though vulnerability and land management determined the scale of the disaster; the 2023 Derna flood in Libya was linked to warmer-than-usual Mediterranean waters.

Attribution has limits that are easy to forget. It is more reliable for heat than for rainfall, more reliable at continental than at local scales, and weakest where observations are sparse — which is usually where the most vulnerable populations live. It also says nothing about who is responsible or who should pay; those are political questions that attribution can inform but cannot answer. Used carefully, it converts a diffuse sense of injustice into a defensible claim; used carelessly, it invites the expectation that every disaster has a single attributable cause.

3.8 Early Warning Systems

Early warning is the most cost-effective adaptation measure available, and it is still missing where it is needed most. The United Nations' Early Warnings for All initiative, launched in 2022, estimated that about half of the world's countries lack a multi-hazard early warning system, and that where systems exist they often fail at the last mile — the warning reaches a capital city but not a village, or arrives in a language and format that the people at risk cannot use (UNDRR and WMO, 2023). The same reports estimate that a twenty-four-hour warning of a coming storm can reduce damage by about thirty per cent.

Bangladesh is the standard example of what is possible. After the Bhola cyclone of 1970 killed an estimated three hundred thousand people, the country built a system of cyclone shelters, volunteer networks and mobile-phone alerts that has cut fatalities dramatically: Cyclone Amphan in 2020, a storm of comparable strength, killed roughly a hundred people. The improvement was not primarily technological. It came from shelters placed where people live, warnings issued in local languages, and drills that made evacuation routine.

Heat is the harder case, because heat kills quietly and without a visible landfall. Ahmedabad's heat action plan, adopted after a 2010 heatwave killed more than a thousand people, combines colour-coded forecasts, cooling centres, altered working

hours and outreach to vulnerable groups; evaluations suggest it has reduced mortality substantially. The lesson generalises: a warning is only an early warning if it arrives with the capacity to act on it.

3.9 Data Quality and Uncertainty

Uncertainty is not the same as ignorance, and the distinction matters for how scientific statements should be read. A measurement can be uncertain in a way that is precisely quantified — plus or minus a tenth of a degree, with a stated confidence interval — and a projection can be uncertain in a way that reflects genuine disagreement among plausible models. The first kind of uncertainty shrinks with better instruments; the second shrinks with better understanding, and sometimes not at all. Confusing the two leads either to false confidence or to unwarranted paralysis.

Climate science distinguishes between internal variability, which is the unpredictable component generated by the system itself; model uncertainty, which is the spread produced by different representations of the same physics; and scenario uncertainty, which reflects the fact that future emissions depend on choices not yet made. Only the first is reduced by more data, and only the third is reduced by policy. When a projection is presented as a range, the width of that range is usually dominated by scenario uncertainty at the end of the century and by model uncertainty in the near term.

There is also a discipline problem that the field has only recently begun to address. A dataset that is used to make a political argument will be scrutinised in a way that a dataset used only for research is not, and the scrutiny often targets the processing rather than the physics. The defence is transparency: open code, documented adjustments, published uncertainty ranges, and an explicit account of what would change the conclusion. Those practices are now standard in the major temperature and sea-level products, and they are the reason the records have survived decades of political attack.

3.10 Using Data for Decisions

The final step is the one most often skipped: converting a probabilistic projection into a decision. The standard approach in engineering is to design for a return period — a flood defence built for the one-in-a-hundred-year event — but that approach assumes a stationary climate, which is precisely what no longer holds. A defence designed for the statistics of the twentieth century will be overtopped more often than its designers intended, not because it was badly built but because the statistics it was built against have moved.

The alternative is robust decision-making: rather than optimising for a single best estimate, identify the options that perform acceptably across a wide range of futures, and prefer those that can be adjusted as information arrives. The Netherlands' Delta Programme, which plans for sea-level rise up to several metres while building in stages that can be accelerated or paused, is the clearest example. The Thames Barrier, designed in the 1970s for a scenario that now looks optimistic, is another: its closure rate has risen from a few times a year to several times a year, and the decision to raise it

further is taken on the basis of monitoring rather than prediction.

What these examples share is a willingness to treat uncertainty as a design parameter rather than an obstacle. That requires institutions capable of revisiting decisions, budgets that can be reallocated, and a public that understands why a plan might change. The technical tools are available; the constraint is almost always institutional, which is why the chapters that follow devote as much attention to governance as to physics.

3.11 Case Study: Argo and the Ocean That Was Hiding

Before 2000, the ocean's interior was sampled the way a continent might be mapped by a few dozen travellers: ships lowered instruments at chosen points, took a reading, and moved on. The coverage was sparse, biased toward shipping lanes, and concentrated in the northern hemisphere. Estimates of ocean heat content therefore carried uncertainties large enough to obscure the trend for a decade at a time, which meant that the largest reservoir of the climate system was also the least measured.

Argo changed that with a design of remarkable simplicity. A float is dropped from a ship; it sinks to a programmed depth, drifts with the current, then rises while measuring temperature and salinity. At the surface it transmits its data and its position by satellite, then sinks again. Each cycle takes about ten days, each float lasts about four years, and the international programme maintains roughly four thousand of them, distributed across every ocean basin. The data are released within a day and are free to use.

The scientific consequences were immediate. Ocean heat content became measurable with an uncertainty small enough to detect the annual increase; the seasonal cycle of the upper ocean became visible; and the deep ocean's role in absorbing heat could be quantified rather than inferred. When the record was extended backward with older ship data and forward with Argo, it showed a steady accumulation of heat that has continued every year since, with the last decade the warmest in the record (Cheng et al., 2024).

Argo also demonstrates a governance lesson. The programme works because countries contribute floats, share data and accept common standards without a central authority that can compel them. It is coordinated by a small international secretariat and sustained by the self-interest of the participants, each of whom gets access to the whole dataset in exchange for contributing a share. That model — voluntary, standardised, open — has proved more durable than many formal treaties.

The gaps that remain are instructive. Ice-covered seas, marginal seas and the deep ocean below two thousand metres are still poorly sampled, and the floats themselves are biased toward regions where deployment is easy. The next generation of instruments, including deep Argo and under-ice floats, is being deployed to close those gaps. The lesson for other domains is that a monitoring system is never finished; it is a commitment to keep looking, renewed each time the technology changes.

3.12 Future Scenarios and Predictions: Systems, Feedbacks and Thresholds

The future of the climate system is not a single number but a set of conditional paths, and the condition is the same one that governs the previous chapter: the cumulative carbon emitted. What this chapter adds is the possibility of abruptness. The historical record shows that the system has changed gradually for most of the last ten thousand years, and that it has also changed abruptly in the more distant past — the Younger Dryas cold snap, the collapse of ice sheets at the end of the last glaciation — when a threshold was crossed. The question for the next century is whether the thresholds now being approached will be crossed slowly or quickly.

The present position is that several of the slower variables have already begun to move. The Atlantic overturning circulation is weaker than at any time in the last millennium according to proxy reconstructions, though the size and significance of the decline remain contested (Caesar et al., 2021). Arctic sea ice is on a trajectory to be essentially absent in September before mid-century under all but the lowest emission scenarios. Warm-water coral reefs have already suffered four global bleaching events, and the interval between them has shortened.

The assumptions that drive the scenarios are therefore not only about emissions but about the response of these slow variables. If the overturning circulation weakens moderately, European winters cool slightly while the tropics warm faster; if it collapses, the regional consequences are severe and the global consequences are still debated. If the Greenland ice sheet is committed to long-term loss, sea level rises for centuries regardless of what happens to emissions after mid-century. These are not alternative theories; they are different positions on a continuum of probability, and the monitoring systems described above are what allow the probabilities to be updated.

Table 3.1 System thresholds and what to watch for

Element	Threshold of concern	Time to observe change	Signal to monitor
Atlantic overturning	Weakening beyond ~15% of baseline	Decades	Florida Strait transport; Labrador Sea density
Arctic summer sea ice	Below 1 million km² in September	Years	NSIDC extent; ice thickness from CryoSat
Greenland ice sheet	Committed loss above ~1.5°C	Centuries	GRACE gravity; surface melt days
Amazon rainforest	Dieback beyond ~20–25% forest loss	Decades	Fire counts; dry-season length; carbon flux
Coral reefs	Frequent bleaching at 1.5°C; collapse at 2°C	Years	Degree-heating weeks; reef surveys

Two conclusions follow for decisions. The first is that monitoring is not a scientific luxury but an early-warning system for the planet: the sooner a threshold is detected, the more options remain. The second is that the standard tools of cost-benefit analysis are poorly suited to a system with thresholds, because they average over outcomes that are not interchangeable. A small probability of an irreversible change is not equivalent to a larger probability of a recoverable one, and any appraisal that treats them as equivalent will systematically underinvest in prevention.

What would change these conclusions? A clear, sustained weakening of the overturning circulation would raise the urgency of every adaptation decision in Europe and West Africa. A decade without further Arctic sea-ice decline would suggest that the system is more resilient than the models assume. Neither is likely to be decisive on its own, and that is the nature of a system observed at the pace of decades: the evidence arrives slowly, and the decisions cannot wait for it.

CHAPTER 4

EXTREME HEAT AND URBAN HEALTH

Theme: extreme heat, urban heat islands and the physiology of heat death

Heat is the deadliest climate hazard in most years, and the least visible. A flood leaves debris, a fire leaves ash, a hurricane leaves a coastline that can be photographed from space. A heatwave leaves bodies in apartments, morgues that fill quietly, and death certificates that list cardiac arrest rather than exposure. The number of people killed by heat is therefore almost always discovered later, by epidemiologists comparing the deaths that occurred with the deaths that would have been expected, rather than by officials counting at the scene.

The physical basis is simple. The human body produces heat continuously and must lose it to the environment. It does so by radiation, convection, evaporation of sweat and, to a small extent, by breathing. When air temperature approaches skin temperature, radiation and convection stop working and evaporation becomes the only route. When humidity is also high, evaporation slows, and the body's core temperature begins to rise. At a wet-bulb temperature near 35°C — the theoretical limit at which sweat can no longer cool the body — even a healthy adult at rest will overheat within hours.

Cities make this worse in ways that are measurable. Concrete and asphalt absorb solar radiation during the day and release it at night, so the urban heat island is strongest after sunset, when the body most needs relief. Buildings block wind, traffic adds waste heat, and the loss of trees removes shade and transpiration. The result is that a heatwave of the same meteorological magnitude can be survivable in a village and lethal in a city, and that the difference is concentrated in the neighbourhoods with the least green cover and the oldest housing.

This chapter examines heat as a physical hazard, as a physiological stress, as an urban design problem and as a public health emergency. It pays particular attention to the question that determines whether a heatwave becomes a disaster: who is exposed, who is warned, and who has somewhere cooler to go.

4.1 Measuring Heat: Temperature, Humidity and Wet-Bulb

A thermometer in the shade measures air temperature, but the body responds to a combination of temperature, humidity, wind and radiation. The heat index, familiar from weather forecasts, approximates the apparent temperature by combining temperature and relative humidity. Wet-bulb temperature goes further: it is the lowest temperature to which a surface can be cooled by evaporation, and it is the variable most closely tied to the body's ability to shed heat. A dry 45°C is dangerous; a humid 35°C wet-bulb is close to unsurvivable for more than a few hours.

The theoretical limit of 35°C wet-bulb was proposed by Sherwood and Huber (2010) on the basis of a resting metabolic rate. Laboratory work since then has found that the critical wet-bulb temperature for healthy young adults at rest is closer to 31°C, and lower still during light activity (Vecellio, Wolf, Cottle and Kenney, 2022). Older people, people with cardiovascular disease, pregnant women and children reach their

limits earlier. The practical implication is that conditions once considered rare are already occurring in South Asia and the Persian Gulf, and that the margin between discomfort and death is narrower than the 35°C figure suggests.

Heat is also cumulative. A single hot day is survivable; a week of hot nights is not, because sleep is when the body recovers. Public health agencies therefore increasingly report heat in terms of consecutive days above a threshold, or degree-days, rather than daily maxima. The distinction matters for warnings: a forecast of 40°C for one afternoon is a different emergency from a forecast of 35°C for seven consecutive days with nights above 28°C.

4.2 The Physiology of Heat Illness

Heat illness progresses along a continuum. Heat cramps are painful muscle spasms caused by salt loss; heat exhaustion produces weakness, dizziness, headache and nausea as blood pools in the skin and the heart struggles to maintain blood pressure; heatstroke occurs when the core temperature exceeds about 40°C and the body's thermoregulation fails. Heatstroke is a medical emergency with a mortality rate that can exceed 50 per cent if cooling is delayed, and it can leave survivors with permanent neurological damage.

The pathway to death is often cardiovascular rather than thermal. During a heatwave, the heart works harder to pump blood to the skin for cooling, while sweating reduces blood volume. People with narrowed coronary arteries, heart failure or kidney disease can die of a heart attack or kidney failure without ever reaching the temperature that defines heatstroke. This is why excess mortality during heatwaves is dominated by cardiovascular and respiratory causes, and why death certificates rarely mention heat.

Certain medications increase risk. Diuretics reduce blood volume; beta-blockers limit the heart's ability to increase output; anticholinergics reduce sweating; some antidepressants impair the perception of thirst. People who live alone may not notice the onset of symptoms, and people with dementia may not be able to seek help. The physiological and the social therefore combine: the same temperature that causes discomfort in a healthy adult can be lethal to an isolated elderly person on the top floor of a building without shade.

4.3 The Urban Heat Island

The urban heat island is one of the best-documented effects of human settlement. Cities are typically 3 to 7°C warmer than their rural surroundings during the day and can be more than 10°C warmer at night. The causes are well understood: dark surfaces absorb more solar radiation, buildings have a high thermal mass and release heat slowly, the geometry of streets traps radiation, vehicles and air-conditioning units release waste heat, and vegetation that would otherwise cool the air through transpiration is scarce.

The intensity varies enormously within a single city. Satellite measurements of land surface temperature in Jakarta, Delhi, Manila and São Paulo show differences of 5 to 10°C between neighbourhoods a few kilometres apart, with the hottest areas corresponding to dense informal settlements, industrial zones and bare ground, and the

coolest to parks, tree-lined streets and water bodies. Because these differences track income, the urban heat island is also a map of inequality: the people with the least capacity to buy cooling live in the hottest places.

Urban form determines how the heat island will change. A city that grows through sprawl with dark roofs and wide asphalt roads will intensify it; a city that protects green space, plants street trees, uses reflective materials and preserves ventilation corridors can keep it in check even as the climate warms. The decisions are made in building codes, zoning rules and investment plans, which is why heat policy is inseparable from urban planning.

4.4 Who Dies in a Heatwave

The epidemiology of heat mortality is remarkably consistent across countries. The elderly account for the majority of deaths, particularly those over 75 and those living alone. People with pre-existing cardiovascular, respiratory, renal or psychiatric conditions are at elevated risk. Outdoor workers, construction labourers, agricultural workers and delivery riders are exposed to the highest temperatures for the longest periods. Homeless people, prisoners, refugees and people in informal settlements are exposed both to heat and to the absence of cooling.

The Chicago heatwave of July 1995 is the canonical study. More than 700 people died in five days, most of them in their homes, and the distribution was not explained by temperature alone. Eric Klinenberg's social autopsy found that neighbourhoods with similar demographics and similar temperatures had very different death rates, and that the difference was associated with social isolation, fear of crime, and the quality of the local social infrastructure — whether people knew their neighbours, whether they trusted the city, whether they were willing to leave their apartments (Klinenberg, 2002).

The European heatwave of August 2003 killed an estimated 70,000 people, with France alone accounting for nearly 15,000. The deaths were concentrated among the elderly, many of whom were left in apartments without air conditioning while their families were on holiday, and the crisis exposed the absence of a public health plan for heat. The 2022 European summer, by contrast, killed about 61,700 people despite a much better warning system, which suggests that warnings alone are insufficient when the underlying vulnerability remains (Ballester et al., 2023).

4.5 Counting the Dead: Excess Mortality

Heat deaths are counted by comparing the number of deaths observed during a heatwave with the number expected from historical averages for the same period. The difference is excess mortality. The method is robust to the misclassification of individual death certificates, because it does not require the certifier to recognise heat as a cause; it only requires that the total number of deaths be counted accurately. This is why epidemiological estimates of heat deaths are usually much higher than official counts.

The method has limitations. Heat can displace deaths that would have occurred within weeks — a phenomenon known as mortality displacement, or harvesting — which means that a heatwave may accelerate deaths rather than cause them. Studies generally find that displacement accounts for a minority of excess deaths, especially for severe heatwaves, but the size of the effect is contested. Excess mortality also captures deaths from other causes that rise during heatwaves, such as heart attacks and kidney failure, which may not be recorded as heat-related.

The global burden is large. The Lancet Countdown estimates that heat-related mortality among people over 65 has increased by about 70 per cent since the early 1990s, and that more than 500,000 deaths a year are associated with heat. The figure is an estimate, not a count, and it depends on assumptions about temperature-mortality relationships that vary by location. But the direction is unambiguous, and the trend is one of the clearest signals that climate change is already affecting human health.

4.6 Heat Action Plans and Early Warning

Ahmedabad, India, adopted one of the first heat action plans in South Asia after a 2010 heatwave killed more than 1,300 people. The plan combines a colour-coded forecast system, cooling centres in public buildings, adjusted working hours for outdoor labourers, water distribution, and outreach to slum communities through community health workers. An evaluation published in 2018 estimated that the plan had prevented about 1,190 deaths a year, though the authors cautioned that the estimate depended on assumptions about the counterfactual (Hess et al., 2018).

The components of an effective plan are now fairly standard. A warning system that triggers specific actions at specific thresholds; a communication strategy that reaches people in their own languages through channels they trust; a registry of vulnerable people, particularly the elderly living alone; designated cooling spaces that are open, accessible and free; and a protocol for hospitals and emergency services to prepare for surges. The weakest link is usually the last mile: the warning exists, but the person at risk does not hear it, cannot act on it, or has nowhere to go.

Plans also fail when they are designed for the wrong hazard. A plan built around daily maximum temperature will miss a heatwave with moderate daytime temperatures and very hot nights, which is precisely the pattern that kills in humid cities. Plans built around a single city will miss rural workers and informal settlements outside the administrative boundary. And plans that are not rehearsed will not be implemented when the emergency arrives, because the agencies involved will be learning their roles in real time.

4.7 Heat at Work: Productivity and Labour

Heat affects economies through labour as well as through death. The International Labour Organization estimates that heat stress already reduces global working hours by about 2.2 per cent, equivalent to 80 million full-time jobs, and that the loss could rise to 2.4 per cent by 2030 under a high-emission scenario. Agriculture accounts for about 60 per cent of the lost hours, followed by construction and industry. The losses are concentrated in South Asia, Sub-Saharan Africa and Southeast Asia,

where outdoor work is common and cooling is scarce.

The mechanism is physiological. Above about 26°C wet-bulb, the body must divert blood to the skin to dissipate heat, which reduces the blood available to muscles and the brain. Physical capacity declines, error rates rise, and the risk of accidents increases. Workers adapt by taking breaks, shifting hours to early morning or night, and drinking more water, but these adaptations have limits, and they are often unavailable to workers paid by output rather than by time.

The economic consequence is not only lost output but lost income. A construction worker who cannot work during the hottest hours loses a fraction of daily wages; a farmer whose crop fails loses a season. Because informal workers rarely have heat insurance or sick leave, the cost is transferred to the household. Heat policy is therefore also labour policy, and the most effective interventions often involve collective bargaining, enforceable rest breaks and access to shade and water on site.

4.8 Cooling, Energy and Equity

Air conditioning saves lives, and its spread is one of the reasons heat mortality has fallen in high-income countries even as heatwaves have become more severe. The International Energy Agency counted about 1.6 billion air-conditioning units in 2016 and projected that the number could reach 5.6 billion by 2050, with most of the growth in developing countries. Cooling already accounts for about a tenth of global electricity consumption, and demand peaks during exactly the hours when the grid is most stressed.

The equity problem is that the people who most need cooling are least able to afford it. In many cities, electricity is unreliable, the upfront cost of an air conditioner is prohibitive, and landlords do not pay for efficiency improvements. Passive cooling — shade, insulation, reflective roofs, cross-ventilation, courtyards, trees — is cheaper and does not increase emissions, but it requires design decisions that are locked in for decades and is difficult to retrofit into informal housing.

There is also a feedback loop. Air conditioning releases waste heat into the street, raising the outdoor temperature and increasing the demand for cooling in neighbouring buildings. The effect is small relative to the heat island but measurable in dense districts. The solution is not to discourage cooling but to combine it with efficiency standards, building codes, urban greening and protection for people who cannot pay — a package that treats cooling as infrastructure rather than as a consumer good.

4.9 Designing Cooler Cities

Urban heat can be reduced by design, and the interventions are well understood. Street trees are the most effective single measure: a mature canopy can lower surface temperature by 10 to 15°C and air temperature by 2 to 4°C within its shade. Reflective roofs and pavements raise albedo and reduce heat absorption. Green roofs and walls insulate buildings and cool the air through transpiration. Water features, parks and ventilation corridors allow cool air to move through the city.

The evidence base has improved. A study of 601 European cities estimated that increasing tree cover to 30 per cent could have prevented about 2,600 heat deaths a year, roughly a third of the total, and that the benefit was largest in southern and eastern Europe (Iungman et al., 2023). Modelling of US cities found that reflective roofs and street trees could reduce afternoon temperatures by 1.5 to 3°C in dense neighbourhoods. The interventions pay for themselves through reduced cooling demand, improved air quality and lower mortality.

The obstacle is not knowledge but implementation. Trees take decades to mature, compete for space with utilities and parking, and require maintenance that municipal budgets often lack. Reflective roofs need building codes that are enforced. Ventilation corridors need planning rules that resist the pressure to build on every open space. The cities that succeed tend to have a dedicated heat office, a long-term budget and a political constituency that treats heat as a public health issue rather than an inconvenience.

4.10 The Limits of Adaptation

Adaptation to heat has real limits. Once wet-bulb temperatures approach the physiological threshold, no amount of shading, cooling or behaviour change can make outdoor work safe, and even indoor spaces without mechanical cooling become lethal. The regions most exposed — the Indus Valley, the Persian Gulf, the North China Plain, parts of West Africa — are also regions with large populations, limited air conditioning and fragile grids. In those places, adaptation buys time but does not remove the hazard.

The second limit is economic. Cooling for everyone would require a massive expansion of electricity generation, much of which would need to be low-carbon to avoid worsening the problem it addresses. The International Energy Agency estimates that efficient cooling could halve the growth in demand relative to a business-as-usual path, but efficiency requires standards, financing and enforcement that many countries lack. The third limit is behavioural: people underestimate their own vulnerability, especially if they have survived previous heatwaves, and warnings that are repeated without visible consequences lose credibility.

These limits do not make adaptation pointless. They define the boundary between what can be managed and what must be prevented. A city that plants trees, insulates homes, protects outdoor workers and opens cooling centres can prevent most heat deaths at 1.5°C or 2°C of warming. At 3°C or 4°C, the same city will face conditions that no amount of planning can fully offset. That is why heat policy must be paired with emission reduction: adaptation and mitigation are not alternatives but two halves of the same decision.

4.11 Case Study: The European Summer of 2022

The summer of 2022 was the hottest in Europe since records began, and it was preceded by a spring drought that had already lowered rivers and dried soils. Temperatures exceeded 40°C in Britain for the first time, reached 47°C in Portugal, and persisted across the Mediterranean for weeks. Wildfires burned more than 800,000 hectares across the continent, the largest area in the European record, and the combined

effect of heat and drought reduced maize and sunflower harvests by double-digit percentages.

The mortality estimate published in Nature Medicine attributed about 61,700 deaths to the summer, with the highest numbers in Italy, Spain, Germany and France (Ballester et al., 2023). The figure was lower than the 70,000 of 2003 in absolute terms, but the summer was hotter, which suggests that adaptation had reduced the death rate per degree even as the absolute burden remained enormous. The authors noted that the estimate depended on the choice of temperature-mortality model and that the true figure could be higher.

The pattern of deaths revealed the limits of adaptation. Most deaths occurred among people over 75, many with cardiovascular or respiratory disease, and many in households without air conditioning. Southern European countries with older housing stock and less air conditioning had higher death rates than northern countries with cooler climates but better building standards. The lesson is that vulnerability is not simply a function of latitude: it is a function of housing, income, age structure and the presence of a plan.

The policy response varied. France, which had built a national heat plan after 2003, activated a colour-coded alert system, opened cooling centres and sent text messages to vulnerable households. Spain and Italy introduced restrictions on outdoor work during the hottest hours. Britain, which had no comparable plan, declared a national emergency for the first time and cancelled appointments at hospitals that were already under pressure. The differences in outcome are difficult to isolate because many factors changed at once, but the comparison suggests that plans reduce deaths when they are specific, funded and rehearsed.

The summer also exposed the interaction between heat and other systems. Railways buckled in Britain, nuclear plants in France reduced output because river water was too warm for cooling, and hospitals cancelled surgeries because operating theatres could not be kept cool. These failures were not caused by the temperature itself but by infrastructure designed for a cooler climate. They illustrate the central point of this chapter: heat becomes a disaster when it encounters systems that were not built to tolerate it.

For planners, the case suggests three priorities. The first is to treat heat as a health emergency with a dedicated budget, not as a weather inconvenience. The second is to identify the people at risk before the heatwave, through registries of elderly and isolated residents, rather than trying to find them during it. The third is to protect the infrastructure that heat disables — power, water, transport, hospitals — because a heatwave that shuts down the grid becomes a different and much larger emergency.

4.12 Future Scenarios and Predictions: Heat in a Warmer World

The historical record shows a clear trend. The number of days with extreme heat has increased on every continent, the frequency of heatwaves has risen by a factor of roughly three since the 1950s, and the hottest year on record has been exceeded repeatedly in the past decade. Heat mortality has risen with it: the Lancet Countdown estimates that heat deaths among people over 65 have increased by about 70 per cent

since the 1990s, and that the trend is accelerating in Europe and South Asia.

Present conditions vary enormously by place. In Delhi, the summer of 2024 produced weeks of temperatures above 45°C and a record electricity demand; in Jakarta, the combination of high humidity and a strong urban heat island produces wet-bulb conditions that are already dangerous for outdoor workers; in Karachi, the 2015 heatwave killed more than 1,200 people in a city where many neighbourhoods lack reliable water and power. These are not future scenarios. They are the baseline from which the projections begin.

The assumptions that drive the projections are therefore about adaptation as much as about emissions. If cities invest in shade, cooling, warning systems and housing, mortality can fall even as temperatures rise. If they do not, mortality will rise faster than the temperature because exposure is growing with urbanisation and ageing. The scenarios below combine the two: they show what happens to heat risk under different emission paths and different levels of adaptive investment.

Table 4.1 Heat exposure in selected cities, present and 2050

City	Days above 35°C now	Days above 35°C in 2050 (SSP2-4.5)	Wet-bulb risk	Critical adaptation
Delhi	~100	~140	High in monsoon	Cooling centres; outdoor work bans
Karachi	~60	~100	Very high	Water supply; power reliability
Jakarta	~30	~70	Very high	Shade; drainage; social housing
Lagos	~20	~60	High	Grid; informal settlement upgrading
Athens	~30	~55	Moderate	Building codes; tree canopy
Phoenix	~110	~150	Low but extreme	Cooling for homeless; grid capacity

Two conclusions follow. The first is that heat risk is not determined by temperature alone: the same number of hot days produces very different mortality depending on housing, water, power and social protection. The second is that the largest gains come from interventions that are boring and cheap — shade, water, warnings, check-ins on elderly neighbours — rather than from technologies that are expensive and unproven. A city that does the boring things well can save thousands of lives at 1.5°C and still save many at 2.5°C, though the margin narrows.

What would change these conclusions? A breakthrough in cheap, efficient cooling would reduce mortality in hot countries but would increase electricity demand and emissions unless it were paired with clean generation. A collapse of the power grid during a heatwave would turn a manageable event into a catastrophe, as happened in Texas in 2021 and in India in 2012. And a wet-bulb event above the physiological threshold would create a situation in which adaptation is impossible and only evacuation remains.

CHAPTER 5

WATER SCARCITY AND DROUGHT

Theme: water scarcity, drought and the politics of a finite resource

In June 2019, the reservoirs that supply Chennai, a city of about nine million people, fell to one per cent of their capacity. The city government declared a water emergency, trucks delivered water to neighbourhoods that had none, and trains carried water from the Cauvery river more than three hundred kilometres to the city. Offices asked employees to work from home. Restaurants closed. The monsoon arrived in October, and within weeks the reservoirs were full again — but the crisis had lasted four months, and it had been predicted for more than a decade.

Chennai is not an arid city. It receives about 1,400 millimetres of rain a year, more than London or Paris. The crisis was not caused by a lack of rainfall in the long term but by a sequence of weak monsoons, the loss of lakes and wetlands to construction, the depletion of groundwater, and a supply system that had not kept pace with growth. The city had built its prosperity on the assumption that water would always be available somewhere nearby, and the assumption failed.

The same assumption is failing across much of the world. Fresh water is a small fraction of the planet's water — about 2.5 per cent — and most of that is locked in glaciers or buried deep underground. The water that is accessible, in rivers, lakes and shallow aquifers, is renewed by rainfall and snowmelt at a rate that is essentially fixed. Humanity uses about half of that renewable supply, and in many regions it uses more, mining groundwater that took thousands of years to accumulate.

This chapter examines water scarcity as a physical, economic and political problem. It distinguishes between drought, which is a temporary shortage of rain, and scarcity, which is a persistent imbalance between demand and supply. It examines the agricultural, urban and industrial demands that drive the imbalance, and the adaptations — storage, efficiency, reuse, desalination, pricing and institutions — that can restore it. It closes with the question that will shape the next century: what happens when a shared river no longer carries enough water for the people who depend on it.

5.1 The Physics and Geography of Fresh Water

The global water cycle moves about 500,000 cubic kilometres of water through evaporation and precipitation each year, but only about 40,000 cubic kilometres falls on land, and only a fraction of that is accessible to humans at a reasonable cost. Rivers and lakes hold about 90,000 cubic kilometres; groundwater holds about 10 million cubic kilometres, but most of it is deep, saline or unrenewable. The renewable freshwater supply that humanity can use is therefore a thin and unevenly distributed layer of the planet's water.

Distribution is as important as quantity. Nine countries hold about half of the world's renewable freshwater, while forty countries have less than 1,000 cubic metres per person per year — the threshold commonly used to define water scarcity. The Middle East and North Africa are the most water-scarce regions, but scarcity is also severe in parts of India, Pakistan, Mexico and South Africa, where growing populations

and economies have outgrown the supply that rainfall provides.

Climate change alters the cycle in three ways. It increases evaporation as temperatures rise, which reduces the water available in soils and reservoirs. It changes precipitation patterns, making rain more concentrated in intense events and less reliable as a steady supply. And it reduces snowpack and glacier storage, which acts as a natural reservoir that releases water slowly through the dry season. The third effect is especially important in the Himalayas, the Andes and the Alps, where millions of people depend on meltwater.

These physical facts do not by themselves produce scarcity. Scarcity is produced by the interaction between supply and the institutions that allocate it. A country can have abundant water and still experience shortages if its infrastructure leaks, its agriculture is inefficient, its cities grow without planning, or its water rights are allocated to uses that no longer make sense. Understanding that distinction is the first step toward treating scarcity as a problem that can be governed rather than as a fate that must be endured.

5.2 Drought: Four Definitions

Drought is usually defined in four ways. Meteorological drought is a prolonged period of below-average rainfall. Agricultural drought is a shortage of soil moisture during the growing season, which can occur even when rainfall is normal if the rain falls at the wrong time or runs off compacted ground. Hydrological drought is a depletion of rivers, reservoirs and groundwater, which usually develops more slowly than the other two. Socioeconomic drought occurs when the shortage begins to affect the supply of goods and services, including food, energy and water.

The four definitions explain why drought is often declared late. A government may see normal rainfall and conclude that there is no drought, while farmers are already losing a season because the rain came in one week instead of eight. Hydrological drought may persist for years after the rain returns, because aquifers and reservoirs recover slowly. The 2022 Horn of Africa drought was declared after five consecutive failed rainy seasons, by which time more than twenty million people were acutely food insecure.

Drought also has a long tail. The economic losses extend beyond the harvest: livestock die, rural incomes collapse, migration accelerates, and children drop out of school. A study of the 2011–2017 drought in the Horn of Africa found that the effects on nutrition and education persisted for years after the rains returned. Drought is therefore not an event that ends when the rain begins; it is a shock whose consequences are transmitted through households and institutions.

5.3 Groundwater: The Invisible Reserve

Groundwater is the largest accessible store of fresh water on Earth and the source of drinking water for about half the world's population. It is also the most vulnerable to overuse, because it is invisible: a farmer who pumps a well cannot see the aquifer declining, and a city that licenses new wells cannot easily measure the cumulative effect. The result is a global pattern of depletion that has been called a

slow-moving crisis.

The Ogallala aquifer under the US Great Plains is the canonical example. It supplies water to about 30 per cent of US irrigated land, and parts of it have lost more than half their saturated thickness. In some areas of Kansas and Texas, the water table has fallen by more than 50 metres, and farmers have abandoned irrigation. The aquifer recharges at a rate of a few millimetres a year, while withdrawals are measured in metres, which means that the resource is being mined rather than used.

Similar patterns appear in the Indo-Gangetic Plain, the North China Plain, Mexico's central plateau, and the Arabian Peninsula. In Jakarta, groundwater extraction has caused land subsidence of up to 25 centimetres a year in some districts, which increases flood risk and has contributed to the decision to move the national capital. The common feature is that the water is being used faster than it is replenished, and that the users have no direct incentive to stop because the cost of depletion falls on someone else, often on future generations.

There are success stories. Arizona's Groundwater Management Act of 1980 required new developments to demonstrate a hundred-year water supply, and it slowed depletion in the most populated areas, though it did not eliminate it. Israel's water economy combines drip irrigation, recycled wastewater and desalination to support agriculture and a growing population with less water per capita than any other developed country. The lesson is that groundwater can be governed, but only with measurement, enforcement and a legal framework that treats the aquifer as a shared resource.

5.4 Rivers Under Stress

The great rivers of the world are the clearest visible measure of water stress. The Colorado River no longer reaches the sea in most years; its flow has declined by about 20 per cent since the early twentieth century, and the two largest reservoirs, Lake Mead and Lake Powell, fell below 30 per cent of capacity in 2022. The river supplies water to about 40 million people and 1.2 million hectares of farmland across seven US states and northern Mexico, under an allocation system designed in 1922 that assumed a flow larger than the river has delivered.

The Yellow River in China has been fully allocated since the 1990s and sometimes fails to reach the sea; the Indus supplies a basin that is home to more than 250 million people and is dependent on snow and glacier melt that is projected to decline; the Murray–Darling in Australia has experienced severe droughts and ecosystem collapse in its lower reaches. In each case, the river is both a water supply and an ecosystem, and the two uses compete for the same flow.

The politics of rivers are upstream-downstream politics. A dam built upstream reduces the flow available downstream; a city that withdraws water for its residents reduces the water available for farmers; a country that claims a share based on historical use may be claiming water that other countries believe is theirs. International river basins cover about half the land surface of the planet and are home to about 40 per cent of its population, which is why water is often described as a source of both cooperation and conflict.

5.5 Water Quality and the Hidden Scarcity

Water can be physically present and still unusable. Contamination by pathogens, nitrogen and phosphorus from agriculture, heavy metals, arsenic and industrial chemicals reduces the supply of safe water as effectively as a drought. The World Health Organization estimates that about 2.2 billion people lack access to safely managed drinking water, and that contaminated water is associated with hundreds of thousands of diarrhoeal deaths a year, most of them among children under five.

Arsenic is a particularly insidious example. Wells drilled in Bangladesh and West Bengal in the 1970s to provide microbiologically safe water inadvertently tapped aquifers contaminated with naturally occurring arsenic. Tens of millions of people were exposed, and the health consequences — skin lesions, cancers, cardiovascular disease — appeared decades later. The episode is a reminder that a technological solution can create a new problem while solving an old one, and that water quality must be monitored as carefully as water quantity.

Salinity is another hidden form of scarcity. In coastal aquifers, over-extraction draws seawater inland; in irrigated regions, rising water tables bring salts to the surface, where evaporation concentrates them. Salinisation has degraded hundreds of millions of hectares of farmland and continues to spread in the Indus basin, the Tigris–Euphrates plain and parts of Australia. Unlike a drought, salinity does not end when the rain returns; it can make land unusable for decades.

5.6 Agriculture: The Largest User

Agriculture accounts for about 70 per cent of global freshwater withdrawals, and in some developing countries the share is closer to 90 per cent. Irrigation underpins about 20 per cent of cultivated land but produces about 40 per cent of global food, which means that any constraint on irrigation has an outsized effect on food security. The efficiency of that use is often low: flood irrigation loses large amounts to evaporation and runoff, and poorly maintained canals leak before the water reaches the field.

The gains from improving efficiency are large but not automatic. Drip irrigation can reduce water use by 30 to 50 per cent compared with flood irrigation while maintaining or increasing yields, but it requires capital, maintenance and a reliable supply of clean water. In India, the world's largest user of groundwater, subsidised electricity for pumping has encouraged over-extraction, and the response has been a mix of subsidies for drip systems, restrictions on new wells and, in some states, direct payments to farmers who switch to less water-intensive crops.

The deeper problem is that water is usually priced below its cost. A farmer who pays nothing for water has no incentive to economise, and a city that charges less than the cost of supply cannot maintain its pipes. Pricing reform is politically difficult because water is a basic need and because farmers are organised. The alternatives — quotas, tradable rights, tiered tariffs that protect basic consumption while charging more for luxury use — are all attempts to impose a price signal without denying people the water they need to survive.

5.7 Urban Water Systems

Cities concentrate demand and lose water before it reaches the tap. Non-revenue water — the difference between what a utility produces and what it bills — averages about 30 per cent globally and exceeds 50 per cent in some cities. The losses come from leaking pipes, illegal connections, metering errors and unbilled public use. In a city with a shortage, repairing the distribution network can be cheaper than building a new reservoir.

Urban growth also consumes the landscapes that once stored water. Chennai's lakes and wetlands were built over as the city expanded; Jakarta's groundwater was pumped because the piped network did not reach most neighbourhoods; Mexico City's aquifers were drained faster than they could recharge. The result is that the city becomes dependent on distant sources, which increases cost, vulnerability and political tension with the regions that supply it.

The responses that work combine supply, demand and storage. Singapore's water strategy — local catchment, imported water, desalination and recycled water, known as the four national taps — is the most cited example, and it has allowed a city-state with no natural aquifer to achieve water security. Windhoek, Namibia, has been recycling wastewater directly into the drinking supply since 1968. These cases show that scarcity is not simply a function of rainfall; it is a function of how deliberately a city manages the cycle.

5.8 Drought, Food and Prices

Drought transmits to the economy through food prices. When a major producing region loses a harvest, global prices rise, and the effect is largest for poor households that spend a large share of their income on food. The 2010–2011 drought in Russia and the subsequent export ban contributed to the global food price spike that year; the 2022 drought in Europe reduced maize and sunflower output and added to the price pressures already caused by the war in Ukraine.

The transmission is not automatic, because trade, storage and substitution can absorb part of the shock. But those buffers are thinner than they appear. Global grain stocks are held by a small number of countries, and exporters sometimes restrict shipments in a crisis, which amplifies the price increase. Importing countries respond by buying more, which amplifies it further. The result is that a regional drought can become a global price event through the interaction of policy responses rather than through the physical shortage alone.

For households, the consequence is a change in diet. When prices rise, families substitute cheaper staples, reduce protein and fresh produce, and sometimes skip meals. The nutritional effect persists long after prices fall, particularly for children, whose growth and cognitive development can be permanently affected. Drought is therefore a food security problem as much as a water problem, and the two cannot be addressed separately.

5.9 Water Conflict and Cooperation

Water has been called a source of war, but the historical record is more complex. The Pacific Institute's Water Conflict Chronology records thousands of water-related tensions, most of them local, and far more instances of cooperation than of armed conflict over water. The Nile, the Indus, the Mekong and the Rhine all have long histories of negotiation, and some of the most durable treaties in the world are water treaties.

That does not mean water is not a source of conflict. It means that the conflict usually takes the form of displacement, dispossession and violence within countries rather than war between them. Dam construction displaces communities; irrigation withdrawals deprive downstream farmers; pollution destroys fisheries; water tariffs provoke protests. The risk of interstate war over water is real but rare, and it is highest where water scarcity interacts with existing political hostility and weak institutions.

The cooperative record offers lessons. The Indus Waters Treaty of 1960 survived three wars between India and Pakistan and has been remarkably resilient, because it created a joint commission with technical authority and a dispute-resolution mechanism. The Nile Basin Initiative, the Mekong River Commission and the Senegal River Organisation show that shared basins can be governed when the parties have an interest in the information and the rules that the institution provides. Water is more often a reason to talk than a reason to fight.

5.10 Adaptation: Storage, Efficiency and Reuse

The portfolio of water adaptations is well known. Storage — dams, reservoirs, aquifer recharge, rainwater harvesting — bridges the gap between wet and dry seasons. Efficiency — drip irrigation, leak repair, industrial recycling — produces more output per cubic metre. Reuse — treating wastewater for irrigation or industry — turns a waste stream into a resource. Desalination — removing salt from seawater or brackish groundwater — creates new supply at a cost in energy and money.

Each option has limits. Dams displace people and silt up; the world has built most of the good sites. Efficiency can be offset by rebound effects, in which farmers use the saved water to expand irrigated area. Desalination is expensive, energy-intensive and produces brine that must be disposed of carefully; it is affordable for wealthy coastal cities but not for inland agriculture. Reuse requires treatment, monitoring and public acceptance, which is often the hardest part.

The most important adaptation is institutional. Water is managed through rights, prices, quotas and basin agreements, and the difference between a well-managed basin and a poorly managed one is often larger than the difference between a wet year and a dry one. Australia's Murray–Darling Basin Plan, which caps total withdrawals and allows water to be traded, is an example of an attempt to manage a basin as a system rather than as a collection of individual claims. It has been criticised for the environmental outcomes it has not delivered, but it demonstrates that the institutional design is at least as important as the engineering.

5.11 Case Study: Chennai's Day Zero

Chennai's water crisis did not arrive without warning. The city depends on four reservoirs and on groundwater, and the reservoirs had been declining for years as urban expansion covered the lakes that once captured monsoon rain. In 2018 the northeast monsoon failed, and by June 2019 the reservoirs held about one per cent of their capacity. The city government declared a water emergency and began to truck water to neighbourhoods, while the state arranged for trains to carry water from the Cauvery.

The crisis affected different residents differently. Households with private wells and storage tanks managed; households in informal settlements, which depended on public taps and tanker deliveries, queued for hours. Schools closed, restaurants reduced service, and information technology companies asked staff to work from home. The economic cost was estimated in the hundreds of millions of dollars, and the political cost was higher: the crisis became a symbol of the failure of urban water management in a rapidly growing city.

The response included short-term measures — tankers, trains, restrictions on non-essential use — and longer-term investments in desalination, sewage treatment and rainwater harvesting. The city also began to restore some of the lakes and wetlands that had been built over, recognising that the cheapest storage was the landscape itself. By 2024, the reservoirs had recovered, but the underlying imbalance had not been resolved: the city still depends on a monsoon that is becoming less reliable and on groundwater that is being depleted.

The case illustrates a pattern that appears in cities around the world. Water crises are rarely caused by a single year of low rainfall; they are caused by the accumulation of decisions that reduced storage, increased demand and deferred maintenance. The crisis is the moment when the accumulated deficit becomes visible. Because the deficit is institutional, the solution must be institutional: a plan that combines supply, demand, storage and governance, and that is implemented before the reservoirs empty.

For other cities, the lessons are specific. First, measure and publicly report the water balance, including losses in the distribution network and the rate of groundwater depletion. Second, protect the landscapes that store water — lakes, wetlands, floodplains and recharge areas — before they are built over. Third, diversify supply, including reuse and desalination where affordable, and price water in a way that funds maintenance. Fourth, identify the households that will be hit first and build a delivery system that reaches them, because the crisis will expose the difference between a city that has water and a city that has access to it.

5.12 Future Scenarios and Predictions: Water in 2050

The historical data show a consistent trend. Global water withdrawals have roughly tripled since 1950, driven by population growth, irrigation and industrial demand. Groundwater depletion has accelerated in the past two decades, and the area of land affected by drought has increased. The World Bank estimates that water scarcity could reduce GDP growth in some regions by up to 6 per cent by 2050, and the United Nations Convention to Combat Desertification estimates that drought could displace

700 million people by 2030.

Present conditions give the baseline. About 4 billion people experience severe water scarcity for at least one month a year, and 2.2 billion lack safely managed drinking water. Agriculture uses about 70 per cent of withdrawals, and in many basins the allocation of water exceeds the renewable supply. The gap is filled by groundwater mining, which cannot continue indefinitely. The question is not whether the deficit will be closed but how: by efficiency, by reallocation, by new supply, or by crisis.

The assumptions that drive the projections are population, diet, technology and governance. A world that converges toward efficient irrigation, recycles more water and limits groundwater withdrawal can meet demand with less water than it uses today. A world that continues current trends will face a shortfall of about 40 per cent between available supply and projected demand by 2030, according to the 2030 Water Resources Group. The scenarios below show what the main basins might look like under different assumptions.

Table 5.1 Water stress in major basins, present and 2050

Basin	Population (million)	Current stress	2050 gap (SSP2-4.5)	Main adaptation
Indus	~300	Extreme	30–40%	Drip irrigation; groundwater regulation
Colorado	~40	Extreme	20–30%	Demand caps; reuse; fallowing
Nile	~280	High	25–35%	Storage; efficiency; basin agreement
Citarum (Java)	~25	Very high	20–30%	Wastewater treatment; reforestation
Murray–Darling	~2.4	High	10–20%	Water trading; environmental flows
Tigris–Euphrates	~60	Very high	30–40%	Dam coordination; irrigation reform

Two conclusions follow. The first is that the gap is largest in basins where agriculture dominates and groundwater is unregulated, which means that the most important interventions are agricultural rather than urban. The second is that the gap can be closed with existing technology; the obstacle is the political cost of reallocating water from one user to another. A basin that caps withdrawals will produce less in the short term, and the farmers who lose water will resist.

What would change these conclusions? A breakthrough in cheap desalination would expand supply for coastal cities but not for inland agriculture. A change in diet toward less water-intensive food would reduce demand substantially, since meat and dairy require far more water per calorie than grains. And a sustained increase in precipitation in some basins, which some climate projections suggest, would relieve pressure in those places — but the same projections show declines in others, and the uncertainty is largest where the stakes are highest.

CHAPTER 6

FLOODS AND SEA LEVEL RISE

Theme: floods, coastal risk and the slow emergency of rising seas

On 29 October 2024, a line of thunderstorms stalled over the province of Valencia in eastern Spain and released more rain in eight hours than the region normally receives in a year. The Turia and Magro river systems filled within minutes. Cars were swept out of underpasses, bridges collapsed, and more than two hundred people died, most of them in towns that had been built on floodplains but had not experienced a flood of that magnitude in living memory. The regional government's alert reached mobile phones after the water had already risen.

Floods are the most frequent and most costly of climate disasters. Between 1995 and 2015, floods accounted for about 47 per cent of all weather-related disasters and affected 2.3 billion people, according to the United Nations Office for Disaster Risk Reduction. They kill in two ways: by drowning and trauma during the event, and by disease, displacement and loss of livelihood in the months that follow. The World Health Organization estimates that floods affect more people than any other natural hazard.

The physical mechanism is simple and getting stronger. Warmer air holds more water vapour, about seven per cent more for each degree of warming, following the Clausius–Clapeyron relation. When that moisture condenses, it falls in heavier bursts. At the same time, land surfaces have been made less absorbent by compaction, paving and the loss of wetlands, so a larger fraction of the rain becomes runoff. The combination means that a storm of the same meteorological magnitude delivers more water to rivers and streets than it did a century ago.

Sea level rise adds a second, slower dimension. The ocean has risen about 20 to 25 centimetres since 1880, and the rate has increased to about 4.5 millimetres a year. That sounds small until it is combined with storm surge, high tide and subsidence: a city that has sunk 20 centimetres and faces a sea 20 centimetres higher experiences a 40-centimetre change in the effective height of its defences. This chapter examines both dimensions, because in most coastal cities they arrive together.

6.1 The Physics of Flooding

Flooding occurs when the volume of water arriving at a place exceeds the capacity of the channel, the soil or the drainage system to carry it away. The volume is determined by rainfall intensity, the area that drains into the channel, the moisture already in the soil, and the speed with which water moves across the surface. A catchment that is dry and vegetated can absorb a storm that would overwhelm the same catchment after a wet month or after the vegetation has been removed.

The intensification of rainfall is one of the most robust findings in climate science. Observations show that extreme precipitation has increased in most regions, and that the increase is consistent with the Clausius–Clapeyron relationship. A warmer atmosphere can hold more water, and when the conditions for condensation are met, more of it falls at once. The effect is not uniform: some regions are becoming drier, and

in those regions the same mechanism can produce shorter, more violent storms separated by longer droughts.

Land use determines how much of that rain becomes flood. Urbanisation typically increases peak runoff by a factor of two to five, because pavement and roofs replace soil and vegetation. Deforestation in upland catchments increases erosion and reduces infiltration. Drainage systems designed for the rainfall of 1970 cannot carry the rainfall of 2024. The result is that floods are produced by the interaction of the atmosphere and the landscape, and that engineering a channel without managing the landscape upstream often moves the problem downstream.

6.2 River Floods and the Great Deltas

The largest concentration of flood risk is in the deltas of Asia. The Ganges–Brahmaputra, the Mekong, the Irrawaddy, the Indus and the Yangtze carry enormous volumes of water and sediment into low-lying plains that are home to hundreds of millions of people. The deltas are fertile because rivers deposit sediment there; they are vulnerable because the same sediment is subsiding, and because the channels that carry the water are constrained by embankments that prevent the floodplain from absorbing the flood.

The Mekong Delta is the clearest example. It supports about 17 million people and produces more than half of Vietnam's rice, but it is sinking as groundwater is pumped and as sediment is trapped behind dams upstream. Saltwater intrusion has advanced tens of kilometres inland during dry seasons, damaging rice fields and forcing farmers to switch to shrimp or to leave. The combination of subsidence, reduced sediment and sea level rise means that the delta is losing height relative to the sea even while the sea itself is rising.

The Indus delta illustrates a different mechanism. Upstream dams and barrages have reduced the flow reaching the delta, which has caused seawater to intrude and mangroves to retreat. The 2022 floods demonstrated that the river still carries enormous volumes in extreme years, but the infrastructure built to control those volumes — embankments, canals, barrages — had been designed for a different pattern of flow and failed in ways that concentrated the damage in particular districts. The lesson is that flood control infrastructure is only as good as the assumptions about flow on which it was designed.

6.3 Coastal Flooding and Sea Level Rise

Coastal flooding has three components: mean sea level, tides and storm surge. Mean sea level is rising because ocean water expands as it warms and because land ice is melting. Tides are predictable, which is why the timing of a storm matters: a surge that arrives at high tide can overwhelm defences that would hold at low tide. Storm surge is the temporary rise in sea level caused by wind and low pressure, and it is the component that does most of the damage in tropical cyclones.

Subsidence is the fourth component and often the largest. Many coastal cities are built on deltas or on reclaimed land, and they sink as groundwater is extracted and as the sediment beneath them compacts. Jakarta has subsided by up to four metres in some

districts over the past three decades; Bangkok, Manila, Ho Chi Minh City and Shanghai have all subsided by tens of centimetres. In these cities, the effective rate of sea level rise is the global rate plus the local rate of sinking, which can be ten times larger.

The consequences are already visible in the frequency of nuisance flooding. Miami Beach now experiences tidal flooding on sunny days several times a year, a phenomenon that barely occurred a generation ago. Jakarta experiences routine flooding during the monsoon and during high tides, even without rain. These events are not catastrophic individually, but they impose costs on transport, sanitation and property values, and they signal that the defences are being overtaken by the water.

6.4 Jakarta: The Sinking Capital

Jakarta is the most studied example of a city losing its race with the sea. About 40 per cent of the city lies below sea level, and some northern districts have subsided by more than four metres since the 1970s. The subsidence is caused mainly by groundwater extraction: only about half the population is connected to the piped water network, and the rest pumps from aquifers. The city has 13 rivers and a network of canals, and it receives about 1,700 millimetres of rain a year, concentrated in a wet season that regularly produces floods.

The 2020 New Year flood displaced about 400,000 people and killed more than 60. The 2013 and 2007 floods were comparable or worse. The city has responded with a sea wall, pumping stations, dredging and a giant offshore dyke project, but the project has been delayed by land acquisition, funding and disputes about its design. In 2022, the government passed a law to move the national capital to Nusantara in East Kalimantan, citing congestion, inequality and environmental vulnerability, including flooding.

The capital move does not solve Jakarta's problem; it relocates the state's administrative functions while leaving about 30 million people in the metropolitan area. The lesson is not that relocation is wrong but that it is not a substitute for managing subsidence. Stopping groundwater extraction, providing piped water to the whole city, restoring natural drainage and building defences are all necessary, and all are slow, expensive and politically difficult. Jakarta is therefore a test case for whether a large city can adapt faster than the ground beneath it sinks.

6.5 Flash Floods and Urban Drainage

Flash floods are the most lethal type of flood because they arrive with little warning. They occur when intense rainfall falls on steep terrain or on urban surfaces that cannot absorb it, and the water can rise within minutes. The Valencia flood of 2024, the Kuala Lumpur flash floods of 2021 and the Seoul floods of 2022 all followed this pattern: a convective storm, a drainage system designed for a milder climate, and a population that did not know which roads had become rivers.

Urban drainage design uses return periods: a pipe or canal is sized for a storm expected once every 5, 10 or 50 years. The assumption is that the climate is stationary and that the return period is stable. Both assumptions have failed. A system designed for a 10-year storm in 1990 may now experience that storm every three or four years, and the design storm itself has become more intense. The result is that cities are being

flooded by storms that their engineers reasonably expected them to survive.

The responses are known but slow. Separating stormwater from sewage reduces contamination and increases capacity. Green infrastructure — permeable pavements, rain gardens, retention ponds, constructed wetlands — absorbs and slows runoff. Zoning that keeps floodplains free of construction reduces exposure. Early warning systems that target drivers and pedestrians, rather than only river levels, save lives in flash floods. The common obstacle is that these measures require coordination across agencies and budgets that are usually organised by sector.

6.6 Who Dies in Floods

Flood mortality is concentrated in a few well-understood groups. People who are poor, elderly, disabled or living in informal settlements are at highest risk; so are people who attempt to cross flooded roads, which is the leading cause of death in flash floods in high-income countries. In low-income countries, drowning in homes and on agricultural land is more common, and women are often at higher risk because of restrictions on mobility and because they are less likely to have been taught to swim.

The 1970 Bhola cyclone in Bangladesh killed an estimated 300,000 to 500,000 people, making it one of the deadliest disasters in history. The 1991 cyclone killed about 138,000. In both cases, the deaths were concentrated among the poor who lived on low-lying islands and had no shelters. The subsequent investment in cyclone shelters, early warning and volunteer networks reduced deaths in later storms by orders of magnitude, which demonstrates that flood mortality is a function of preparedness rather than of the hazard alone.

The indirect mortality is larger and less visible. Floods contaminate water supplies, destroy crops, disrupt health services and displace populations, and these effects produce deaths from diarrhoeal disease, malnutrition and untreated chronic illness in the weeks and months after the water recedes. The 2010 Pakistan floods, which affected about 20 million people, were followed by a surge in waterborne disease and child malnutrition. Counting only the bodies found during the flood therefore understates the human cost.

6.7 Levees, Dams and Polders

Engineering has protected many cities from flooding, and it has also created new risk. Levees and embankments keep water out of the protected area, but they raise the water level in the channel and increase the consequences if they fail. The Netherlands has built the most sophisticated flood defence system in the world — the Delta Works, a network of dams, storm surge barriers, dikes and pumps that protects about a quarter of the country below sea level — and it maintains that system with a national programme and a statutory budget.

New Orleans is the counter-example. The city's levees failed during Hurricane Katrina in 2005, killing about 1,800 people and causing more than US$125 billion in damage. Investigations found that the failure was not caused by the storm exceeding the design height but by defective construction, inadequate maintenance and a design that had not accounted for the subsidence of the land and the loss of coastal wetlands that

had once absorbed surge. The lesson is that a defence is a system, and that its reliability depends on maintenance and on the condition of the landscape around it.

Polders — areas enclosed by dikes and drained by pumps — are a model for coastal cities with the resources to maintain them. They require continuous pumping, regular inspection and a political commitment that outlasts electoral cycles. Tokyo's underground reservoirs, Rotterdam's water plazas and Singapore's ABC waterways show a different approach: instead of keeping water out, they create space for it to go during extreme events. The choice between resistance and accommodation depends on the local geography and on how much risk a society is willing to accept.

6.8 Insurance, Retreat and Managed Realignment

Insurance is the financial technology that allows a society to absorb flood losses, and it is retreating from the most exposed places. In the United States, the National Flood Insurance Program has been insolvent for years and has been forced to raise premiums, which has made some properties unaffordable. Private insurers have withdrawn from parts of California and Florida, where wildfire and hurricane risk have increased faster than premiums could adjust.

When insurance becomes unavailable, the market begins to price the risk, and property values fall. That process is already visible in some coastal communities, where homes are sold at a discount or abandoned. The alternative to a disorderly retreat is a managed one: buyouts of the most exposed properties, restoration of floodplains, and relocation assistance for households that cannot afford to move on their own. The Netherlands has used managed realignment for decades, returning some polders to the rivers to reduce flood levels elsewhere.

Retreat is politically difficult because it is seen as giving up. But in many places it is the only option that is economically rational, and delaying it increases the eventual cost. The communities that have managed retreat best are those that began early, compensated households fairly, and combined relocation with investment in the places people moved to. Those that have managed it worst waited until a disaster forced the issue, which is the most expensive and least equitable way to adapt.

6.9 Ecosystem-Based Protection

Natural systems protect coasts at a fraction of the cost of engineering. Mangrove forests reduce wave height and attenuate storm surge; coral reefs break waves before they reach the shore; sand dunes and beaches absorb energy; wetlands store floodwater and release it slowly. A study of 59 coastal defence projects found that mangrove belts can reduce wave heights by 60 to 70 per cent over a few hundred metres, and that the benefit is greatest for small storms, which are the most frequent.

The protection is not absolute. Mangroves cannot stop a tsunami or a category 5 storm surge, and they require space and suitable conditions to survive. Coral reefs are themselves dying from warming and acidification, so the protection they provide is declining at the moment it is most needed. Dunes migrate and require space to migrate; seawalls prevent that migration and cause the beach to erode. Ecosystem-based protection works best as part of a layered strategy, not as a replacement for all

engineering.

The economics are favourable. Restoring mangroves is cheaper than building seawalls, and it provides additional benefits: fisheries nursery habitat, carbon storage, timber and tourism. Indonesia has the largest mangrove area in the world and has committed to restoring hundreds of thousands of hectares, though the record of restoration projects has been mixed because planting has sometimes occurred in places where mangroves cannot survive. The lesson is that ecosystem-based adaptation requires the same rigour as engineering: site selection, monitoring and maintenance.

6.10 Warning, Evacuation and Recovery

Most flood deaths are preventable with warning and evacuation, and the countries that have invested in both have seen mortality fall dramatically. Bangladesh's cyclone preparedness programme, built after the disasters of 1970 and 1991, combines forecasts, shelters, volunteers and drills, and it has reduced deaths in comparable storms from hundreds of thousands to dozens. The programme works because it is embedded in communities, not because it relies on technology alone.

Effective warning requires four elements: a forecast that is accurate enough to justify action; a message that reaches the people at risk in a language and format they understand; a population that knows what to do and has somewhere to go; and a decision-making system that is willing to act before the event is certain. The failure of any one element can turn a warning into a false alarm or a missed evacuation. The Valencia flood of 2024 failed at the second and third elements: the warning arrived late and the population did not know how quickly the water would rise.

Recovery determines whether a flood becomes a long-term disaster. Households that receive timely compensation, temporary housing and support to rebuild recover faster; those that do not may remain displaced for years. The 2004 Indian Ocean tsunami and the 2011 Japan earthquake both showed that recovery is fastest when reconstruction is combined with livelihood support and when local institutions are strengthened rather than bypassed. Floods are events; recovery is a process, and the process determines the final cost.

6.11 Case Study: Jakarta and the Sea

Jakarta's flood problem is the product of three histories. The first is geological: the city sits on a delta of thirteen rivers, built on soft sediment that compacts under the weight of construction. The second is hydraulic: colonial and post-colonial engineers built canals to drain the city, but the canals were designed for a smaller population and a cooler climate. The third is political: the city grew rapidly in the second half of the twentieth century, and much of that growth occurred in areas that were already floodplains or that were reclaimed from swamps.

The result is a city that sinks. Groundwater extraction, driven by the absence of a reliable piped supply, has lowered the water table and compacted the aquifers. In northern Jakarta, land has subsided by up to four metres since the 1970s, and the subsidence continues at rates of 1 to 15 centimetres a year. Because the sea is also rising, the relative rate of sea level rise in the city is several times the global average.

The 2020 New Year flood, the 2013 flood and the 2007 flood each displaced hundreds of thousands of people.

The response has been a combination of engineering and retreat. The city has built pumping stations, dredged canals, raised embankments and begun a sea wall. A giant offshore dyke, the National Capital Integrated Coastal Development project, was designed to enclose the bay and create a reservoir for water supply, but it has been delayed by land acquisition, financing and concerns about its environmental impact on fishing communities. In 2022, the government began moving the capital to Nusantara, citing Jakarta's congestion, inequality and environmental vulnerability.

The capital move is the largest planned relocation in Indonesia's history, and it is not a solution to Jakarta's flooding. It moves the administrative apparatus, not the 30 million people in the metropolitan area, and it may reduce the political attention that the city needs. The government has paired the move with investment in Jakarta's water supply and flood defences, but the scale of the investment is uncertain and the subsidence continues.

The case illustrates a general problem: cities can adapt to sea level rise if they have resources, institutions and time, but the three are rarely available together. Jakarta has resources, but its institutions are fragmented between the national government, the provincial government and the surrounding municipalities. It has time, but less than it thinks, because subsidence is faster than the global rate. And it has a plan, but the plan requires a level of coordination that has not yet been achieved.

For other coastal cities, the lessons are clear. Stop groundwater extraction before the land sinks; provide piped water to everyone, because otherwise people will pump. Protect and restore mangroves and wetlands, which cost less than seawalls and provide additional benefits. Build defences in stages that can be raised as the sea rises. And plan for retreat from the most exposed areas, because some of them cannot be defended at any acceptable cost.

6.12 Future Scenarios and Predictions: Coasts in 2100

The historical record is unambiguous. Global mean sea level has risen by about 20 to 25 centimetres since 1880, and the rate has accelerated from about 1.4 millimetres per year in the early twentieth century to about 4.5 millimetres per year today. The frequency of coastal flooding has increased even where the sea level rise is modest, because flooding depends on the combination of sea level, tide, surge and waves, and because subsidence and coastal development have increased exposure.

Present conditions show the scale of the exposure. About 600 million people live in low-elevation coastal zones less than 10 metres above sea level, and about 1 billion live within 100 kilometres of a coast. The value of assets in these zones is enormous: in the United States alone, coastal property worth trillions of dollars is exposed to storm surge. In developing countries, the exposure is concentrated in informal settlements that have no insurance and little public protection.

The assumptions that drive the projections are the emission path, the response of ice sheets, and the rate of subsidence. Under the low-emission scenario, sea level is projected to rise by about 0.3 to 0.4 metres by 2100; under the high-emission scenario,

by 0.6 to 0.8 metres; and if the West Antarctic ice sheet becomes unstable, by more than a metre. The difference matters less for the global average than for the frequency of extreme events: a 0.5-metre rise can turn a once-in-a-century flood into a once-in-a-decade flood, and in some places into an annual one.

Table 6.1 Coastal exposure in selected cities

City	Subsidence (cm/yr)	People exposed to 1-in-100 flood	2050 sea level (SSP2-4.5)	Adaptation
Jakarta	1–15	~2.5 million	+0.15 m	Sea wall; water supply; relocation
Manila	1–5	~1.5 million	+0.17 m	Drainage; mangrove restoration
Bangkok	1–3	~1 million	+0.16 m	Polders; groundwater control
Ho Chi Minh City	1–3	~1 million	+0.16 m	Embankments; delta management
Miami	0.2–0.5	~0.4 million	+0.18 m	Pumps; elevation; managed retreat
Alexandria	0.3–0.6	~0.5 million	+0.17 m	Sea wall; beach nourishment

Two conclusions follow. First, the dominant variable in many cities is not global sea level rise but local subsidence, which is caused by human decisions and can be slowed or stopped. Second, the cost of adaptation rises faster than the sea, because each increment of rise requires a larger and more expensive defence, and because the probability of failure increases as the design threshold is approached.

What would change these conclusions? A collapse of the West Antarctic ice sheet would make the high-end projections much more likely and would shift the policy focus from protection to retreat. A breakthrough in cheap, large-scale pumping or in floating construction could allow some cities to remain in place longer. And a reduction in emissions would slow the rise, but it would not stop it, because the ocean and the ice sheets respond over centuries. For coasts, mitigation determines the speed of the emergency; adaptation determines whether the city survives it.

CHAPTER 7

FOOD SECURITY AND AGRICULTURE

Theme: food security, crop physiology and the fragility of the global harvest

In May 2022, the government of India banned wheat exports with a single notification. The trigger was a heatwave in March that had arrived earlier and lasted longer than any recorded spring heat in the wheat belt, and had cut the harvest by an estimated five per cent. The ban was small in global terms — India accounts for a modest share of world wheat trade — but it arrived when the war in Ukraine had already removed a large part of the Black Sea supply, and prices that had been rising for a year rose further.

In July 2023, India restricted rice exports, and the effect was larger. India accounts for about forty per cent of the global rice trade, and the restriction, combined with an El Niño that was reducing production in other Asian countries, pushed rice prices to their highest level since 2008. Panic buying in the Philippines, Indonesia and West Africa amplified the increase. The rice was not physically scarce in most importing countries; the expectation of scarcity was enough to empty shelves.

These episodes show how food security fails. It is not only a matter of how much the world grows; it is a matter of how the harvest is distributed, stored, traded and priced. A heatwave in one region becomes a price shock in another through export restrictions, speculative buying and the thinness of global stocks. The poorest households, which spend half their income on food, absorb the shock first, and they absorb it by eating less and eating worse.

This chapter examines the physical effects of climate on crops, livestock and fisheries, and the economic and political mechanisms that turn those effects into hunger. It pays particular attention to the crops that feed most of humanity — wheat, maize and rice — and to the smallholders who grow much of the food consumed in low-income countries. It closes with the adaptations that are known to work and the reasons they are not spreading fast enough.

7.1 How Heat Affects Crops

Plants grow within a narrow temperature range, and the range differs by crop and by stage of development. Wheat and rice perform best around 20 to 25°C; maize around 25 to 30°C. Above the optimum, photosynthesis slows, respiration increases, and the plant partitions less energy into grain. The effect is not linear: a few days above 35°C during flowering can reduce grain set by a large fraction, even if the rest of the season is favourable.

Night temperature matters as much as daytime heat. Cool nights allow the plant to recover from the day's water loss and to complete the metabolic processes that fill grain. When nights remain above 25°C, respiration continues through the night, consuming the carbohydrates that would otherwise become yield. Studies in the Indo-Gangetic Plain have found that night temperature is a stronger predictor of rice yield than daytime maximum, which is one reason climate projections that report only daytime averages underestimate the risk.

The combined effect of warming and CO2 is complex. Higher CO2 increases photosynthesis in C3 crops such as wheat, rice and soy, an effect known as CO2 fertilisation, which can offset part of the heat penalty. But the effect is smaller in field conditions than in greenhouses, because nutrients and water become limiting, and it reduces the concentration of protein, zinc and iron in the grain. A crop that yields slightly more but is less nutritious is not necessarily an improvement for the people who eat it.

7.2 Water, Soil and the Limits of Irrigation

Agriculture consumes about 70 per cent of global freshwater withdrawals, and irrigation is the largest single use. In regions where rainfall is unreliable, irrigation is what makes the difference between one harvest and two, or between survival and failure. But irrigation depends on water that is increasingly contested: aquifers are being depleted, rivers are fully allocated, and snowmelt that feeds many systems is declining.

Soil is the other invisible input. A healthy soil holds water, cycles nutrients and supports the microbial communities that make them available to plants. Erosion, compaction and the loss of organic matter reduce that capacity, which makes crops more vulnerable to drought even when rainfall is unchanged. The global loss of topsoil is estimated at 24 billion tonnes a year, and the degradation of arable land is one of the reasons yields have stagnated in some regions despite improved varieties.

These constraints interact. A field with depleted soil and no irrigation cannot benefit from a heat-tolerant variety; a farmer who cannot afford fertiliser cannot compensate for nutrient dilution by adding more. The result is that the same climate shock produces very different losses on different farms, and that adaptation must address soil, water and seeds together rather than one at a time.

7.3 Pests, Pathogens and Pollinators

Warmer temperatures expand the range of many agricultural pests and change their life cycles. The fall armyworm, a moth native to the Americas, spread across Africa and then Asia in the 2010s and has become a major pest of maize; its expansion was aided by trade and by warmer winters that allowed it to survive in new regions. Aphids, whiteflies and locusts respond to temperature and rainfall in ways that are difficult to predict but increasingly favourable to outbreaks.

Plant diseases follow the same pattern. Fungal pathogens that were once limited to warm regions are moving poleward and upward; coffee leaf rust, which devastated Central American plantations in 2012–2013, was associated with higher temperatures and more variable rainfall. The response is often more pesticide, which raises costs and creates resistance, or the abandonment of susceptible crops, which reduces diversity.

Pollinators are a separate vulnerability. About three-quarters of the world's leading food crops depend at least partly on animal pollination, and pollinators are themselves sensitive to heat, drought and the loss of habitat. Studies in tropical regions have found that bee diversity declines with warming, and that the decline reduces yields of crops such as coffee, cocoa and many fruits and vegetables. The effect is easy to overlook because it is gradual, but it accumulates in the same way that soil loss does.

7.4 Extreme Events and Harvest Failure

Most of the year-to-year variability in global production comes from extreme events rather than from gradual warming. A drought during the grain-filling period, a flood at planting, a heatwave during flowering or a frost at the wrong moment can remove a large share of a national harvest. Because global stocks are thin relative to consumption, a series of such events in different regions can produce a global crisis.

The 2007–2008 food price crisis is the classic example. Drought in Australia, poor harvests in Europe and the United States, and the diversion of maize to biofuel combined to draw down stocks, and when major exporters restricted shipments, prices doubled. The result was riots in more than thirty countries and a reversal of poverty reduction in many others. The physical shortfall was modest; the policy response turned it into a catastrophe.

The 2022–2023 period repeated the pattern with different triggers. Heat in India, drought in the Horn of Africa, the war in Ukraine and an El Niño that was forecast to reduce rice output in Asia combined to raise prices and reduce availability in importing countries. Each event was within the range of natural variability; together they exceeded the capacity of the trade system to buffer, which is exactly what climate change is expected to make more frequent.

7.5 Livestock and Pastoralism

Livestock keep about 1.3 billion people employed and provide protein, traction and savings for pastoral communities. Heat affects animals directly: dairy cows reduce milk production when the temperature-humidity index exceeds about 72, poultry suffer mortality in hot weather, and grazing animals lose condition when rangelands dry out. The 2022 drought in the Horn of Africa killed millions of animals, destroying the savings of pastoral households.

Pastoralism is a system adapted to variability. Herders move with the rains, keep mixed herds of different species, and use social networks to share animals and information. Climate change stresses that system by making rainfall less predictable, reducing the number of good seasons, and increasing the frequency of droughts that occur before herds have recovered from the previous one. The result is not a single disaster but a slow erosion of the asset base.

The livestock sector is also a major source of emissions, contributing about 12 per cent of global greenhouse gases through enteric fermentation, feed production and land-use change. That creates a tension in policy: reducing livestock numbers would reduce emissions and free land for other uses, but it would also affect the livelihoods and nutrition of the poorest households. The solution is not a single global target but a set of context-specific changes: better feed, better breeds, better grazing management and a shift in high-income countries toward less meat-intensive diets.

7.6 Fisheries and Aquaculture

Fisheries provide about 17 per cent of the animal protein consumed globally, and more than three billion people depend on fish as a significant source of protein and micronutrients. Climate change affects fish through warming, acidification, oxygen loss and the movement of species toward cooler water. The result is that stocks are shifting away from the tropical regions where dependence is highest, and toward the poles where fishing fleets are less able to follow.

Coral reefs are the most vulnerable fishery habitat. Reef fish depend on the structure and productivity of live coral, and when reefs bleach and die, the fish community changes: herbivorous species take over, larger predatory fish decline, and catch rates fall. In Southeast Asia and the Pacific, where reef fisheries support millions of coastal households, the loss is both nutritional and economic. Warming of 2°C is projected to eliminate most of the world's reefs.

Aquaculture is the fastest-growing food sector and can offset part of the decline in capture fisheries, but it has its own vulnerabilities. Coastal ponds are exposed to sea level rise, storm surge and saline intrusion; the feed supply depends on wild fish and on crops that are themselves affected by climate; and disease outbreaks spread more easily in warm water. Aquaculture can be part of the adaptation, but it is not a substitute for protecting the ecosystems that underpin wild stocks.

7.7 Prices, Trade and Stocks

The global food trade buffers local production shocks, but it also transmits them. A country that experiences a drought imports more, which raises prices for everyone else; an exporting country that restricts shipments to protect its own consumers amplifies the shortage. The result is that the price effect of a climate shock can be larger than the physical effect, and that the poorest countries, which are most dependent on imports, pay the highest price.

Stocks are the first line of defence, and they are thinner than they appear. Global grain stocks are held by a small number of countries and by private traders who have no obligation to release them in a crisis. Official stocks are sometimes low because holding them is expensive and because governments prefer to intervene in markets when prices rise. The 2007–2008 crisis revealed that the stocks meant to stabilise the market were insufficient once exporters began to hoard.

Trade policy is therefore climate policy. A world that maintains open markets, limits export restrictions and invests in storage can absorb a series of regional shocks without a global crisis. A world that responds to every shortage with export bans and panic buying converts regional harvest failures into global price spikes, and the spikes fall hardest on the households with the least capacity to absorb them.

7.8 Smallholders and the Poorest Consumers

Smallholder farms produce about a third of the world's food and a much larger share in low-income countries, but they manage only a small fraction of the investment and irrigation. They are the most exposed to climate shocks and the least able to absorb them, because they lack insurance, credit, storage and access to markets. A single failed season can force a household to sell assets, withdraw children from school and borrow

at high interest, which reduces its capacity to cope with the next season.

Women are disproportionately affected. In many countries they are responsible for water, fuel and food processing, and they often have weaker land rights and less access to extension services. When a drought increases the time needed to collect water or fuel, girls are often the first to be withdrawn from school. The result is that a climate shock does not simply reduce income; it changes the trajectory of a household, sometimes permanently.

Social protection is the most effective instrument for preventing that trajectory. Cash transfers, school feeding, employment guarantee schemes and crop insurance can keep households from selling productive assets during a shock. Ethiopia's Productive Safety Net Programme, India's Mahatma Gandhi National Rural Employment Guarantee Act and Mexico's Progresa-Oportunidades have all been shown to reduce the impact of droughts on consumption, and they are often more cost-effective than emergency assistance after the crisis.

7.9 Adaptation: Seeds, Soil, Water and Insurance

The portfolio of agricultural adaptation is well established. Heat- and drought-tolerant varieties, earlier planting, crop diversification, improved irrigation, soil conservation and agroforestry can reduce losses under moderate warming. Climate-smart agriculture, a term that covers many of these practices, has been adopted widely in development policy, though the evidence for its effectiveness varies by context and by how it is measured.

Genetic improvement is the most powerful tool and the slowest to deploy. Breeding a new variety takes eight to twelve years, and the climate that the variety will face when it is released is not the climate for which it was selected. Gene banks hold the diversity that breeders need, but they are underfunded, and the traits that matter — heat tolerance at flowering, deep roots, resistance to new pests — are often found in landraces that are being lost as farmers adopt modern varieties.

Insurance and finance are the enabling conditions. A farmer who cannot borrow cannot invest in irrigation or improved seed; a farmer without insurance may avoid risk by planting low-yielding but reliable crops. Index-based insurance, which pays out when rainfall or temperature crosses a threshold, has been promoted as a solution, but its uptake has been limited by basis risk, cost and trust. The most successful programmes combine insurance with credit, extension and access to markets, because insurance alone does not solve the underlying problem of low productivity.

7.10 The Food System as a Whole

Food security is not only a question of production. It depends on storage, transport, processing, retail and waste. About a third of the food produced is lost or wasted, and reducing that waste is equivalent to increasing production without using more land or water. It depends on diets: about a third of crop calories are fed to animals, and a shift toward less meat-intensive diets in high-income countries would free land and water for human food.

It depends on the political economy of food. Subsidies that encourage water-intensive crops in dry regions, tariffs that protect inefficient producers, and land tenure systems that exclude smallholders all reduce the system's capacity to adapt. The barriers are rarely technical; they are institutional and distributional, which is why the chapters on governance and justice are as relevant to food as the chapters on agronomy.

The final point is that the food system and the climate system are coupled. Agriculture, forestry and land use account for about a fifth of global greenhouse gas emissions, and food loss and waste add more. A transformation that reduces emissions from agriculture — through better fertiliser management, less deforestation, healthier diets and less waste — would also make the food system more resilient, because it would reduce the pressure on land and water that makes it fragile.

7.11 Case Study: The 2023 Rice Shock

Rice is the most important food crop in the world by the number of people who depend on it. About 3.5 billion people obtain a fifth or more of their calories from rice, and most of them live in Asia, where the crop is often produced by smallholders and consumed close to where it is grown. The global rice trade is thin relative to production — only about ten per cent of output crosses a border — which means that a small disruption in a major exporter can move prices for everyone.

In 2023, that is what happened. India, which accounts for about forty per cent of global rice exports, restricted shipments after a poor monsoon and rising domestic prices. The restriction coincided with an El Niño that was reducing production in Thailand, Vietnam and the Philippines, and with a global market that was already nervous after the war in Ukraine. Rice prices rose by about 30 per cent in a few months, and importing countries responded by buying more, which pushed prices higher.

The effects were felt far from the harvest. In West Africa, where rice is a staple and imports account for a large share of consumption, governments subsidised purchases and released stocks. In the Philippines, the government capped rice prices, which reduced supply and created queues. In Indonesia, the state logistics agency increased imports and distributed rice through social assistance. None of these countries had experienced a harvest failure; they were experiencing the transmission of a policy decision through a thin market.

The case shows that food security is a systems problem. The physical shock was a weak monsoon, which is within the range of natural variability. The crisis was produced by the interaction of that shock with export restrictions, thin stocks, panic buying and the concentration of trade in a few exporters. Climate change increases the probability of the physical shock, but the severity of the crisis depends on how the trade system responds.

The lessons are specific. First, importing countries should hold strategic reserves and diversify their suppliers, rather than relying on the cheapest source. Second, exporting countries should coordinate restrictions rather than impose them unilaterally, because unilateral restrictions invite retaliation and panic. Third, international institutions should monitor stocks and prices in real time, so that governments can distinguish a physical shortage from a speculative one. Fourth, social protection should

be in place before the crisis, because the households that are pushed into hunger by a price spike cannot wait for a new programme to be designed.

7.12 Future Scenarios and Predictions: Feeding the World in 2050

The historical evidence is that warming has already reduced yields relative to what they would have been without it. A meta-analysis of field experiments found that each degree of warming reduces global wheat yields by about 6 per cent, maize by 7.4 per cent, rice by 3.2 per cent and soybeans by 3.1 per cent, before adaptation (Zhao et al., 2017). The losses are larger in tropical regions, where crops are already close to their heat limits, and smaller in temperate regions, where longer growing seasons can partly offset the heat.

Present conditions give the baseline. Global food production has kept pace with population growth for most of the past half-century, but the rate of yield improvement has slowed, and about 735 million people were undernourished in 2022. The food system is also responsible for about a third of global greenhouse gas emissions, which means that it must reduce emissions while increasing production — a combination that is difficult but not impossible.

The assumptions that drive the projections are population growth, diet, technology and trade. A world that stabilises population, reduces food waste, shifts toward healthier diets and invests in adaptation can feed a larger population with less land and water. A world that continues current trends will face rising prices, increasing dependence on imports and a growing number of climate-driven food emergencies. The scenarios below show the range.

Table 7.1 Crop yield sensitivity and 2050 projections

Crop	Yield change per °C	2050 change (SSP2-4.5)	Main adaptation	Key region
Wheat	−6%	−10 to −15%	Heat-tolerant varieties; earlier sowing	South Asia; Europe
Maize	−7.4%	−15 to −20%	Drought tolerance; diversification	Sub-Saharan Africa; US
Rice	−3.2%	−5 to −10%	Flood-tolerant varieties; irrigation	South and Southeast Asia
Soybean	−3.1%	−5 to −8%	Crop rotation; heat tolerance	Brazil; US; Argentina
Coffee	−10 to −20%	−25 to −50%	Shade; altitude shift; new varieties	Ethiopia; Colombia; Vietnam
Livestock	−5 to −10% (milk)	−10 to −20%	Shade; breeds; feed quality	South Asia; Sahel

Two conclusions follow. First, the crops that feed the poorest people are the most sensitive to heat, and the regions where they are grown are the ones where adaptation is least funded. Second, the projected losses are not catastrophic in the global aggregate, but they are severe in particular places, and they interact with poverty, conflict and weak institutions to produce famine rather than shortage.

What would change these conclusions? A breakthrough in heat-tolerant varieties, or in the efficiency of photosynthesis, would reduce the losses, but it would take a decade or more to reach farmers. A shift toward plant-based diets in high-income countries would reduce demand for feed grains and free land for human food. And a sustained increase in investment in irrigation, storage and rural roads would reduce the post-harvest losses that currently account for a large share of the gap. The most important variable, however, is emissions: the difference between 1.5°C and 3°C is the difference between a food system that can adapt and one that cannot.

CHAPTER 8

ECOSYSTEMS AND BIODIVERSITY LOSS

Theme: ecosystem collapse, extinction risk and the services that nature provides

In the first half of 2024, the Great Barrier Reef experienced its fifth mass bleaching event in eight years. Surveys by the Australian Institute of Marine Science found that about 73 per cent of surveyed reefs had bleached, with the highest levels in the southern section, which had previously escaped the worst heat. Bleaching is not death: a bleached coral has expelled the algae that feed it and can recover if the water cools quickly. But the interval between events has become shorter than the time a reef needs to recover, and that is what turns a recurring disturbance into a permanent decline.

The reef is the most visible example of a general pattern. Warming is reorganising the living world: species are moving toward the poles and up mountains, the timing of seasons is shifting, and the disturbances that ecosystems evolved with — fire, drought, storms — are becoming more frequent and more severe. The result is not a uniform decline but a reshuffling, in which some species thrive, many decline, and the relationships among them change in ways that are difficult to predict.

The consequences for people are not sentimental. Ecosystems provide food, fibre, clean water, pollination, flood protection, carbon storage and medicine. They regulate the climate and the water cycle, and they underpin the livelihoods of billions of people. When a reef dies, the fish that depend on it decline, and the coastal communities that depend on the fish lose protein and income. When a forest dries out, it stops absorbing carbon and starts releasing it, which accelerates the warming that caused the drying.

This chapter examines the main mechanisms by which climate change affects ecosystems, the thresholds beyond which the changes become irreversible, and the services that are at stake. It pays particular attention to three systems that are both highly vulnerable and highly consequential: coral reefs, tropical forests and permafrost. It closes with the conservation and restoration strategies that can buy time, and with an honest assessment of what they cannot save.

8.1 The Biodiversity Baseline

The living world is already under pressure from land conversion, overexploitation, pollution and invasive species, and climate change is arriving on top of those pressures. The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services estimated in 2019 that about one million species are threatened with extinction, and that the average abundance of native species in most major terrestrial biomes has fallen by at least 20 per cent since 1900. The Living Planet Index, which tracks monitored vertebrate populations, recorded an average decline of 69 per cent between 1970 and 2018 (WWF, 2022).

These numbers describe a system already in deficit. Climate change adds a new set of pressures: heat stress, altered rainfall, ocean acidification, rising sea level and more frequent extreme events. The interaction is what makes the situation dangerous. A

forest fragment that is already degraded by logging has less capacity to survive a drought; a coral reef that is already polluted has less capacity to recover from bleaching; a species whose range has been reduced by land clearing has fewer places to move when its remaining habitat becomes too warm.

The baseline also matters for what can be saved. Ecosystems that retain their structure and diversity are more resilient than those that have been simplified, and the difference can be large. A intact mangrove forest attenuates storm surge; a cleared one does not. A diverse grassland recovers from drought; a monoculture may not. Protecting what remains is therefore as important as restoring what has been lost, and it is usually cheaper.

8.2 Range Shifts and Phenology

Species respond to warming by moving. On land, they move up mountains and toward the poles; in the ocean, they move toward cooler water and greater depths. A meta-analysis of observed shifts found that terrestrial species have moved to higher elevations at an average of about 11 metres per decade and toward the poles at about 17 kilometres per decade, while marine species have shifted at about 72 kilometres per decade (Chen et al., 2011). The rates are fast in ecological terms and slow in comparison with the rate at which the climate is changing.

The problem is that many species cannot move fast enough, and some have nowhere to go. A species that lives on a mountain top has no higher ground; a species that lives in a lake has no other lake; a species that lives on a low island has no landward refuge if the coast is developed. The result is that range shifts benefit mobile generalists and penalise specialists, which changes the composition of communities even when total biodiversity is maintained.

Phenology — the timing of seasonal events — is shifting even faster. Earlier springs advance the emergence of insects, the flowering of plants and the arrival of migratory birds, but the shifts are not synchronised across species. A bird that arrives at its breeding ground on its traditional date may find that the insects it feeds on have already peaked. These mismatches are difficult to observe and can reduce breeding success without any obvious change in the landscape.

8.3 Coral Reefs

Coral reefs occupy less than one per cent of the ocean floor but support about a quarter of marine species and provide food, income and coastal protection for hundreds of millions of people. They are also among the most sensitive ecosystems to warming, because the symbiotic algae that live in coral tissue are expelled when the water exceeds their thermal tolerance. The result is bleaching, and if the heat persists, the coral dies.

The frequency of bleaching has increased sharply. Before the 1980s, mass bleaching was rare; since 1998 there have been four global bleaching events, and the interval between them has shortened from about twelve years to about six. The Great Barrier Reef has bleached five times since 2016, and the 2024 event affected reefs that had previously been considered refuges because they lived in cooler water. Global coral cover has declined by about half since the 1950s, and the decline has accelerated.

The projections are stark. At 1.5°C of warming, about 70 to 90 per cent of warm-water reefs are projected to decline; at 2°C, more than 99 per cent. These are not gradual declines; they are the loss of the physical structure that supports the ecosystem, followed by the collapse of the fish community. Some corals may survive in cooler refuges, and some may adapt, but the reef as a functioning system is unlikely to persist in the form that has existed for thousands of years.

8.4 Tropical Forests and the Amazon

Tropical forests store about half of the carbon in terrestrial vegetation and support a large share of the world's biodiversity. They also influence rainfall over vast regions: the Amazon recycles moisture that falls as rain across South America, and the Congo basin does the same for central Africa. When forests are cleared or degraded, both the carbon and the rainfall are lost, and the two losses reinforce each other.

The Amazon is the most studied case. About 17 per cent of the forest has been cleared, and a further fraction is degraded by logging and fire. Climate models and observational studies suggest that beyond a threshold of about 20 to 25 per cent deforestation, the forest may lose its ability to recycle moisture and begin to convert to savannah in the eastern and southern parts. The 2023–2024 drought, the most severe in the instrumental record, lowered river levels to historic lows and created conditions for fires that burned millions of hectares.

The forest is also a carbon bomb. If it dies back, the carbon it stores will be released, accelerating warming and making the dieback more likely. The Amazon therefore contains a positive feedback that is not fully represented in most climate models, and the uncertainty is asymmetric: the risk of a rapid transition is small in any given decade but the consequences are irreversible on human timescales. Protecting the forest is therefore both a mitigation measure and an adaptation measure, and it is one of the cheapest available.

8.5 Permafrost and the Tundra

Permafrost — ground that remains frozen for at least two consecutive years — covers about a quarter of the Northern Hemisphere and stores an estimated 1,400 to 1,600 billion tonnes of carbon, roughly twice the carbon in the atmosphere. As the Arctic warms three to four times faster than the global average, the ground thaws, and microbes decompose the organic matter that has been frozen for millennia, releasing carbon dioxide and methane.

The release is already measurable. Arctic rivers carry increasing amounts of dissolved organic carbon; thaw slumps and thermokarst lakes are appearing in regions that were previously stable; and winter carbon dioxide emissions from northern soils have increased. The processes are slow relative to fossil fuel emissions, but they are large and difficult to reverse, and they are not included in the carbon budgets that governments use to plan emission reductions.

Thawing permafrost also damages infrastructure. Roads, pipelines, buildings and airports built on frozen ground deform when the ground subsides, and the cost of repair is already significant in Alaska, Canada and Russia. For communities that depend on ice

roads and frozen ground for transport, the loss of permafrost is a direct threat to access to food, fuel and medical care.

8.6 Freshwater Ecosystems

Freshwater ecosystems — rivers, lakes, wetlands and aquifers — are among the most degraded and the most important for people. They provide drinking water, irrigation, fisheries, transport and flood protection, and they support a disproportionate share of biodiversity. Climate change affects them through warming, changed flow, reduced snowmelt, increased evaporation and more frequent droughts and floods.

Warming lakes stratify earlier and more strongly, which reduces the mixing that brings nutrients from the bottom and oxygen from the surface. The result is low-oxygen zones that kill fish and release phosphorus and methane from sediments. Rivers fed by glaciers change from snowmelt-dominated to rain-dominated flow, which reduces summer flow precisely when demand is highest. Wetlands dry out, releasing the carbon they stored and losing their capacity to buffer floods.

Freshwater species are declining faster than terrestrial or marine species. The Living Planet Index for freshwater populations shows an average decline of about 83 per cent since 1970, driven by dams, pollution, water abstraction and invasive species, with climate change now adding to those pressures. The loss matters directly to people: inland fisheries provide protein for hundreds of millions of people, and the loss of river connectivity blocks the migration of fish that communities depend on.

8.7 Ocean Acidification and Deoxygenation

The ocean absorbs about a quarter of the carbon dioxide that humans emit, and when carbon dioxide dissolves in seawater it forms carbonic acid. The result is that surface ocean pH has fallen from about 8.2 to 8.1 since the pre-industrial period, a 30 per cent increase in hydrogen ion concentration. The change is large by geological standards, and it is occurring far faster than at any time in the past 65 million years.

Acidification affects organisms that build shells and skeletons from calcium carbonate. Corals, oysters, mussels, pteropods and some plankton are particularly sensitive, especially in the cold waters where carbonate is least available. Experiments show reduced calcification, thinner shells and slower growth, and field observations in the Southern Ocean and the Arctic already show effects. The consequences propagate through food webs, because the organisms affected are often at the base of the chain.

Deoxygenation is a related problem. Warmer water holds less dissolved oxygen, and nutrient runoff from agriculture increases the growth of algae that consume oxygen when they decompose. The result is an expansion of low-oxygen zones in coastal waters and the open ocean, which forces fish into shallower water and reduces the habitat available to species that need cold, oxygen-rich water. The combination of warming, acidification and deoxygenation is sometimes called the triple threat to marine life.

8.8 Extinction Risk and Climate Velocity

Extinction is the endpoint of a process that begins with range contraction, population decline and loss of genetic diversity. Climate change contributes to that process by reducing habitat, changing the timing of life cycles and increasing the frequency of extreme events. The species most at risk are those with small ranges, low mobility, specialised habitat requirements and long generation times — which includes many amphibians, mountain species, island endemics and large mammals.

The concept of climate velocity measures how fast a species would have to move to stay within its preferred temperature range. In flat terrain, the velocity is high because a small change in temperature corresponds to a large change in latitude; in mountainous terrain, it is lower because the species can move up a short distance. The concept helps explain why species in plains and on low islands are more vulnerable than species in mountains, and why connectivity between habitats is essential.

Projections of extinction risk depend on assumptions about dispersal and adaptation, and the range is wide. A widely cited analysis found that at 2°C of warming, about 18 per cent of insects, 16 per cent of plants and 8 per cent of vertebrates would lose more than half their geographic range, and that at 3.2°C the figures rise to 49 per cent, 44 per cent and 26 per cent (Warren et al., 2018). Range loss is not extinction, but it is the precursor, and it is much easier to measure.

8.9 Ecosystem Services and Human Dependence

The value of ecosystems to people is usually described as services: provisioning services such as food, fibre and fresh water; regulating services such as climate, flood and disease regulation; supporting services such as soil formation and nutrient cycling; and cultural services such as recreation and spiritual significance. The categories are a useful accounting device, but they understate the extent to which the services are interdependent and irreplaceable.

Pollination is a clear example. About three-quarters of the leading food crops depend at least partly on animal pollinators, and the value of that service is estimated in the hundreds of billions of dollars. If pollinators decline, the effect is not a uniform reduction in food supply but a shift toward crops that do not need them — wind-pollinated grains — which reduces dietary diversity and increases the cost of fruits, vegetables and nuts.

Coastal protection is another. Mangroves, reefs and dunes reduce wave energy and storm surge, and the loss of these systems increases the cost of coastal defence. A study of the 2004 Indian Ocean tsunami found that villages behind intact mangroves suffered fewer casualties and less damage than villages on exposed coasts, though the effect depended on the thickness of the belt and the angle of the waves. Ecosystem services are not a luxury; they are part of the infrastructure.

8.10 Conservation, Restoration and Their Limits

Conservation can buy time. Protecting large, connected areas allows species to move; reducing non-climate pressures such as pollution, overfishing and invasive species increases resilience; restoring degraded habitats can recover some of the services that have been lost. The Kunming–Montreal Global Biodiversity Framework,

adopted in 2022, set a target of protecting 30 per cent of land and sea by 2030, and many countries have committed to it.

Restoration is harder than protection. Planting trees is easy; growing a forest that stores carbon, supports biodiversity and provides livelihoods is not. Restored mangroves often fail because they are planted in the wrong place or because the hydrology that sustains them has been altered. Restored coral reefs are limited by the same warming that killed them. Restoration can recover structure within a decade or two, but recovery of biodiversity and function takes much longer, and some systems cannot be restored at all.

The limits are ultimately set by temperature. Conservation can protect species from land conversion, hunting and pollution, but it cannot protect them from a climate that exceeds their tolerance. That is why the biodiversity crisis and the climate crisis are the same crisis, and why the targets for one cannot be met without the targets for the other. Protecting 30 per cent of the planet is necessary but not sufficient; the other 70 per cent, and the atmosphere above it, determine whether the protected areas remain viable.

8.11 Case Study: The Great Barrier Reef

The Great Barrier Reef is the largest living structure on Earth, extending more than 2,300 kilometres along the Queensland coast and comprising about 3,000 individual reefs. It is a UNESCO World Heritage site, a source of income for tourism worth billions of dollars, and a cultural landscape for the Traditional Owners who have managed its waters for thousands of years. It is also the best-documented example of an ecosystem being pushed beyond its thermal limits.

The reef has bleached five times since 2016: in 2016, 2017, 2020, 2022 and 2024. The 2016 event killed about half the corals in the northern section; the 2017 event affected the central section; the 2020 event was the most widespread on record; and the 2024 event affected reefs that had previously escaped because they lived in cooler water. Between events, coral cover recovered in some places, particularly in the southern section, but the recovery has been repeatedly interrupted.

The reef is also affected by other pressures. Runoff from agriculture carries sediment, nitrogen and pesticides into coastal waters, which reduces water quality and encourages algal growth. Crown-of-thorns starfish outbreaks consume coral; cyclones break it; and fishing removes herbivorous fish that would otherwise control algae. Management has addressed some of these pressures — runoff targets, fishing closures, starfish control — but it cannot address the heat.

The Australian government has invested in adaptation: coral breeding programmes, assisted migration of heat-tolerant corals, shading and cooling experiments, and research on cloud brightening. These are small-scale and uncertain, and they cannot protect 2,300 kilometres of reef. The most effective adaptation is emission reduction, which is outside the jurisdiction of the reef's managers, and that is the central dilemma of coral conservation: the people with the authority to protect the reef do not control the variable that determines its future.

The case has wider lessons. First, repeated disturbances are more damaging than isolated ones, because recovery time is essential. Second, local pressures compound climate stress, so reducing them buys time even if it cannot stop the decline. Third, adaptation technologies are not a substitute for mitigation, and presenting them as one delays the action that matters. Fourth, the loss of a reef is not only an ecological event; it is an economic and cultural event for the communities that depend on it.

8.12 Future Scenarios and Predictions: The Living World in 2100

The historical record shows that ecosystems are already changing. The Living Planet Index recorded a 69 per cent average decline in monitored vertebrate populations between 1970 and 2018; global coral cover has halved since the 1950s; and the area of primary forest lost each year remains above 10 million hectares. These declines predate the worst of the warming, which means that the climate pressure is arriving on an already weakened system.

Present conditions are a mixture of loss and resilience. Some ecosystems retain the capacity to recover: the southern Great Barrier Reef recovered much of its coral cover after the 2016–2017 events, and some forests regrow after logging. Others have crossed thresholds: the Amazon is emitting more carbon than it absorbs in some regions, and parts of the Arctic are now net sources of carbon rather than sinks. The difference between the two groups is the key to understanding what can be saved.

The assumptions that drive the projections are the emission path, the rate of land-use change and the effectiveness of conservation. Under a low-emission path, many species and ecosystems can persist, though in altered form. Under a high-emission path, the projections converge on widespread loss: coral reefs as we know them disappear, tropical forests decline, and a large fraction of species face range loss. The scenarios below summarise the range.

Table 8.1 Ecosystem thresholds and projected change

Ecosystem	Threshold of concern	Change at 2°C	Change at 3°C	Service at risk
Coral reefs	1.5°C; frequent bleaching	70–90% decline	>99% decline	Fisheries; coastal protection
Amazon forest	20–25% deforestation + warming	Dieback in east/south	Large-scale conversion	Carbon storage; rainfall
Boreal forest	Fire and pest outbreaks	Increased burning	Shift to shrub/grass	Carbon; timber
Permafrost	Thaw onset	10–20% thaw	30–50% thaw	Carbon; infrastructure
Mangroves	Sea level > migration rate	10–30% loss	30–50% loss	Coastal defence; nurseries
Freshwater	Flow and temperature change	20–30% species loss	40–60% species loss	Water; inland fisheries

Two conclusions follow. First, the difference between 2°C and 3°C is the difference between a world where coral reefs survive in reduced form and one where they are largely gone, and between a forest that stores carbon and one that releases it. Second, the loss is not evenly distributed: tropical ecosystems, which hold the most biodiversity and support the most vulnerable people, are the most exposed.

What would change these conclusions? A rapid reduction in deforestation would buy time for the Amazon and reduce emissions. A breakthrough in coral restoration, or the discovery of heat-tolerant symbionts, could preserve some reefs, but it would not restore the physical structure at scale. And a change in the fire regime — through better management and less ignition — would reduce the conversion of forest to savannah. None of these substitutes for emission reduction, but together they can reduce the rate of loss while the climate policy is negotiated.

CHAPTER 9

HEALTH IMPACTS OF CLIMATE CHANGE

Theme: climate-sensitive disease, health systems and the epidemiology of a warming world

In 2024, the Pan American Health Organization recorded more than 12 million cases of dengue in the Americas, the highest number ever reported and roughly three times the previous record. Brazil alone reported more than six million suspected cases. The epidemic was driven by a combination of factors: warmer temperatures that shortened the incubation period of the virus inside the mosquito, heavier rainfall that created breeding sites, urbanisation that brought people and mosquitoes together, and the movement of people and goods that carried the virus across borders.

Dengue is one of the clearest examples of a disease that is climate-sensitive. The Aedes aegypti mosquito that transmits it thrives in warm, humid cities, and its range has expanded as temperatures have risen. The virus replicates faster in warmer mosquitoes, which means that a mosquito that would have been infectious after twelve days may become infectious after seven. The result is that a city which was too cool for sustained transmission a generation ago may now experience regular outbreaks.

The health effects of climate change are broader than infectious disease. Heat kills directly and indirectly; air pollution from wildfire smoke and fossil fuel combustion damages lungs and hearts; droughts and floods reduce food production and contaminate water; and extreme events disrupt the health systems that would otherwise treat the sick. The World Health Organization estimates that climate change will cause about 250,000 additional deaths per year between 2030 and 2050 from heat, malnutrition, malaria and diarrhoea alone, a figure that excludes many of the pathways described in this chapter.

This chapter examines those pathways, the populations they affect most, and the capacity of health systems to respond. It pays particular attention to the difference between direct effects, which are relatively easy to attribute to a heatwave or a flood, and indirect effects, which operate through ecosystems, food systems and infrastructure and are therefore harder to measure but often larger.

9.1 Direct Effects: Heat, Injury and Displacement

The direct effects of climate on health are the easiest to see. Heat causes heatstroke, cardiovascular collapse and kidney injury, and it exacerbates pre-existing conditions; the European summer of 2022 was associated with about 61,700 deaths. Floods cause drowning, trauma and electrocution, and they displace people into temporary shelters where disease spreads easily. Storms cause injuries during the event and infections in the aftermath, when wounds are exposed to contaminated water.

The direct effects are concentrated in specific moments, which makes them visible and therefore politically salient. A heatwave produces a spike in deaths that can be measured within weeks; a flood produces images that circulate globally. But the concentration also creates a bias in how the health burden is understood: the deaths that occur during an event are counted, while the deaths that occur months later, from the

disruption of health services, the loss of income or the deterioration of chronic disease, are not.

Displacement is the bridge between direct and indirect effects. People who lose their homes to a flood or a fire may live in crowded shelters, lose access to medication and treatment, and face months of uncertainty. The health consequences of displacement — infectious disease, mental distress, interruption of chronic care — often exceed the direct consequences of the event that caused it.

9.2 Vector-Borne Diseases

Malaria, dengue, chikungunya, Zika and yellow fever are all sensitive to temperature and rainfall. The mosquitoes that transmit them have temperature optima for survival, reproduction and virus development, and their ranges are bounded by cold and by dryness. Warming expands the suitable range, lengthens the transmission season and accelerates the development of the pathogen inside the vector.

Malaria is the most consequential. About 249 million cases and 608,000 deaths were reported in 2022, mostly in sub-Saharan Africa and mostly among children under five. The relationship between malaria and climate is complex: too much rain can wash away breeding sites, and too little can eliminate them; the effect depends on the species of mosquito and the local ecology. But the overall trend is toward a larger transmission zone, particularly in the highlands of East Africa and the Andes, where warmer temperatures allow the parasite to complete its cycle.

Dengue has expanded even faster. The Aedes aegypti mosquito has spread to more than 150 countries, and the virus now circulates in urban areas where it was previously absent. The 2024 epidemic in the Americas was the largest on record, and similar surges occurred in Southeast Asia and the Western Pacific. The expansion is driven by urbanisation and travel as well as by climate, which is why the response must combine vector control, surveillance and health system preparedness rather than climate policy alone.

9.3 Waterborne and Foodborne Disease

Floods and droughts both increase the risk of waterborne disease, through different mechanisms. Floods overwhelm sanitation systems, contaminate wells and carry pathogens into homes; droughts concentrate pathogens in scarce water and reduce the water available for hygiene. Cholera, typhoid, hepatitis A and diarrhoeal diseases all follow this pattern, and children under five are the most vulnerable.

The 2022 Pakistan floods produced a surge in diarrhoeal disease and malaria, and the 2010 Haiti earthquake and subsequent cholera outbreak showed how quickly a contaminated water supply can become a national emergency. In drought-affected regions, the problem is often not the absence of water but its quality: as sources dry up, people use whatever water is available, including water that is contaminated with sewage or agricultural runoff.

Foodborne disease is also climate-sensitive. Warmer temperatures increase the growth rate of bacteria such as Salmonella and Campylobacter in food, and flooding can contaminate crops with pathogens. The European Centre for Disease Prevention and

Control has documented increases in foodborne infections during heatwaves, and the risk is highest in countries with limited cold-chain infrastructure. The result is a health burden that is rarely attributed to climate but is nonetheless part of it.

9.4 Air Pollution and Wildfire Smoke

Air pollution is the largest environmental health risk in the world, causing about seven million deaths a year according to the World Health Organization. Much of that pollution comes from the same fossil fuels that drive climate change, which means that reducing emissions produces immediate health benefits — the co-benefit that is often the strongest argument for climate action in the short term.

Wildfire smoke is a growing component. The 2023 Canadian fires burned more than eighteen million hectares and sent smoke across North America and the Atlantic, exposing tens of millions of people to particulate matter concentrations that exceeded health guidelines. The 2019–2020 Australian fires exposed millions of people to smoke for weeks and were associated with thousands of hospital admissions and hundreds of deaths. Smoke is particularly dangerous because it contains fine particles that penetrate deep into the lungs and enter the bloodstream.

The health effects of air pollution are not limited to the respiratory system. Fine particles are associated with cardiovascular disease, stroke, diabetes and adverse birth outcomes. Ozone, which forms more readily at higher temperatures, damages lung tissue and reduces crop yields. The combination means that climate change increases air pollution at the same time that air pollution increases climate change, through the deposition of black carbon on snow and ice.

9.5 Nutrition and Child Development

The health effects of climate change on nutrition are indirect and profound. Higher temperatures reduce crop yields, lower the nutrient content of grains and raise food prices, which reduces the quantity and quality of food that poor households can afford. The result is an increase in undernutrition, which weakens immune systems and increases vulnerability to infectious disease.

Children are the most affected. Undernutrition during the first thousand days of life — from conception to age two — causes stunting, which is irreversible and reduces cognitive development, school performance and lifetime earnings. The World Health Organization estimates that climate change will cause an additional 95,000 child deaths per year from undernutrition by 2050, and that the burden will fall disproportionately on South Asia and sub-Saharan Africa.

The CO2 fertilisation effect adds a subtle dimension. Higher carbon dioxide concentrations increase the yield of C3 crops but reduce the concentration of protein, zinc and iron in the grain. Experiments on rice, wheat and soy show declines of 5 to 10 per cent in protein and micronutrients under elevated CO2, which means that a harvest of the same weight may be less nutritious. For populations that depend on a single staple crop, this 'hidden hunger' can matter as much as a decline in calories.

9.6 Mental Health

The mental health effects of climate change are increasingly recognised. Acute events produce post-traumatic stress, anxiety and depression, particularly among people who lose homes, livelihoods or family members. Slow-onset changes — drought, sea level rise, the loss of a familiar landscape — produce a more diffuse distress that researchers describe as solastalgia, the pain of seeing one's home environment change beyond recognition.

The 2019–2020 Australian bushfires were followed by a documented increase in psychological distress, and the 2021 floods in Germany and Belgium produced similar effects. Farmers in drought-affected regions show elevated rates of depression and suicide, and the effect is not confined to the people directly affected: entire communities experience the loss of social networks and economic security that accompanies environmental change.

The mental health burden is undercounted because it is not captured by disaster statistics and because stigma prevents people from seeking help. It is also undertreated: mental health services are scarce in most low- and middle-income countries, and they are rarely included in disaster preparedness plans. The consequence is that the psychological effects of climate change persist long after the physical damage has been repaired.

9.7 Health Systems Under Stress

Health systems are themselves vulnerable to climate change. Hospitals are damaged by floods and storms; power outages disable ventilators and refrigeration; roads are cut, preventing patients from reaching care; and staff are affected by the same events as their patients. The result is that the demand for health care rises at the same time as the capacity to provide it falls.

The 2022 European heatwave illustrated the problem. Hospitals cancelled surgeries because operating theatres could not be kept cool; emergency departments were overwhelmed by heat-related admissions; and ambulance response times increased. In Puerto Rico, Hurricane Maria in 2017 caused a prolonged failure of the electricity system that was associated with an estimated 2,975 excess deaths, many of them from chronic conditions that could not be treated.

Preparedness requires investment in infrastructure, supplies and planning. Hospitals need backup power, cooling, water storage and flood protection. Health systems need surge capacity, stockpiles of medicines and vaccines, and plans for continuity of care during disruptions. They also need surveillance systems that can detect outbreaks early, and a workforce trained to recognise climate-sensitive disease. Most of these measures are inexpensive relative to the cost of failure, but they are rarely prioritised until an emergency occurs.

9.8 Vulnerable Populations

The health burden of climate change is not distributed evenly. Children, pregnant women, older people, people with disabilities, outdoor workers and people living in poverty are at higher risk, and the same groups are less likely to have access to care. The World Health Organization estimates that children under five bear about 88 per cent of the disease burden from climate change, because their immune systems are immature, their bodies are small, and they depend on adults for protection.

Pregnancy is a particularly vulnerable period. Heat exposure is associated with preterm birth, low birth weight and stillbirth, and the effects are strongest among women who work outdoors or live in housing without cooling. Air pollution has similar effects. The 2023 Lancet Countdown reported that heatwaves affected pregnant women in all regions, and that the health systems in most countries were not prepared to protect them.

Disability and age interact with poverty. A person who uses a wheelchair may be unable to evacuate a building without an elevator; a person with dementia may not understand a heat warning; an older person living alone may not be able to open a window or reach a cooling centre. These vulnerabilities are not inevitable; they are the result of design choices and social arrangements that can be changed.

9.9 Surveillance and Early Warning

Early warning is the most effective health adaptation. Heat warning systems that trigger cooling centres, outreach to vulnerable people and changes in hospital staffing can reduce deaths substantially, as the Ahmedabad experience demonstrates. Disease surveillance systems that detect outbreaks early can mobilise vector control and treatment before the epidemic peaks, and climate forecasts can be used to predict where conditions are favourable for transmission.

The integration of climate and health data is improving. The Lancet Countdown publishes annual indicators that track exposure, vulnerability and adaptation across countries, and the World Health Organization has developed a climate and health monitoring framework. Several countries now operate heat-health warning systems, and some are developing malaria early warning systems that use rainfall and temperature forecasts to predict outbreaks.

The gap is in implementation. A warning system is only useful if it reaches the people at risk and if they have the capacity to act on it. In many countries, the warning is issued at the national level but not translated into local action; in others, the data exist but are not shared between health and meteorological agencies. Closing that gap is less a matter of technology than of institutional coordination and trust.

9.10 Co-Benefits and the Health Case for Climate Action

The health argument for climate action is stronger than the environmental argument, because it is immediate and personal. Reducing fossil fuel combustion reduces air pollution, which prevents millions of deaths a year. Increasing walking and cycling reduces obesity and cardiovascular disease while reducing emissions. Improving building insulation reduces energy demand and protects people from heat and cold. Shifting toward plant-rich diets reduces emissions and improves health.

These co-benefits are often larger than the costs of the policies that produce them, particularly in countries with high air pollution. A study of the health benefits of meeting the Paris targets in China estimated that the reduction in air pollution would prevent hundreds of thousands of premature deaths a year, a benefit that is not counted in the conventional cost of decarbonisation.

The health frame also changes the politics. Climate policy is often presented as a cost to be borne for the sake of future generations; health policy is presented as a benefit to be delivered now. The same measures — clean energy, efficient buildings, active transport, healthy diets — can be justified in either language, and the health language often travels further because it speaks to what people experience in their own bodies.

9.11 Case Study: The 2024 Dengue Epidemic

Dengue is a viral infection transmitted by Aedes mosquitoes, and it has been expanding for decades. The virus causes fever, severe headache, joint pain and, in a small fraction of cases, haemorrhagic fever that can be fatal. There is no specific treatment, and the vaccines that exist are only partially effective and are not recommended for everyone. Control depends on reducing mosquito breeding sites and on timely diagnosis and supportive care.

The 2024 epidemic in the Americas was the largest on record. More than 12 million cases were reported, with Brazil accounting for about half. The epidemic followed a year of record heat and heavy rainfall, which expanded the habitat of the mosquito and accelerated the replication of the virus. It also followed the relaxation of vector control programmes during the COVID-19 pandemic, which had left fewer resources for surveillance and spraying.

The health system response varied. Brazil deployed mobile testing teams and expanded hospital capacity; Argentina and Paraguay declared epidemiological alerts; and the Pan American Health Organization coordinated surveillance and laboratory support. But the epidemic exposed the limits of reactive response: by the time cases are detected, the mosquito population has already expanded, and the measures that reduce transmission — eliminating standing water, covering tanks, using repellent — depend on household behaviour that is difficult to sustain.

The case illustrates the connection between climate and health systems. The weather created the conditions for transmission, but the scale of the epidemic depended on the capacity of health systems to detect, treat and control it. Countries with strong surveillance and vector control reported fewer severe cases and deaths than countries with weaker systems, even when the climate conditions were similar.

The lessons are practical. First, vector control must be continuous, not episodic; the mosquito does not wait for an outbreak to be declared. Second, surveillance must integrate climate data, so that health authorities can anticipate where conditions are favourable and act before cases appear. Third, health systems need surge capacity for the diseases that climate change is expanding, including staff, beds, tests and medicines. Fourth, communities must be engaged, because the most effective interventions — eliminating standing water, using bed nets and repellents — happen in homes, not in hospitals.

9.12 Future Scenarios and Predictions: Health in 2050

The historical data show that the health effects of climate change are already measurable. Heat-related mortality among people over 65 has increased by about 70 per cent since the 1990s; the transmission season for dengue has lengthened in many regions; and the area burned by wildfire has increased, exposing more people to smoke. These trends are not projections; they are observations, and they are consistent with the physical changes described in earlier chapters.

Present conditions give the baseline. About 3.5 billion people live in areas suitable for dengue transmission; malaria transmission is expanding in highland regions; and air pollution remains the largest environmental health risk, causing about seven million deaths a year. Health systems in low-income countries are the least prepared, with fewer beds, fewer health workers and less capacity to detect outbreaks.

The assumptions that drive the projections are the emission path, population growth, ageing, urbanisation and the strength of health systems. A world that reduces emissions and invests in health systems can prevent much of the projected burden; a world that does neither will face a burden that grows faster than the climate because the population is ageing and urbanising at the same time. The scenarios below summarise the range.

Table 9.1 Climate-sensitive health risks, 2050

Risk	Climate driver	Change at 2°C	Change at 3°C	Main adaptation
Heat mortality	Heatwaves	+50–100%	+150–300%	Warning; cooling; housing
Malaria	Warming; rainfall	Expanding highlands	Wider range	Bed nets; treatment; surveillance
Dengue	Warming; urban water	+100–200%	+200–400%	Vector control; surveillance
Undernutrition	Crop yields; prices	+50–100k deaths/yr	+100–200k deaths/yr	Social protection; nutrition
Air pollution	Fossil fuels; fires	Falling with policy	Rising without policy	Clean energy; smoke warnings
Displacement health	Floods; drought	+50–100 million exposed	+100–200 million exposed	Shelter; health services

Two conclusions follow. First, the health burden is dominated by indirect pathways — nutrition, infectious disease, air pollution — rather than by the direct effects that make headlines. Second, the burden is preventable with known interventions: clean energy, vector control, water and sanitation, heat warnings, and strong health systems. The gap between what is known and what is done is the central problem of climate and health.

What would change these conclusions? A breakthrough in dengue vaccines or in gene-drive mosquito control could reduce transmission substantially. A rapid expansion of clean energy would reduce air pollution and heat simultaneously. And a strengthening of primary health care in low-income countries would improve the response to every climate-sensitive disease. None of these is a substitute for emission reduction, but each would reduce the human cost of the warming that is already committed.

CHAPTER 10

ECONOMIC IMPACTS AND FINANCIAL RESILIENCE

Theme: the economics of climate damage, insurance and the cost of delay

In August 2017, Hurricane Harvey stalled over Houston and dropped more than 1,500 millimetres of rain in five days. The damage was estimated at US$125 billion, most of it from flooding rather than wind. A few weeks later, Hurricane Irma struck the Caribbean and Florida, and Hurricane Maria struck Puerto Rico, causing damage estimated at US$77 billion and US$90 billion respectively. The three storms together produced losses of about US$300 billion, of which only about a third was insured.

The 2017 season illustrates a pattern that is becoming more common. The physical damage is only the first layer of economic loss. Businesses that close do not pay wages; workers who lose wages spend less; local governments that lose tax revenue cut services; insurers that pay claims raise premiums; and households that lose their savings reduce consumption for years. The indirect losses can exceed the direct damage, and they fall hardest on the people with the least cushion.

Climate economics has moved from the margins of the discipline to its centre because the damage is no longer hypothetical. The estimates vary widely, but they converge on a conclusion: the cost of inaction is larger than the cost of action, and the difference grows with every year of delay. The debate is no longer about whether to act but about how to allocate the cost, who bears the risk, and how to finance the transition.

This chapter examines the economic channels through which climate change affects households, firms and states. It distinguishes between direct damage and indirect loss, between stock and flow, and between the risks that can be insured and those that cannot. It closes with the financial instruments that can build resilience — reserve funds, parametric insurance, catastrophe bonds, and the public balance sheets that ultimately backstop them.

10.1 Direct Damage and Indirect Loss

Disaster economics distinguishes between damage, which is the cost of repairing or replacing physical assets, and loss, which is the cost of lost economic activity. A factory that is destroyed represents damage; the production that the factory would have generated during the months it is rebuilt represents loss. The two are often reported together, but they behave differently: damage is a one-time cost, while loss can persist until the factory is rebuilt and the market recovers.

The distinction matters for policy. Compensation that covers damage but not loss leaves businesses unable to pay wages and households unable to replace income, which prolongs the disruption. The 2011 Thailand floods damaged industrial estates that produced a large share of the world's hard drives; the damage was repaired in months, but the global production shortfall lasted longer and raised computer prices for a year. The loss was transmitted through supply chains to firms and consumers who never saw the flood.

Indirect losses are difficult to measure because they are counterfactual: they describe what would have happened without the disaster. Economists estimate them using input-output models, computable general equilibrium models, or econometric comparisons of affected and unaffected regions. The estimates vary, but they consistently find that indirect losses are larger in economies with tighter linkages, weaker social protection and less fiscal space to respond.

10.2 Productivity, Labour and Heat

Heat reduces economic output through labour productivity. The International Labour Organization estimates that heat stress already reduces global working hours by about 2.2 per cent, equivalent to 80 million full-time jobs, and that the loss could reach 2.4 per cent by 2030. Agriculture accounts for about 60 per cent of the lost hours, followed by construction, manufacturing and transport.

The mechanism is physiological and behavioural. Above a certain wet-bulb temperature, workers must slow down, take breaks and drink water; in the absence of those adaptations, heat exhaustion and heatstroke occur. Firms respond by shifting hours, installing cooling, or automating tasks, but each response has a cost. For workers paid by output, the cost is borne directly as lost income; for firms, it appears as lower productivity and higher capital expenditure.

The distributional effect is important. Outdoor workers, agricultural labourers and construction workers are the most exposed, and they are often the least able to insure against lost income. In countries where a large share of employment is informal, heat does not appear in official productivity statistics, because the work is not measured in the first place. The result is that the economic cost of heat is systematically underestimated in the places where it is highest.

10.3 Food Prices and Inflation

Climate shocks to food production are transmitted to the wider economy through prices. When a major producer loses a harvest, global prices rise, and the effect reaches consumers through retail prices and, in poor countries, through the cost of the food basket that dominates household budgets. Central banks have begun to include climate shocks in their inflation models, because food and energy price spikes can destabilise expectations even when core inflation is stable.

The 2022 food price crisis is an example. The war in Ukraine was the immediate trigger, but it arrived on top of a series of climate-related harvest failures in Europe, South Asia and the Horn of Africa. Food prices rose by more than 20 per cent globally, and the increase was largest in countries that import a high share of their calories. The result was a rise in poverty and a decline in dietary quality that persisted after prices fell.

The inflationary effect is not only a problem for households; it is a problem for governments. Food and energy subsidies are politically difficult to remove and fiscally expensive to maintain, and they tend to expand precisely when revenues are falling because the economy is contracting. The result is a pro-cyclical fiscal burden that can turn a climate shock into a debt crisis, particularly in countries with limited access to

international capital markets.

10.4 Sovereign Risk and the Cost of Capital

Climate change affects the cost at which governments and firms borrow. Credit rating agencies have begun to incorporate climate vulnerability into sovereign ratings, and the effect is already visible in the spreads that some countries pay on their debt. A country that is exposed to repeated disasters, dependent on climate-sensitive exports and lacking fiscal buffers is a riskier borrower than one that is not, even if its current debt level is the same.

The mechanism operates through several channels. Disasters reduce tax revenue and increase expenditure at the same time, which worsens the fiscal balance. Repeated disasters can reduce potential growth by destroying capital and discouraging investment. And the expectation of future losses can raise the risk premium that investors demand, which increases the cost of servicing debt and reduces the resources available for adaptation.

The result is a vicious circle for the most vulnerable countries. They face the highest adaptation costs, the highest cost of capital and the least fiscal space, and the combination can make adaptation unaffordable even when the returns are high. International financial institutions have begun to address the problem through concessional finance, debt swaps and pre-arranged emergency lending, but the scale remains small relative to the need.

10.5 Insurance and the Limits of Risk Transfer

Insurance is the mechanism that allows losses to be spread across many policyholders and over time. It works when the risk is measurable, the events are independent, and the premiums cover the expected losses plus a margin. Climate change violates all three conditions: the risk is changing, the events are becoming correlated across regions, and the losses are rising faster than premiums can adjust.

The result is retreat. Insurers have withdrawn from parts of California and Florida, where wildfire and hurricane risk have increased; they have raised premiums in coastal areas; and they have reduced coverage for floods and storm surge. In the United States, the National Flood Insurance Program has been insolvent for years and has been forced to raise premiums, which has made insurance unaffordable for some homeowners and reduced property values.

When insurance fails, the cost shifts to governments and households. Governments provide disaster relief, which is often untargeted and slow; households draw down savings, borrow at high interest, or abandon property. The shift is regressive, because wealthy households can self-insure and poor households cannot. The result is that climate risk becomes a mechanism for transferring wealth from those who cannot bear it to those who can.

10.6 Stranded Assets and Transition Risk

The transition away from fossil fuels creates a different kind of financial risk: assets that were expected to produce revenue may become uneconomic before they are fully depreciated. Coal plants, oil fields, gas pipelines and the infrastructure that supports them could be stranded if demand falls faster than expected or if policy restricts their use. The value of those assets is reflected in the balance sheets of companies, banks and pension funds, which means that a rapid transition could create losses across the financial system.

The risk is not hypothetical. Coal companies in the United States and Europe have written down assets as gas and renewables have displaced coal; oil majors have faced pressure from investors to disclose their climate assumptions; and central banks have begun to stress-test banks against transition scenarios. The magnitude of the risk depends on the speed of the transition and on whether it is orderly or disorderly.

The policy implication is that transition risk and physical risk must be managed together. A disorderly transition, in which policy changes abruptly and assets are written down suddenly, can destabilise financial institutions; a delayed transition, in which physical risk continues to accumulate, can destabilise them even more. The goal is an orderly transition that signals policy direction early, phases out the most exposed assets gradually and protects the workers and communities that depend on them.

10.7 The Cost of Inaction

Estimating the cost of inaction requires comparing a future with climate change to a future without it, and the comparison depends on assumptions about growth, adaptation and discounting. Early integrated assessment models, such as the DICE model developed by William Nordhaus, produced estimates of a few per cent of global GDP by the end of the century, and those estimates were criticised for understating the risk of catastrophic and irreversible impacts.

More recent estimates are larger. A 2024 study in Nature found that a high-emission pathway would reduce global income by about 19 per cent by 2050 relative to a no-climate baseline, with damages of about US$38 trillion a year, and that the losses would fall disproportionately on countries with low historical emissions (Kotz, Levermann and Wenz, 2024). The Swiss Re Institute estimated that climate change could reduce global GDP by 10 per cent by 2050 in a severe scenario, and the World Bank has estimated that poor countries could lose 1.2 to 3.3 per cent of GDP by 2030.

The estimates differ because they use different assumptions, but they share a conclusion: the cost of inaction is larger than the cost of action, and the difference is growing. The economics of climate change is therefore not primarily about whether to act but about how quickly, how fairly and how effectively. The longer the delay, the larger the physical risk and the more abrupt the transition that will be required to avoid it.

10.8 Who Bears the Loss

The economic burden of climate change is distributed unevenly across countries, sectors and households. Low-income countries are more exposed because their economies depend on agriculture, their infrastructure is less resilient, and their fiscal capacity to respond is smaller. Within countries, poor households are more exposed because they live in more hazardous places, work outdoors, and have fewer savings and less insurance.

The distributional effect is compounded by the fact that the countries and households that contributed least to emissions are often the most exposed. This is the basis of the loss and damage debate in international climate negotiations, which led to the creation of a fund at COP27 in 2022. The fund is small relative to the need, and its governance and disbursement remain contested, but it establishes the principle that climate losses are a matter of justice, not only of charity.

The household-level distribution is equally important. A disaster that destroys a poor household's home can push it into poverty permanently, because it has no savings, no insurance and no access to credit. A wealthier household can absorb the same loss and rebuild. The result is that a single event can widen inequality for a generation, and that social protection is as important as physical infrastructure in determining who recovers.

10.9 Financial Instruments for Resilience

Several instruments can transfer or finance climate risk. Sovereign catastrophe bonds pay out when a predefined event occurs, providing rapid liquidity after a disaster. Parametric insurance does the same at a smaller scale, using rainfall or wind speed as a proxy for loss. Contingent credit lines allow governments to borrow quickly when a disaster is declared. Reserve funds accumulate savings in good years for use in bad ones.

Each instrument has strengths and weaknesses. Catastrophe bonds are expensive and require sophisticated legal and financial infrastructure; parametric insurance suffers from basis risk, because the index may not match the actual loss; contingent credit depends on the government's creditworthiness; and reserve funds require fiscal discipline that is difficult to maintain. The most effective approach combines several instruments and layers them so that small events are covered by reserves, medium events by insurance and large events by sovereign or international support.

The deeper issue is that risk transfer does not reduce risk; it only spreads it. A country that insures against floods still experiences the flood, and the premium it pays reduces the resources available for prevention. The most cost-effective financial strategy is therefore to invest in reducing risk first, and to use insurance for the residual risk that cannot be eliminated. That sequencing — prevention before transfer — is the core principle of financial resilience.

10.10 The Macroeconomics of a Warming World

At the macroeconomic level, climate change affects growth through several channels: it destroys capital, reduces productivity, diverts investment from productive uses to reconstruction, and increases uncertainty, which discourages long-term

investment. The combined effect is a reduction in potential output that persists long after the event that caused it.

The effect is not uniform across countries. Economies that depend on agriculture, tourism or outdoor labour are more exposed; economies with diversified production and strong institutions are more resilient. The result is that climate change is likely to widen the gap between rich and poor countries, reversing some of the convergence that globalisation produced in the late twentieth century.

The policy response has three pillars. The first is mitigation, which reduces the magnitude of future damage. The second is adaptation, which reduces vulnerability to the damage that is already committed. The third is financial resilience, which ensures that households, firms and governments can absorb the shocks that cannot be prevented. The three are complementary: mitigation without adaptation leaves people exposed to unavoidable impacts; adaptation without mitigation eventually exceeds its limits; and both without financial resilience leave the poorest to bear the cost.

10.11 Case Study: The 2017 Atlantic Hurricane Season

The 2017 Atlantic hurricane season was one of the most destructive on record. Hurricane Harvey made landfall in Texas in August and stalled, dropping more than 1,500 millimetres of rain over Houston and causing damage estimated at US$125 billion. Hurricane Irma followed in September, devastating several Caribbean islands and causing damage estimated at US$77 billion. Hurricane Maria struck Puerto Rico two weeks later, destroying the electricity grid and causing damage estimated at US$90 billion.

The total damage was about US$300 billion, of which only about a third was insured. The gap was concentrated in Puerto Rico and the smaller Caribbean islands, where insurance penetration is low and where the losses were largest relative to the size of the economy. In Puerto Rico, the storm caused an estimated 2,975 excess deaths in the months after landfall, most of them from the failure of the electricity and health systems rather than from the storm itself.

The economic effects persisted for years. In Puerto Rico, the population declined by several per cent as residents moved to the mainland; businesses closed; and the reconstruction of the grid took more than a year and cost billions of dollars. In Texas, the recovery was faster because the state had resources, insurance and access to federal assistance, but the damage to housing was concentrated in low-income neighbourhoods that lacked flood insurance.

The season illustrates the difference between insured and uninsured loss, and between the capacity of rich and poor jurisdictions to recover. It also illustrates the importance of infrastructure: the failure of the Puerto Rican grid turned a hurricane into a public health emergency, and the failure of the Texas power system during a winter storm four years later showed the same pattern in a different season. In both cases, the physical hazard was within the range of what the infrastructure was supposed to withstand; the failure was in the design, maintenance and governance of the system.

The lessons are financial as well as physical. First, insurance penetration is a determinant of recovery, and it is lowest where the need is greatest. Second, infrastructure resilience is a form of financial protection, because it reduces the liability that governments and households must bear after an event. Third, the poorest households need grants and social protection, not loans, because a loan against future income is not a recovery strategy for someone who has lost their livelihood. Fourth, the cost of reconstruction is rising, and the willingness of donors and insurers to pay is not.

10.12 Future Scenarios and Predictions: The Economic Cost to 2050

The historical record shows that economic losses from climate-related disasters have increased. The total damage from weather-related disasters, adjusted for inflation, has grown by about 5 to 7 per cent a year since the 1970s, faster than exposure alone would predict. Insured losses have grown even faster, because more assets are concentrated in exposed areas and because insurance has expanded into new markets.

Present conditions give the baseline. Global losses from weather-related disasters are estimated at US$200 to US$300 billion a year, of which about half is insured. The uninsured half falls on households, firms and governments, and it is concentrated in low- and middle-income countries. The cost of adaptation is estimated at US$100 to US$300 billion a year by 2030 for developing countries, and the finance available is a fraction of that.

The assumptions that drive the projections are the emission path, the rate of adaptation, and the discount rate used to compare present and future costs. A low discount rate makes future damages look larger and justifies more spending now; a high discount rate makes them look smaller and justifies delay. The choice of discount rate is therefore an ethical choice as much as an economic one, because it determines how much weight is given to people who are not yet born.

Table 10.1 Economic losses under three pathways, 2050

Indicator	Low emissions	Current policies	High emissions	Main driver
Global GDP loss vs no-climate baseline	−3 to −5%	−8 to −12%	−15 to −20%	Labour; agriculture; capital
Annual damage (US$ trillion)	1.5–2.5	4–6	7–10	Extreme events; slow onset
Uninsured losses (share)	40%	50%	60%	Insurance retreat
People pushed into poverty (million)	50–100	100–200	200–400	Food prices; displacement
Adaptation finance needed (US$ bn/yr)	100	200	300+	Infrastructure; social protection

Two conclusions follow. First, the difference between the low and high pathways is measured in percentage points of global GDP, which is to say in trillions of dollars and in hundreds of millions of people pushed into poverty. Second, the losses are not a future that arrives in 2050; they accumulate year by year, and the countries that are least able to adapt are already paying the highest share.

What would change these conclusions? A rapid diffusion of cheap clean energy would reduce both the physical damage and the transition risk, and it is already happening faster than most models assumed. A breakthrough in adaptation finance — through reform of multilateral institutions, debt relief or new levies — would reduce the burden on the poorest countries. And a change in the discount rate used by governments would make long-term damages look larger and justify more investment now. None of these is certain, but the direction of the economics is: delay costs more than action.

CHAPTER 11

MIGRATION AND DISPLACEMENT

Theme: climate migration, displacement and the politics of moving people

Tuvalu is a country of nine coral atolls with a population of about eleven thousand people and a highest point of less than five metres above sea level. At high tide, the capital, Funafuti, is flooded in places; at spring tides, seawater bubbles up through the ground. The government has been negotiating the terms of its own disappearance for more than a decade, and in November 2023 it signed an agreement with Australia, the Falepili Union, that provides for the movement of Tuvaluans to Australia under a special visa arrangement while preserving Tuvalu's sovereignty and maritime boundaries.

Tuvalu is the clearest case, but it is not the largest. Most climate migration is internal, rural-to-urban, and temporary. People move from a village to a city after a drought, from a floodplain to higher ground after a flood, or from a coast to an inland town after salt water contaminates the wells. They move within their own country, often return when conditions improve, and rarely cross an international border. The image of a wave of climate refugees crossing borders is misleading; the reality is millions of small movements, most of them within national boundaries.

The scale is difficult to measure because climate is rarely the only reason for moving. A farmer who leaves a drought-affected village may be moving because of debt, because his children have no school, or because a relative in the city has offered work. Climate change is one pressure among many, and it is often the pressure that tips a decision that was already being considered. That makes attribution difficult and prediction uncertain, but it does not make the phenomenon any less real.

This chapter examines the drivers of climate migration, the difference between voluntary and forced movement, and the destinations that receive people. It pays particular attention to the places that cannot adapt in place — small islands, low deltas and drought-prone rangelands — and to the legal and political frameworks that determine whether movement is managed or chaotic. It closes with the question that receiving communities face: how to absorb people without creating new vulnerability.

11.1 Definitions: Migration, Displacement and Relocation

Migration is a broad term that includes voluntary movement in search of opportunity and forced movement in response to threat. Displacement usually refers to movement caused by a specific event, such as a flood or a conflict, and it implies that people intend to return. Planned relocation refers to the organised movement of a community to a new site, usually with government support. The distinctions matter because they determine which legal protections apply and which forms of assistance are available.

Climate change can produce all three. A flood displaces people temporarily; a drought may push people to migrate permanently; a village that is repeatedly flooded may be relocated by the state. The same household may experience several of these in sequence: displacement to a shelter, return to a damaged home, migration to a city, and

eventually relocation to a new settlement. The categories are useful for policy but they do not describe a linear process.

The legal framework is incomplete. The 1951 Refugee Convention defines a refugee as a person fleeing persecution on the basis of race, religion, nationality, social group or political opinion; it does not cover people fleeing environmental change. There is no binding international instrument that grants climate migrants a right to enter another country or to receive protection. Some countries have created humanitarian visas, and regional agreements have begun to address the gap, but the protection available to a person whose island is disappearing depends on where they happen to be born.

11.2 Rapid-Onset and Slow-Onset Drivers

Rapid-onset events — floods, storms, wildfires — displace people immediately, often in large numbers. The Internal Displacement Monitoring Centre recorded about 26.4 million disaster displacements in 2023, the majority from floods and storms. Most of these displacements are short-lived: people return home within weeks or months, once the water recedes and the damage is repaired. But repeated displacement erodes the assets and social networks that make recovery possible.

Slow-onset changes — sea level rise, desertification, salinisation, glacier retreat — displace people gradually, and the movement is harder to see. A household whose well becomes saline may first buy water, then reduce its herd, then send a son to the city, and finally abandon the farm. By the time the village is recognised as uninhabitable, most of its population may already have left. The slow-onset process is less visible than a flood but often more permanent.

The two interact. A community weakened by repeated drought may be unable to recover from a flood; a coastal city that is subsiding may be overwhelmed by a storm surge that would have been manageable a generation ago. The sequence of shocks matters as much as the magnitude of any single event, which is why vulnerability is best understood as a trajectory rather than a state.

11.3 How Much Migration Is Climate-Related?

Estimating the number of people who move because of climate change is difficult because climate is rarely the sole cause. Researchers use household surveys, satellite data on vegetation and rainfall, and econometric models to identify the effect of climate variables on migration decisions, controlling for income, conflict, land tenure and social networks. The estimates vary widely, but they consistently find that climate shocks increase migration, particularly from rural areas and particularly among households that depend on agriculture.

The World Bank's Groundswell reports projected that 216 million people could move within their own countries by 2050 in six regions under a high-emission scenario, with most of the movement in Sub-Saharan Africa, South Asia and Latin America. The same reports found that most of the movement could be avoided through adaptation — irrigation, crop diversification, social protection and urban planning — which is why they are framed as a call to action rather than a prediction.

The number of people who cross international borders for climate reasons is much smaller, and it is likely to remain so. International migration is expensive, requires documents and networks, and is often deterred by border controls. The people who are most exposed to climate risk are usually the least able to migrate internationally, which is why the most vulnerable often stay in place. This phenomenon — the immobility of the most exposed — is one of the most important and least discussed features of climate migration.

11.4 Pastoralists, Farmers and the Sahel

Pastoralists are among the most climate-exposed groups because their livelihoods depend on grazing that varies with rainfall. In the Sahel, herders have long moved seasonally between dry and wet season pastures, and those movements were governed by customary agreements with farming communities. As rainfall has become more variable and pastures have degraded, the movements have become longer, more frequent and more contested.

The result is conflict over land and water, particularly where the state is weak and where farmers have expanded into corridors that herders once used. The conflict is often described as ethnic or religious, and those identities matter, but the underlying driver is frequently a dispute over access to a shrinking resource. Climate change does not create the conflict, but it intensifies the competition that makes it more likely.

Farmers respond to drought in a different way. They may migrate temporarily to cities, send a family member to work, diversify into non-farm activities, or sell livestock to buy food. The strategies are cumulative, and each one reduces the household's capacity to return to farming. A household that sells its animals and its tools may not be able to restart even if the rains return, which is why drought can produce permanent de-agrarianisation in regions that have experienced it repeatedly.

11.5 Small Island States and Coastal Communities

Small island developing states are the most exposed to sea level rise because their land area is small, their elevation is low, and their populations are concentrated on the coast. Tuvalu, Kiribati, the Marshall Islands and the Maldives have average elevations of a few metres, and their freshwater supplies depend on thin lenses of groundwater that are vulnerable to saltwater intrusion. For these states, adaptation in place has physical limits.

The responses have three tracks. The first is coastal defence: sea walls, reclamation, and raised infrastructure. The second is adaptation within the island: rainwater harvesting, desalination, salt-tolerant crops and the relocation of settlements to higher ground. The third is migration: agreements with other countries that allow citizens to move, work and eventually settle. The Falepili Union between Tuvalu and Australia is the most recent example, and it combines a visa pathway with Australian recognition of Tuvalu's maritime boundaries and a commitment to assist in the event of a major disaster.

The politics of these agreements is delicate. Island states do not want to be seen as giving up, and they do not want to lose their sovereignty or their maritime zones, which are vast relative to their land area. Australia and New Zealand have been willing to accept limited numbers of migrants, but not to recognize a legal category of climate refugee. The result is a patchwork of bilateral arrangements that provides a pathway for some people but not a right for all.

11.6 Floods, Displacement and Return

Floods displace more people than any other climate hazard. The 2022 Pakistan floods displaced about eight million people, many of whom lived in temporary camps for months; the 2010 floods in the same country displaced about 14 million. In Bangladesh, cyclones and floods regularly displace hundreds of thousands, and the people who are displaced are disproportionately poor, landless and living in informal settlements.

Most flood displacement is temporary, but return is not always possible. A household whose home has been destroyed may have nowhere to return to, particularly if it rented or occupied land informally. A household whose land has been eroded by a river may have lost its livelihood permanently. And a household that has been displaced repeatedly may decide not to return, even if the house is repaired, because the risk of another flood is too high.

The quality of displacement matters as much as its duration. People in camps face overcrowding, poor sanitation and limited access to health care and education; people who stay with relatives may be less visible but equally vulnerable. The decision to return is often driven by the availability of work and housing rather than by the level of risk, which means that the poorest are the most likely to return to hazardous places.

11.7 Cities as Destinations

Most climate migrants move to cities, and most cities are not prepared for them. New arrivals often settle in informal areas that are themselves exposed to floods, heat and landslides, and they lack access to water, sanitation, health care and formal employment. The result is that migration transfers vulnerability from the countryside to the city rather than resolving it.

Cities that plan for migration do better. Providing land for affordable housing, extending water and sanitation to informal settlements, and creating work opportunities reduces the risk that newcomers will be pushed into hazardous areas. Some cities have adopted policies that explicitly recognize climate migrants, including access to services and temporary shelter. But most have not, and the arrival of large numbers of people in a short period can overwhelm services and create political tension.

The receiving communities matter as much as the migrants. A city that absorbs people successfully is one where the labour market is growing, the infrastructure is adequate and the social fabric is inclusive. A city that is already stressed by unemployment, housing shortages and weak services may experience conflict. The difference is largely a matter of governance and investment, which means that migration policy is also urban policy.

11.8 Remittances and Transnational Adaptation

Migration is not only a failure of adaptation; it is also a strategy of adaptation. Households that send a member to work elsewhere receive remittances that can be used to invest in irrigation, housing, education and health. In 2023, remittances to low- and middle-income countries reached about US$669 billion, more than foreign direct investment and more than official development assistance. For many countries, remittances are the largest source of external finance.

Remittances can increase resilience. A household that receives regular transfers can buy food during a drought, rebuild after a flood, and keep children in school. They can also finance adaptation: a new roof, a water tank, a solar system, or a small business that is less dependent on rainfall. The effect depends on the reliability of the transfers and on how they are used, but the evidence generally shows that remittances reduce vulnerability.

There is a darker side. Migration can separate families, increase the burden on women who remain behind, and expose migrants to exploitation and dangerous work. The remittances may also create dependence on a single source of income that is itself vulnerable to economic shocks in the destination country. The policy challenge is to maximize the benefits of migration while reducing its costs, which requires safe migration pathways, portable social protection and financial systems that reduce the cost of transfers.

11.9 Legal and Political Frameworks

The legal framework for climate migration is fragmented. The Refugee Convention does not cover environmental displacement; the International Organization for Migration is not a protection agency; and the UN Framework Convention on Climate Change established a task force on displacement but not a right to admission. The result is that most climate migrants depend on ordinary immigration rules, which are designed for other purposes and are often restrictive.

Some progress has been made. The Global Compact for Safe, Orderly and Regular Migration, adopted in 2018, recognizes climate and environmental factors as drivers of migration and commits states to cooperate on admission and protection. The Nansen Initiative and its successor, the Platform on Disaster Displacement, have developed guidance on planned relocation and cross-border displacement. Several countries have created humanitarian visas or temporary protection regimes for people affected by disasters.

The political obstacle is that admission is a sovereign decision, and publics in receiving countries are often hostile to migration. The result is that governments prefer to fund adaptation in countries of origin, which is cheaper and politically easier, rather than to open legal pathways. Adaptation is necessary, but it cannot protect everyone, and the people it cannot protect need somewhere to go. The absence of a legal pathway does not prevent movement; it only makes the movement irregular, dangerous and expensive.

11.10 Planned Relocation

Planned relocation is the organised movement of a community to a new site, usually because the original site has become unsafe. It is the most difficult form of adaptation, because it involves the loss of land, livelihoods, social networks and cultural identity. Relocation projects have a long history of failure when they have been imposed without consultation, compensated inadequately, or sited in places that are themselves vulnerable.

The successful cases share several features. Communities are consulted early and repeatedly; compensation covers land, housing and livelihood; the new site has access to water, employment and services; and the process is paced to allow people to move when they are ready. Fiji's planned relocation of coastal villages, supported by a national framework, is one of the more studied examples; it has been slower than the government intended, but it has generally respected community consent.

The hardest question is who decides. A community may prefer to stay even when the government considers the site unsafe; a government may consider relocation necessary even when the community disagrees. The tension is unavoidable, and it is resolved differently in different political systems. The principle that has emerged from international guidance is that relocation should be a last resort, that it should be voluntary wherever possible, and that it should preserve the rights and dignity of the people who move. Those principles are easier to state than to implement.

11.11 Case Study: Tuvalu and the Falepili Union

Tuvalu consists of nine atolls and reef islands scattered across about 1.3 million square kilometres of ocean, with a total land area of about 26 square kilometres. The population is about 11,000, and the highest point is less than five metres above sea level. The country has no rivers, no significant groundwater beyond thin freshwater lenses, and no high ground to which it could retreat. Its territorial waters, by contrast, are vast, and its exclusive economic zone is its most valuable asset.

The physical pressures are already measurable. Sea level at Funafuti has risen by about 3.4 millimetres a year since records began, close to the global average, and the frequency of flooding at high tide has increased. Saltwater intrusion has contaminated wells and reduced the productivity of taro pits, a staple crop. Coral bleaching has damaged reefs that provide food and coastal protection. The government has built sea walls and raised some roads, but the costs are high relative to the national budget.

The Falepili Union, signed with Australia in November 2023, is the most comprehensive bilateral arrangement of its kind. It provides a special visa pathway for Tuvaluans to live and work in Australia, with an initial cap and growth over time; it commits Australia to assist Tuvalu in the event of a major natural disaster or a security threat; and it recognizes Tuvalu's maritime boundaries and statehood even if the land is lost to the sea. The agreement is reciprocal: Australia gains strategic influence in the Pacific, and Tuvalu gains a pathway for its people.

The agreement has been criticized. Some Tuvaluans argue that it encourages emigration and undermines the incentive to adapt in place; others argue that it is the only realistic option for a country that cannot defend itself against the sea. The government has framed it as an insurance policy rather than a surrender, and it has continued to invest in coastal defence and adaptation. The debate reflects a broader tension in climate policy: whether to invest in places that may become uninhabitable, or to plan for the movement of people before the crisis arrives.

The case has implications beyond Tuvalu. It shows that legal pathways for climate migration can be created through bilateral agreements, even in the absence of a global framework. It shows that the recognition of statehood and maritime boundaries can be separated from the physical territory, which is a legal innovation with implications for other island states. And it shows that the choice between adaptation and migration is not binary: a country can do both, and the balance shifts as the physical risk increases.

11.12 Future Scenarios and Predictions: Migration to 2050

The historical data show that climate shocks already move people. The Internal Displacement Monitoring Centre recorded about 26 million disaster displacements in 2023, and the World Bank estimates that 216 million people could move within their own countries by 2050 under a high-emission scenario. These figures are not predictions of a single wave; they describe millions of individual decisions, most of them short-distance and temporary.

Present conditions give the baseline. About 3.5 billion people live in contexts that are highly vulnerable to climate change, and about 1 billion live in low-elevation coastal zones. The countries that are most exposed are also the ones with the least capacity to manage migration: they have weak registration systems, limited social protection and few legal pathways to other countries.

The assumptions that drive the projections are the emission path, the rate of adaptation, and the openness of receiving countries. A world that invests in adaptation in places of origin can reduce the number of people who are forced to move; a world that opens legal pathways can make the movement safer and more orderly. A world that does neither will face a larger number of people moving irregularly, with higher costs for migrants and for the countries that receive them.

Table 11.1 Projected climate migration by 2050

Region	Internal migrants (million)	Main driver	Adaptation priority	Policy risk
Sub-Saharan Africa	~85	Drought; pastoralism	Water; social protection	Conflict; weak states
South Asia	~40	Floods; heat; sea level	Urban planning; embankments	Informal settlements
Latin America	~17	Drought; storms	Rural livelihoods; insurance	Urban violence
East Asia and Pacific	~48	Sea level; storms	Coastal defence; relocation	Land acquisition
North Africa	~19	Drought; water	Irrigation; employment	Political instability
Small island states	~2–3	Sea level rise	Relocation; legal pathways	Sovereignty; culture

Two conclusions follow. First, the scale of climate migration is large but manageable if it is anticipated; the danger is not the number of people but the absence of systems to receive them. Second, most migration will be internal, which means that the responsibility falls primarily on national and municipal governments, not on international institutions.

What would change these conclusions? A rapid reduction in emissions would reduce the number of people who are forced to move, but it would not eliminate the movement that is already committed by the warming that has occurred. A major expansion of adaptation finance would allow more people to remain in place. And a shift in public opinion in receiving countries would make legal pathways more feasible, which would reduce the human cost of movement even if the number of movers did not change.

CHAPTER 12

CONFLICT, SECURITY AND STABILITY

Theme: climate, conflict and the security of states and communities

The Lake Chad basin was once one of the largest inland water bodies in Africa. In the 1960s it covered about 25,000 square kilometres; today it covers roughly a tenth of that, having shrunk through a combination of reduced rainfall, increased irrigation and population growth. The retreating water left behind a landscape of dry lakebed, competing claims to fishing grounds and pasture, and a population that depends on an ecosystem that no longer supports it. The region is now home to one of the world's longest-running insurgencies and about 10.7 million people in need of humanitarian assistance.

The connection between the shrinking lake and the insurgency is real but not simple. Boko Haram did not emerge because the lake dried; it emerged from a combination of political marginalisation, economic neglect, religious radicalisation and weak institutions. What the environmental change did was to remove the livelihoods that might have provided an alternative, to displace people into camps where recruitment is easier, and to weaken the local economy that might have sustained resistance. Climate change acted as a threat multiplier, not as a cause.

The same pattern appears in other regions. Drought in the Sahel has intensified competition between herders and farmers; water scarcity in the Nile basin has raised tensions between Egypt, Sudan and Ethiopia; food price spikes have preceded riots in dozens of countries. In each case, the environmental stress interacts with political, economic and social conditions that determine whether the outcome is cooperation, protest or violence.

This chapter examines that interaction. It reviews what the research actually shows about climate and conflict — a literature with strong claims and serious disagreements — and then examines the pathways through which environmental stress can contribute to violence. It pays particular attention to the role of institutions, because the same drought that produces famine in one country can produce protest in another and nothing at all in a third. The chapter closes with the question of what security institutions can do, and what they cannot.

12.1 What the Research Shows

The modern literature on climate and conflict began with a series of quantitative studies that found a statistical relationship between temperature and violence. A widely cited meta-analysis by Hsiang, Burke and Miguel (2013) reported that each standard deviation increase in temperature was associated with a 14 per cent increase in intergroup conflict and a 4 per cent increase in interpersonal violence, across studies covering many countries and periods.

The finding was influential and contested. Critics, notably Buhaug (2015), argued that the studies suffered from selection bias, that they aggregated very different forms of violence, and that the effect sizes were small relative to political and economic factors. Later work, including a review in Nature by Mach et al. (2019), concluded that

climate has affected armed conflict risk in some contexts but that the evidence is strongest for low-level violence and weakest for civil war, and that confidence in the overall relationship is limited.

The disagreement is not about whether climate can contribute to conflict; it is about how much, where, and under what conditions. The consensus that has emerged is that climate is a risk multiplier rather than an independent cause. It acts on societies that are already vulnerable, through mechanisms that are political and economic rather than meteorological, and its effect is largest where institutions are weak and where livelihoods depend on climate-sensitive resources.

12.2 Pathways: Livelihoods, Prices and Migration

The first pathway is economic. When a drought destroys a harvest or a flood destroys a factory, incomes fall and unemployment rises. Young men who cannot find work are easier to recruit into armed groups; households that cannot feed themselves are more willing to accept protection from whoever offers it. The effect is not automatic, but it is one of the most consistent findings in the literature: negative income shocks increase the risk of violence.

The second pathway is food prices. When prices rise, urban populations protest, and governments that respond with repression can escalate the protest into conflict. The 2010–2011 food price spike coincided with riots in more than thirty countries and with the political transitions of the Arab Spring, though the causal relationship is debated and varies by country. The mechanism is plausible and well documented: food is the largest expenditure for poor households, and a sudden increase can make the difference between coping and collapse.

The third pathway is migration. The movement of people can create tension in receiving areas, particularly when it is rapid, when it is perceived as unfair, and when the state is unable to provide services. But migration can also be a source of resilience, through remittances and labour. The outcome depends on the capacity of receiving communities and institutions to absorb people, which is why the same inflow can produce integration in one place and conflict in another.

12.3 The Sahel and the Lake Chad Basin

The Sahel is the region where the climate-conflict relationship is most often invoked and most difficult to disentangle. Rainfall has declined since the 1970s, though it has partially recovered in some areas; temperatures have risen faster than the global average; and the population has grown rapidly. Pastoralists and farmers compete for land and water, and the state is weak across much of the region.

The conflict in Mali, Burkina Faso and Niger has killed thousands and displaced millions. The causes are political and military as much as environmental: the collapse of the Libyan state in 2011 flooded the region with weapons; jihadist groups exploited local grievances; and governments responded with strategies that alienated rural populations. Climate stress contributed by undermining livelihoods and by making it harder for the state to deliver services, but it did not create the conflict on its own.

The Lake Chad basin is a similar case. The lake's shrinkage has been driven by reduced rainfall, increased irrigation and population growth, and it has destroyed fishing and farming livelihoods across four countries. The resulting poverty and displacement created a reservoir of recruits for Boko Haram and its offshoots, and the military response has displaced millions more. The lesson is not that the lake caused the insurgency; it is that environmental change can create the conditions in which insurgency becomes possible.

12.4 Syria and the Politics of Attribution

The Syrian conflict is often cited as the first climate war. Between 2006 and 2010, a severe drought hit the country's northeastern agricultural region, destroying crops and livestock and pushing as many as 1.5 million people toward the cities. The drought was the worst in the instrumental record, and studies have found that climate change made it more likely.

The causal chain from drought to war is contested. Some analyses argue that the drought, combined with the Assad government's failure to manage water and agriculture, created a rural crisis that fed the uprising. Others, notably Selby et al. (2017), argue that the drought was one factor among many, that the migration was not as large or as sudden as sometimes claimed, and that the regime's repressive response to peaceful protest was the proximate cause of the war. The second view does not deny the role of climate; it insists on a more careful account of how it operated.

The Syrian case illustrates a general problem in climate-conflict research: the temptation to tell a simple story that connects a weather event to a war. The stories are politically useful because they dramatise the stakes, and they are analytically dangerous because they obscure the political decisions that turn vulnerability into violence. A government that responds to drought with redistribution and relief may face protest; a government that responds with repression may face war.

12.5 Water Disputes and Cooperation

Water is the resource most often associated with climate conflict, and the record is more cooperative than the headlines suggest. The Pacific Institute's Water Conflict Chronology records thousands of water-related incidents, most of them local, and far more treaties than wars. The Indus Waters Treaty survived three wars between India and Pakistan; the Nile Basin Initiative has kept dialogue alive among ten countries with competing claims; and the Senegal River Organisation has managed a shared basin for decades.

That does not mean water is not a source of tension. The Grand Ethiopian Renaissance Dam on the Blue Nile has been the subject of a decade of negotiations among Egypt, Sudan and Ethiopia, and the dispute remains unresolved. Egypt depends on the Nile for almost all of its fresh water and has treated the river as a existential resource; Ethiopia argues that it has the right to develop its own hydropower. The dispute has not become a war, but it has shaped regional politics and delayed cooperation on other issues.

The factors that determine whether water disputes escalate are political, not hydrological. Countries that have institutions for sharing information, dispute-resolution mechanisms and a history of cooperation are more likely to manage scarcity without violence. Countries that lack them, or that are already hostile, are more likely to treat water as a zero-sum resource. The lesson is that water treaties are a form of adaptation, and that their erosion is a security risk.

12.6 Fisheries, Maritime Boundaries and the Arctic

Climate change is opening new areas of the ocean and creating new disputes. As Arctic sea ice retreats, shipping routes, fishing grounds and hydrocarbon reserves become accessible, and the states bordering the Arctic have competing claims to the continental shelf. The Arctic Council has provided a forum for cooperation, but the withdrawal of some members after the war in Ukraine has weakened it.

In the tropics, warming is driving fish stocks toward cooler water and across maritime boundaries. Countries whose fleets have historically fished in their own waters may find that the stocks have moved into the waters of a neighbour, which creates disputes over access and enforcement. The resulting tensions are rarely violent, but they can escalate, particularly where fishing is a major source of employment and where enforcement capacity is weak.

Fisheries also illustrate the role of international institutions. Regional fisheries management organisations set quotas and allocate access, and their effectiveness depends on the willingness of states to comply. Climate change makes their task harder because the stocks they manage are moving, and because the scientific advice on which they rely is uncertain. The result is a governance gap that is filled by unilateral action, which is precisely the outcome that institutions are supposed to prevent.

12.7 State Capacity and Fragility

The most important variable in the climate-conflict relationship is the capacity of the state. A state that can deliver relief, maintain order and provide services during a shock can prevent the spiral from drought to famine to violence. A state that cannot — because it is poor, corrupt, illegitimate or already at war — is where the spiral is most likely to occur.

Fragile states are also the most exposed to climate change. They tend to be agriculture-dependent, to have weak infrastructure and to lack the fiscal capacity to respond to disasters. The combination creates a feedback loop: climate shocks weaken the state, and a weakened state is less able to respond to the next shock. The result is that climate change can contribute to state failure, and state failure can make climate impacts much worse.

The policy implication is that climate adaptation and peacebuilding are the same project in fragile states. Building irrigation, restoring rangelands and providing social protection are adaptation measures, but they are also conflict-prevention measures, because they reduce the competition over resources that fuels violence. The reverse is also true: a peace agreement that does not address access to land and water is unlikely to hold.

12.8 Security Institutions and Climate Risk

Military and intelligence institutions have begun to treat climate change as a security issue. The United States Department of Defense has described climate change as a threat multiplier that will increase demand for military missions, damage installations and destabilise regions. NATO has incorporated climate considerations into its planning, and several countries have appointed climate security advisers.

The military's role is controversial. Some argue that framing climate as a security threat militarises a problem that requires development and diplomacy; others argue that the military is one of the few institutions with the logistics capacity to respond to disasters and that its planning assumptions shape how governments understand risk. The balance depends on the country and on whether security institutions are accountable to civilian authority.

The most useful contribution that security institutions can make is often analytical and logistical rather than kinetic. Early warning systems, scenario planning, disaster response capacity and the protection of critical infrastructure are all within the remit of security institutions and all contribute to resilience. The risk is that a security framing crowds out the development and governance responses that address the root causes of vulnerability.

12.9 Food, Prices and Political Instability

The relationship between food prices and political instability is one of the best-documented in the social sciences. The 2007–2008 price spike was followed by riots in more than thirty countries, and the 2010–2011 spike coincided with the Arab Spring. The mechanism is straightforward: when the price of the staple food rises beyond what households can afford, the result is hunger, anger and, often, protest.

The effect is not deterministic. Some countries with high food prices experienced no unrest, and some unrest occurred in countries with relatively low prices. The difference is explained by the presence of other grievances, the capacity of the state to respond and the willingness of the security forces to repress. Food prices are a trigger, not a cause; they ignite conflicts that are already smouldering.

Climate change makes the trigger more likely by increasing the frequency of harvest failures and by increasing the volatility of prices. It also makes the response more difficult, because governments that are already fiscally stressed may be unable to subsidise food or to provide relief. The result is that the same price shock can be absorbed in one country and produce a crisis in another.

12.10 Prevention, Early Warning and Peacebuilding

Conflict prevention is cheaper than conflict response, and climate adaptation can be a form of prevention. Early warning systems that identify areas at risk of drought or flood can trigger relief before people are forced to migrate or to compete for resources. Dialogue mechanisms that bring herders and farmers together can resolve disputes before they become violent. Land reform that clarifies tenure can reduce the uncertainty that fuels conflict.

The evidence for these interventions is mixed but encouraging. The United Nations Development Programme's climate security projects in the Sahel have supported local mediation and early warning, and some evaluations report reductions in conflict. The challenge is scale: the interventions are small relative to the size of the problem, and they depend on local institutions that are often weak.

The deepest lesson is that climate security is not primarily a military problem. It is a problem of development, governance and justice. The countries that are most vulnerable to climate-driven conflict are those where people lack secure access to land, water and food, and where the state is unable or unwilling to provide them. Addressing those conditions is the most effective security policy available, and it is also the most neglected.

12.11 Case Study: The Lake Chad Basin

Lake Chad sits at the junction of Chad, Niger, Nigeria and Cameroon, and it has supported human settlement for thousands of years. In the 1960s it covered about 25,000 square kilometres, making it one of the largest lakes in Africa. By the 2000s it had shrunk to about 2,500 square kilometres, a decline of roughly 90 per cent, driven by reduced rainfall, increased irrigation and the growing demands of a population that rose from a few million to more than 30 million in the basin.

The shrinking lake destroyed livelihoods. Fishing communities lost their catch; pastoralists lost access to water and pasture; farmers lost the flood-recession land that had produced their crops. The result was a population that was poorer, more mobile and more dependent on whatever work was available. At the same time, the region was marginalised politically and economically, with weak state presence and few public services.

Boko Haram emerged in this context. The group was founded in north-eastern Nigeria in the early 2000s and became violent in 2009 after a government crackdown. Its ideology was religious, but its recruitment drew on grievances that were economic and political: unemployment, corruption, neglect and the collapse of traditional livelihoods. The shrinking lake did not create the group, but it created the conditions in which the group could recruit and operate.

The conflict has killed more than 35,000 people and displaced more than two million. The military response, which includes forces from Nigeria, Chad, Niger and Cameroon, has retaken territory but has not restored the livelihoods that would allow people to return. Humanitarian access is limited, and the region has one of the highest rates of food insecurity in the world. The lake itself continues to fluctuate, and the inter-basin water transfer project that was proposed to restore it has not been built.

The case illustrates the limits of a purely security response. Military action can reduce the capacity of an insurgency, but it cannot restore the livelihoods that make recruitment attractive. A durable solution would require investment in irrigation, fisheries, pastoralism and local governance, combined with a political process that addresses the grievances that the insurgency exploited. Those are development measures, but they are also security measures, which is the central point of this chapter.

12.12 Future Scenarios and Predictions: Conflict Risk to 2050

The historical record does not show a simple trend of climate-driven war. The number of armed conflicts has fluctuated, and the end of the Cold War produced a decline that had little to do with climate. What the record does show is that climate shocks are associated with higher risk of low-level violence, riots and local conflict, particularly in places that are already fragile.

Present conditions give the baseline. About 2 billion people live in countries affected by conflict, and a large share of them are in regions that are highly exposed to climate change. The Sahel, the Horn of Africa, the Middle East and parts of South Asia combine high climate exposure with weak institutions and existing conflicts, which is the combination that the research identifies as most dangerous.

The assumptions that drive the projections are the emission path, the rate of development and the quality of governance. A world that reduces emissions and invests in institutions can reduce the risk; a world that does neither will face a larger number of climate shocks arriving in places that are less able to absorb them. The scenarios below summarise the range, with the caveat that conflict is inherently uncertain and that the confidence in any quantitative projection is low.

Table 12.1 Climate exposure and conflict risk in fragile regions

Region	Climate exposure	Institutional capacity	Conflict risk at 2°C	Priority
Sahel	Very high	Low	High	Land tenure; pastoral corridors
Lake Chad	Very high	Low	Very high	Livelihoods; local governance
Horn of Africa	Very high	Low	High	Drought relief; early warning
Nile basin	High	Medium	Medium	Water agreement; storage
South Asia	High	Medium	Medium	Heat; food prices; migration
Central America	High	Medium	Medium	Drought; rural employment

Two conclusions follow. First, the risk of climate-driven conflict is concentrated in places that are already fragile, which means that prevention must be targeted rather than universal. Second, the most effective prevention is development: secure land rights, functioning water institutions, social protection and accountable local government reduce the competition that turns scarcity into violence.

What would change these conclusions? A major peace agreement in a fragile region could reduce the risk substantially, as could a sustained improvement in governance. A prolonged global recession could reduce the resources available for adaptation and increase the risk. And a rapid reduction in emissions would reduce the frequency of the shocks that trigger conflict, though it would not eliminate the underlying vulnerabilities that make conflict possible.

CHAPTER 13

INFRASTRUCTURE, SHELTER AND ENERGY RESILIENCE

Theme: infrastructure, shelter and energy resilience

This chapter examines infrastructure, shelter and energy resilience. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

13.1 Critical Infrastructure and Power Grids

As Raj Patel (2022) argues, hunger reflects power, not scarcity — climate shocks deepen those power gaps.

Patel argues that hunger reflects power and access rather than global scarcity. Climate shocks deepen existing gaps in entitlement. The framing shifts attention from production to distribution. Applied to critical infrastructure and power grids, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For infrastructure operators and utilities, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The strength of the claim about critical infrastructure and power grids varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~100 million people are affected by climate disasters annually (IFRC, 2022). A central estimate without its range is not more precise, only less honest. Within infrastructure operators and utilities, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. As far as infrastructure operators and utilities is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. In the work of infrastructure operators and utilities, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The news cycle supplies a concrete marker: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). Taken alone the episode is weather; taken with the record it is a data point on a slope. As far as infrastructure operators and utilities is concerned, the item is useful mainly as a date stamp on a trend.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. For infrastructure operators and utilities, the operational reading is to record which limit applies to each decision before the decision is taken.

13.2 Water and Sanitation Systems

Peter Gleick (2020) puts it plainly: water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. For water and sanitation systems, the bearing is direct: the constraint is institutional as much as physical.

Uncertainty in water and sanitation systems is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within infrastructure operators and utilities, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as infrastructure operators and utilities is concerned, equity is therefore a design criterion rather than a footnote. Measures that

reach the most exposed first tend to perform better for everyone else as well.

The economics of water and sanitation systems are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). In the work of infrastructure operators and utilities, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: ~100 million people are affected by climate disasters annually (IFRC, 2022). For infrastructure operators and utilities, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

Current reporting adds detail rather than a new direction: Italy’s Emilia-Romagna floods (May 2023) caused 15 deaths and €8.5B damage after months of drought (Italy Civil Protection). The item is useful because it fixes a date to a trend that otherwise drifts. In the work of infrastructure operators and utilities, the report is a reminder that the response is being designed against a target that keeps moving.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. Within infrastructure operators and utilities, the question is which of these decisions can be taken before the next event rather than during it.

13.3 Transport and Communication Networks

As Kevin Anderson (2020) argues, no new fossil exploration fits a 1.5°C budget — it is already spent.

Anderson's carbon-budget arithmetic implies that no new fossil extraction is compatible with 1.5°C. The budget is small enough that existing infrastructure already exhausts it. The claim is arithmetical and depends on the chosen probability of overshoot. Read against transport and communication networks, the point is that intention without capacity changes little.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Averages describe the size of the problem, not who carries it.

Decisions taken by infrastructure operators and utilities — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as infrastructure operators and utilities is concerned, that is the difference between a plan that can be defended and one that merely exists.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. In the work of infrastructure operators and utilities, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). For infrastructure operators and utilities, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

An honest account of transport and communication networks distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. Within infrastructure operators and utilities, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

A recent marker points the same way: California's atmospheric rivers in winter 2022-23 ended drought but caused $4.6B flood damages (NOAA, CA DWR). What the report adds is timing, not a new mechanism. For infrastructure operators and utilities, what matters is not the record but the lead time it implies.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. As far as infrastructure operators and utilities is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

13.4 Building Standards and Cascading Failures

As Francesco La Camera (2023) argues, renewables dominate new power, but deployment must triple by 2030 — finance is the choke point.

IRENA documents that renewables dominate new generating capacity but deployment must roughly triple by 2030 to meet agreed targets. Finance, grids and permits are the binding constraints. Cost is no longer the main obstacle in most markets. The implication for building standards and cascading failures is a question of sequence: what has to be in place before the event, not after it.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. In the work of infrastructure operators and utilities, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~100 million people are affected by climate disasters annually (IFRC, 2022). For infrastructure operators and utilities, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Not all statements about building standards and cascading failures carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. Within infrastructure operators and utilities, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The aggregate picture hides the distribution, and in building standards and cascading failures the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). As far as infrastructure operators and utilities is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

The news cycle supplies a concrete marker: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). The event is best read as a measurement, not as a

verdict. Within infrastructure operators and utilities, the item is useful mainly as a date stamp on a trend.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. In the work of infrastructure operators and utilities, the question is which of these decisions can be taken before the next event rather than during it.

13.5 Case Study: Infrastructure, Shelter and Energy Resilience

One documented episode makes the argument concrete: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. Within infrastructure operators and utilities, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence infrastructure operators and utilities can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

13.6 Future Scenarios and Predictions: Infrastructure, Shelter and Energy Resilience

For infrastructure, shelter and energy, the near term is physical: by 2040 heat buckles 10–15% more roads and rails per heatwave, blackouts rise 30% when air-conditioning demand meets low water for hydropower, and 1-in-20 floods overtop drains.

By 2050, aging grids and housing face combined heat, flood and wind stress, and by 2100 a high-emissions path requires new coastal buildings to be elevated at least 1 m and redesigned for 45 °C days (IPCC, 2022; IEA, 2023).

Heat-resistant asphalt, elevated substations, solar-plus-battery mini-grids, and flood-resilient schools that double as shelters keep services running.

Table 13.1 Future scenarios and predictions for infrastructure, shelter and energy resilience (2030-2100)

Horizon	What changes for this chapter
2030	outages +20%; road damage +10%
2050	combined heat-flood-wind stress
2100	+1 m elevation; 45 °C design days

Watch outage hours per season, road buckling reports, and the flood level at your nearest substation; infrastructure failure thresholds are high-confidence, but actual outages depend on maintenance and backup capacity.

CHAPTER 14

CLIMATE LITERACY AND EDUCATION

Theme: climate literacy and education

The subject of this chapter is climate literacy and education. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

14.1 What Is Climate Literacy

John Cook (2020) puts it plainly: misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. For the meaning of climate literacy, the practical consequence is to name the decision that would have to change.

How a problem is framed decides what counts as a solution, and the meaning of climate literacy is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of schools and adult education, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

Any assessment of the meaning of climate literacy will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. For schools and adult education, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The evidence base behind the meaning of climate literacy is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within schools and adult education, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it

is decoration.

Not all statements about the meaning of climate literacy carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~100 million people are affected by climate disasters annually (IFRC, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. As far as schools and adult education is concerned, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The news cycle supplies a concrete marker: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). It matters less as a record than as evidence that the baseline has already moved. In the work of schools and adult education, the report is a reminder that the response is being designed against a target that keeps moving.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. In the work of schools and adult education, the operational reading is to record which limit applies to each decision before the decision is taken.

14.2 Core Concepts and Age-Appropriate Education

As Katharine Hayhoe (2014) argues, climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For core concepts and age-appropriate education within climate literacy and education, her message directs the analysis toward actionable, locally resonant communication and education strategies. Applied to core concepts and age-appropriate education, the argument sets a test: does the measure still work when capacity is stretched?

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already

shifting: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). None of those signals is a perfect measure, but together they indicate direction. For schools and adult education, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The evidence base for core concepts and age-appropriate education is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within schools and adult education, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The strength of the claim about core concepts and age-appropriate education varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~100 million people are affected by climate disasters annually (IFRC, 2022). A central estimate without its range is not more precise, only less honest. As far as schools and adult education is concerned, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question core concepts and age-appropriate education is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of schools and adult education, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Current reporting adds detail rather than a new direction: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). Treated as a single observation it invites over-reading; treated as one of many it is informative. For schools and adult education, what matters is not the record but the lead time it implies.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. For schools and adult education, the question is which of these decisions can be taken before the next event rather than during it.

14.3 Formal and Informal Education

As John Cook (2020) argues, misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. For formal and informal education, the bearing is direct: the constraint is institutional as much as physical.

The record on formal and informal education is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how schools and adult education should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within schools and adult education, the test is whether the arrangement survives contact with a bad week.

Uncertainty in formal and informal education is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as schools and adult education is concerned, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about formal and informal education are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of schools and adult education, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

Anyone deciding about formal and informal education is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For schools and adult education, the test of

a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

A recent marker points the same way: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). It is one line in a longer ledger, and the ledger is what matters. Within schools and adult education, the item is useful mainly as a date stamp on a trend.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. Within schools and adult education, the operational reading is to record which limit applies to each decision before the decision is taken.

14.4 Teacher Training and Curriculum Design

Katharine Hayhoe (2014) puts it plainly: climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For teacher training and curriculum design within climate literacy and education, her message directs the analysis toward actionable, locally resonant communication and education strategies. Read against teacher training and curriculum design, the point is that intention without capacity changes little.

An honest account of teacher training and curriculum design distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~100 million people are affected by climate disasters annually (IFRC, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. As far as schools and adult education is concerned, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Discussion of teacher training and curriculum design usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: economic losses from climate disasters reach ~$200–300 billion per year (World Bank,

2022). Read that way, the question for schools and adult education is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. In the work of schools and adult education, the implication is a shorter list of decisions than the analysis suggests.

The relevant question for teacher training and curriculum design is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For schools and adult education, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

Evidence on teacher training and curriculum design rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~100 million people are affected by climate disasters annually (IFRC, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within schools and adult education, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). Set beside the record, the episode indicates direction without fixing the rate. As far as schools and adult education is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. As far as schools and adult education is concerned, the question is which of these decisions can be taken before the next event rather than during it.

14.5 Case Study: Climate Literacy and Education

A single episode illustrates the mechanism: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. As far as schools and adult education is concerned, the record supplies both the warning and the timetable.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which schools and adult education controls.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

14.6 Future Scenarios and Predictions: Climate Literacy and Education

For climate literacy and education, the near term is measured in understanding: by 2040 misinformation spikes during every disaster, yet schools teaching climate literacy record 30–40% better preparedness scores, while 25% of adults cannot read a risk map.

By 2050, workers in heat-exposed sectors need retraining as jobs shift, and by 2100 gaps widen where curricula ignore local hazards; literacy becomes as practical as reading a tide table (IPCC, 2022; UNESCO, 2020).

Plain-language risk maps, student citizen science, teacher training, and village noticeboards with verified forecasts turn knowledge into action.

Table 14.1 Future scenarios and predictions for climate literacy and education (2030-2100)

Horizon	What changes for this chapter
2030	preparedness +30–40% in literate schools
2050	retraining for heat-exposed jobs
2100	gaps widen where curricula ignore hazards

Watch school preparedness drills, the share of adults who can interpret a flood map, and how fast rumors are corrected after a disaster; literacy outcomes are measurable, though attribution to any single lesson is weak.

CHAPTER 15

DATA, STATISTICS AND MISINFORMATION

Theme: data, statistics and misinformation in the climate era

This chapter deals with data, statistics and misinformation in the climate era. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

15.1 Why Data Matters and Types of Data

As John Cook (2020) argues, misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. In the terms of why data matter and which data are used, the claim shifts attention from prediction to preparation.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Averages describe the size of the problem, not who carries it. Decisions taken by statistical agencies and the media — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as statistical agencies and the media is concerned, that is the difference between a plan that can be defended and one that merely exists.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of statistical agencies and the media, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. For statistical agencies and the media, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The relevant question for why data matter and which data are used is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance,

redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within statistical agencies and the media, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The news cycle supplies a concrete marker: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). The item is useful because it fixes a date to a trend that otherwise drifts. For statistical agencies and the media, what matters is not the record but the lead time it implies.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. As far as statistical agencies and the media is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

15.2 Descriptive Statistics and Uncertainty

As Katharine Hayhoe (2014) argues, climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For descriptive statistics and uncertainty within data, her message directs the analysis toward actionable, locally resonant communication and education strategies. For descriptive statistics and uncertainty, the practical consequence is to name the decision that would have to change.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~100 million people are affected by climate disasters annually (IFRC, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. In the work of statistical agencies and the media, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The economics of descriptive statistics and uncertainty are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). For statistical agencies and the media, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Any assessment of descriptive statistics and uncertainty will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. Within statistical agencies and the media, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in descriptive statistics and uncertainty the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as statistical agencies and the media is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Current reporting adds detail rather than a new direction: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). What the report adds is timing, not a new mechanism. Within statistical agencies and the media, the item is useful mainly as a date stamp on a trend.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. In the work of statistical agencies and the media, the question is which of these decisions can be taken before the next event rather than during it.

15.3 Correlation vs Causation

John Cook (2020) puts it plainly: misinformation uses a handbook of doubt — learning its tricks inoculates audiences.

Cook's research shows that inoculation against misinformation works better than refutation alone. Audiences who understand the techniques behind denial are harder to mislead later. The effect is modest but replicable. Applied to the difference between correlation and causation, the argument sets a test: does the measure still work when capacity is stretched?

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. For statistical agencies and the media, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). None of those signals is a perfect measure, but together they indicate direction. Within statistical agencies and the media, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~100 million people are affected by climate disasters annually (IFRC, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. As far as statistical agencies and the media is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. In the work of statistical agencies and the media, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

A recent marker points the same way: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). The event is best read as a measurement, not as a verdict. As far as statistical agencies and the media is concerned, the report is a reminder that the response is being designed against a target

that keeps moving.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. For statistical agencies and the media, the operational reading is to record which limit applies to each decision before the decision is taken.

15.4 Graphs, Charts and Misleading Statistics

As Katharine Hayhoe (2014) argues, climate change is here and now, and not in some distant time or place — the choices we make today will have a significant impact on our future.

Hayhoe's statement comes from her 2021 book 'Saving Us' and post-AR5 outreach that frames climate change as a present, local reality. 'Here and now' counters the common perception that impacts are distant in time or geography, which she identifies as a psychological barrier to action. 'The choices we make today' refers to both individual and collective decisions on energy, land use, and consumption that shape emissions pathways. The phrase 'significant impact on our future' underscores the inertia of the climate system, where today's emissions lock in decades of warming. Hayhoe emphasizes that hope must be grounded in agency, not optimism by default. She argues that trusted messengers — teachers, health workers, faith and community leaders — are more persuasive than abstract data. For graphs within data, her message directs the analysis toward actionable, locally resonant communication and education strategies. For graphs, the bearing is direct: the constraint is institutional as much as physical.

Anyone deciding about graphs, charts and misleading statistics is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within statistical agencies and the media, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as statistical agencies and the media is concerned, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in graphs, charts and misleading statistics depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). In the work of statistical agencies and the media, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. For statistical agencies and the media, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

The news cycle supplies a concrete marker: 2023 was the hottest year on record, with global temperatures 1.45 ±0.12 C above pre-industrial levels (WMO State of Global Climate 2023, confirmed Jan 2024). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. In the work of statistical agencies and the media, what matters is not the record but the lead time it implies.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. Within statistical agencies and the media, the question is which of these decisions can be taken before the next event rather than during it.

15.5 Case Study: Data, Statistics and Misinformation

The case that anchors this chapter is documented and recent: Pakistan's Jacobabad hit 51°C in May 2022, among the highest reliably recorded temperatures globally (WMO). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. In the work of statistical agencies and the media, the sequence is worth reconstructing before the next season rather than after it.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which statistical agencies and the media controls.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

15.6 Future Scenarios and Predictions: Data, Statistics and Misinformation

For data, statistics and misinformation, the near term is a race between signals and noise: by 2040, 60–70% of online climate claims mix real and outdated data, and gaps between satellites and ground stations delay warnings by 6–12 hours.

By 2050, AI-generated false imagery and cherry-picked baselines make verification a daily skill, and by 2100 poor data trust costs 10–20% extra in misdirected spending where decisions ignore uncertainty (IPCC, 2021; WMO, 2023).

Check primary sources (IPCC, WMO, NOAA/NASA), triangulate two or three datasets, and report ranges with uncertainty bars rather than single numbers.

Table 15.1 Future scenarios and predictions for data, statistics and misinformation in the climate era (2030-2100)

Horizon	What changes for this chapter
2030	60% mixed claims; warnings delayed 6 h
2050	AI false imagery; daily verification
2100	10–20% wasted spending where trust fails

Watch whether a claim cites a baseline year, whether it matches the nearest ground station, and whether it survives a second source; data quality is checkable, unlike vague predictions.

CHAPTER 16

HOUSEHOLD PREPAREDNESS: WATER, FOOD, SHELTER AND RISK

Theme: household preparedness for water, food, shelter and local risk

What follows concerns household preparedness for water, food, shelter and local risk. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

16.1 Local Hazard Mapping and Personal Risk Matrix

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For local hazard mapping and personal risk matrix within household preparedness for water, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. The implication for local hazard mapping and personal risk matrix is a question of sequence: what has to be in place before the event, not after it.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~2 billion people lack safe water (WHO & UNICEF, 2023). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within households and community groups, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

How a problem is framed decides what counts as a solution, and local hazard mapping and personal risk matrix is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). As far as households and community groups is concerned, the

practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

The record on local hazard mapping and personal risk matrix is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how households and community groups should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. In the work of households and community groups, the implication is a shorter list of decisions than the analysis suggests.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The news cycle supplies a concrete marker: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). Treated as a single observation it invites over-reading; treated as one of many it is informative. Within households and community groups, the item is useful mainly as a date stamp on a trend.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. Within households and community groups, the operational reading is to record which limit applies to each decision before the decision is taken.

16.2 Daily Water Needs, Storage and Purification

W. Neil Adger (2006) puts it plainly: vulnerability is not only exposure to hazards but the social architecture of entitlements, power, and capacity that determines who adapts.

Adger reframes vulnerability as a property of social arrangements rather than of hazards. Entitlements, power and access to resources decide who can prepare, who can leave and who recovers. The point is structural, not semantic. In the terms of daily water needs, the claim shifts attention from prediction to preparation.

It helps to ask what question daily water needs, storage and purification is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical,

because the second framing admits trade-offs that the first hides. The scale is already visible: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). As far as households and community groups is concerned, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on daily water needs, storage and purification rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. In the work of households and community groups, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within households and community groups, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

Current reporting adds detail rather than a new direction: India's wheat yields fell 10% in 2022 heatwave, prompting export ban affecting global prices (FAO, India IMD). It is one line in a longer ledger, and the ledger is what matters. As far as households and community groups is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. As far as households and community groups is concerned, the question is which of these decisions can be taken before the next event rather than during it.

16.3 Rainwater Harvesting and Greywater Reuse

As David Lobell (2011) argues, warming since 1980 has already cut wheat and maize yields, masking gains from technology.

Lobell's analysis separated climate effects from technology and management in global crop records. Warming since 1980 reduced wheat and maize yields relative to a counterfactual without it. Adaptation and carbon fertilisation offset part, but not all, of the loss. For rainwater harvesting and greywater reuse, the practical consequence is to name the decision that would have to change.

The evidence base behind rainwater harvesting and greywater reuse is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of households and community groups, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

The economics of rainwater harvesting and greywater reuse are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Within households and community groups, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Most disagreements about rainwater harvesting and greywater reuse are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as households and community groups is concerned, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in

advance is what separates analysis from advocacy.

A recent marker points the same way: UK's Sept 2022 heat record was shattered in July 2022 at 40.3°C, first time over 40°C (UK Met Office). Set beside the record, the episode indicates direction without fixing the rate. In the work of households and community groups, what matters is not the record but the lead time it implies.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. In the work of households and community groups, the operational reading is to record which limit applies to each decision before the decision is taken.

16.4 Nutritional Needs and Food Storage

As Chuang Zhao (2017) argues, each degree of warming lowers staple yields by ~6%, with wheat most vulnerable to heat.

Zhao and colleagues combined field warming experiments with models to estimate yield responses per degree. Staples fall by roughly six per cent per degree of local warming, with wheat most sensitive. The estimate applies to present-day locations and practices. Applied to nutritional needs and food storage, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: ~2 billion people lack safe water (WHO & UNICEF, 2023). For households and community groups, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: global mean temperature has risen ~1.1–1.2°C since 1850–1900 (IPCC, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within households and community groups, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

Discussion of nutritional needs and food storage usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Read that way, the question for households and community groups is not whether the trend is real but

which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. As far as households and community groups is concerned, the practical question is what would have to change first.

The evidence base for nutritional needs and food storage is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~2 billion people lack safe water (WHO & UNICEF, 2023). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. In the work of households and community groups, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The news cycle supplies a concrete marker: Southeast Asia's April 2024 heatwave saw Thailand at 44.2°C, Myanmar 48.2°C, schools closed (Thai Met Dept, WMO). Read against the longer record, it is an illustration rather than a proof. For households and community groups, the item is useful mainly as a date stamp on a trend.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. For households and community groups, the question is which of these decisions can be taken before the next event rather than during it.

16.5 Case Study: Household Preparedness — Water, Food, Shelter and Risk

One documented episode makes the argument concrete: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. For households and community groups, the episode fixes the date at which the argument stops being hypothetical.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence households and community groups can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

16.6 Future Scenarios and Predictions: Household Preparedness — Water, Food, Shelter and Risk

For household preparedness, the near term is practical: by 2040 households without three days of water and food face 2–3 times more crisis visits, homes below 2 m flood yearly, and kitchens fail 30% more often in heat.

By 2050, insurance and aid arrive slower than neighbors, and by 2100 a high-emissions path requires rain tanks and cool rooms in half of all homes (IFRC, 2022; WHO, 2023).

Go-bags, 15 liters of water per person per day for three days, grain banks, a cool shelter, and a neighborhood mutual-aid roster cover the first 72 hours.

Table 16.1 Future scenarios and predictions for household preparedness for water, food, shelter and local risk (2030-2100)

Horizon	What changes for this chapter
2030	3-day kit cuts losses 20–30%
2050	aid lags; neighbors first responders
2100	rain tanks/cool rooms in 50% of homes

Watch your stored water days, the flood mark on your street, and how long the last outage lasted; household readiness is directly measurable and cheap to improve.

CHAPTER 17

HEALTH, EMERGENCY RESPONSE AND SPECIAL POPULATIONS

Theme: health, emergency response and protecting vulnerable populations

This chapter examines health, emergency response and protecting vulnerable populations. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

17.1 First Aid Basics and Heat Illness

Diana Liverman (2020) puts it plainly: climate justice means accounting for who is most responsible, who is most vulnerable, and that those least responsible for emissions bear the greatest burdens.

Liverman's climate justice scholarship, including her 2019 testimony to the Arizona House and IPCC authorship, underpins her statement on responsibility and vulnerability. 'Accounting for who is most responsible, who is most vulnerable' calls for attributing emissions historically and geographically, not just current national totals. 'Those least responsible bear the greatest burdens' points to smallholder farmers, small islands, and low-income urban communities facing the earliest and harshest impacts despite low per-capita emissions. Liverman argues that responses must address land tenure, gender, and access to resources, not just technical fixes. She emphasizes that top-down prescriptions often fail when they ignore local knowledge and power imbalances. For first aid basics and heat illness within health, her lens requires assessing interventions by who benefits, who pays, and who decides, integrating equity metrics into planning and evaluation. Read against first aid basics and heat illness, the point is that intention without capacity changes little.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~7 million air-pollution deaths occur annually (WHO, 2023). For emergency services and clinics, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The strength of the claim about first aid basics and heat illness varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). A central estimate without its range is not more precise, only less honest. Within emergency services and clinics, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For that reason, average-based targets can be met while the most exposed end up worse off. As far as emergency services and clinics is concerned, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~7 million air-pollution deaths occur annually (WHO, 2023). None of those signals is a perfect measure, but together they indicate direction. In the work of emergency services and clinics, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The news cycle supplies a concrete marker: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). What the report adds is timing, not a new mechanism. As far as emergency services and clinics is concerned, the report is a reminder that the response is being designed against a target that keeps moving.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. For emergency services and clinics, the operational reading is to record which limit applies to each decision before the decision is taken.

17.2 Wound Care, Infection and Medication Management

As Saleemul Huq (2020) argues, it is adapt or die — adaptation is survival, and loss-and-damage must be permanent in the Paris Agreement for vulnerable nations.

Huq's statement was delivered repeatedly at COPs from 2015 to 2023 and reflects his leadership at ICCCAD in Dhaka and IPCC authorship from AR3 to AR5. 'Adapt or die' was his stark shorthand that for low-lying Bangladesh and small islands, adaptation is not optional but survival. 'Loss-and-damage must be permanent in the Paris Agreement' refers to his advocacy for a third pillar beyond mitigation and adaptation to address unavoidable harms. The phrase captures that vulnerable nations face impacts they did not cause, raising justice and finance obligations. Huq linked local experience — cyclones, floods, salinity — to global negotiations, insisting that affected communities have voice. His work bridges research on community-based adaptation with policy for climate finance and compensation. For wound care within health, his perspective centers equity, local leadership, and the limits of adaptation when warming exceeds coping capacity. The implication for wound care is a question of sequence:

what has to be in place before the event, not after it.

Uncertainty in wound care, infection and medication management is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within emergency services and clinics, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. As far as emergency services and clinics is concerned, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Anyone deciding about wound care, infection and medication management is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~7 million air-pollution deaths occur annually (WHO, 2023). In the work of emergency services and clinics, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. For emergency services and clinics, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

Current reporting adds detail rather than a new direction: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). The event is best read as a measurement, not as a verdict. In the work of emergency services and clinics, what matters is not the record but the lead time it implies.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. Within emergency services and clinics, the question is which of these decisions can be taken before the next event rather than during it.

17.3 Chronic Disease and Elderly Care

As Maria Neira (2021) argues, climate change is the single biggest health threat facing humanity — the burning of fossil fuels is killing us.

Neira's statement was made as Director of the WHO Department of Environment, Climate Change and Health around the 2021 COP26 Special Report. 'Single biggest health threat facing humanity' elevates climate from environmental to health emergency, citing heat, air pollution, food and water insecurity, and infectious disease. 'Burning of fossil fuels is killing us' directly links emissions source to mortality — 7 million annual deaths from air pollution, hundreds of thousands from heat, plus wider burdens. Neira argues health professionals are trusted messengers who can make climate personal and urgent. The WHO report she launched called for health systems to be climate-resilient and low-carbon, and for climate policy to be judged on health co-benefits. For chronic disease and elderly care within health, her framing justifies prioritizing prevention, health surveillance linked to climate monitoring, and equity in access to care. In the terms of chronic disease and elderly care, the claim shifts attention from prediction to preparation.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Averages describe the size of the problem, not who carries it. Decisions taken by emergency services and clinics — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. As far as emergency services and clinics is concerned, that is the difference between a plan that can be defended and one that merely exists.

The relevant question for chronic disease and elderly care is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~7 million air-pollution deaths occur annually (WHO, 2023). In the work of emergency services and clinics, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt

(UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For emergency services and clinics, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

An honest account of chronic disease and elderly care distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Numbers carry confidence intervals, and those intervals should travel with them into any decision. Within emergency services and clinics, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

A recent marker points the same way: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. For emergency services and clinics, the item is useful mainly as a date stamp on a trend.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. As far as emergency services and clinics is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

17.4 Children, Disabilities and Special Needs

Kristie Ebi (2018) puts it plainly: if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with co-benefits for productivity and equity. For children within health, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. For children, the practical consequence is to name the decision that would have to change.

Any assessment of children, disabilities and special needs will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~7 million air-pollution deaths occur annually (WHO, 2023). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. In the work of emergency services and clinics, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

Institutional performance in children, disabilities and special needs depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). For emergency services and clinics, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Not all statements about children, disabilities and special needs carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. Within emergency services and clinics, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

The aggregate picture hides the distribution, and in children, disabilities and special needs the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as emergency services and clinics is concerned, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

The news cycle supplies a concrete marker: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). On its own it proves little; placed against the trend it shows the direction of travel. Within emergency services and clinics, the report is a reminder that the response is being designed against a target that keeps moving.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still

open. In the work of emergency services and clinics, the question is which of these decisions can be taken before the next event rather than during it.

17.5 Case Study: Health, Emergency Response and Special Populations

A single episode illustrates the mechanism: Peru's Cyclone Yaku (Mar 2023) brought unprecedented rains to arid north, 70 deaths (SENAMHI, Reuters). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. Within emergency services and clinics, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence emergency services and clinics can measure in its own records.

In the work of emergency services and clinics, the transferable lesson is procedural: identify which decision failed first, and fix that decision rather than the symptom.

17.6 Future Scenarios and Predictions: Health, Emergency Response and Special Populations

For emergency response and vulnerable populations, the near term is counted in minutes: by 2040 ambulance delays rise 20% during heat and floods, clinics lose power 10–15% of storm days, and mortality among older and disabled people runs 2–3 times higher without outreach.

By 2050, simultaneous heat, flood and disease events stretch responders beyond capacity, and by 2100 a high-emissions path overwhelms formal services without community-based teams (WHO, 2023; IFRC, 2022).

Volunteer responder teams, door-to-door registries of vulnerable residents, clinic backup power, and mobile health units close the gap in the first hours.

Table 17.1 Future scenarios and predictions for health, emergency response and protecting vulnerable populations (2030-2100)

Horizon	What changes for this chapter
2030	ambulance delays +20%; outages 10–15%
2050	compound events exceed capacity
2100	formal services overwhelmed without volunteers

Watch response times, the number of vulnerable residents with a named contact, and clinic backup-power hours; response capacity is testable through drills, not guesswork.

CHAPTER 18

SOCIAL CAPITAL, MUTUAL AID AND GOVERNANCE

Theme: social capital, mutual aid and community governance

The subject of this chapter is social capital, mutual aid and community governance. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

18.1 What Is Social Capital and Trust

As Daniel P. Aldrich (2015) argues, social networks, especially bridging ties, are the strongest predictor of post-disaster survival and recovery.

Aldrich finds that social capital, and especially bridging ties across groups, predicts post-disaster survival better than physical defences. Networks deliver information, shelter and credit before agencies arrive. Bonding ties alone can also exclude outsiders. For the meaning of social capital and trust, the bearing is direct: the constraint is institutional as much as physical.

Uncertainty in the meaning of social capital and trust is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of civil society and local associations, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about the meaning of social capital and trust are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~7 million air-pollution deaths occur annually (WHO, 2023). For civil society and local associations, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind the meaning of social capital and trust is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within civil society and local associations, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not

evidence; it is decoration.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

The news cycle supplies a concrete marker: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). It is one line in a longer ledger, and the ledger is what matters. In the work of civil society and local associations, what matters is not the record but the lead time it implies.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. In the work of civil society and local associations, the operational reading is to record which limit applies to each decision before the decision is taken.

18.2 Network Mapping, Bridging and Bonding

As Naomi Klein (2014) argues, climate change changes everything about how we live, consume, and govern.

Klein connects climate policy to the economic model that produced it. She argues that market instruments alone cannot deliver the pace required. The claim is a political argument rather than an empirical finding. Read against network mapping, the point is that intention without capacity changes little.

Discussion of network mapping, bridging and bonding usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~7 million air-pollution deaths occur annually (WHO, 2023). Read that way, the question for civil society and local associations is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. For civil society and local associations, that is the difference between a plan that can be defended and one that merely exists.

The evidence base for network mapping, bridging and bonding is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~100 million people are affected by climate disasters annually (IFRC, 2022). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within

civil society and local associations, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

An honest account of network mapping, bridging and bonding distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~7 million air-pollution deaths occur annually (WHO, 2023). Numbers carry confidence intervals, and those intervals should travel with them into any decision. In the work of civil society and local associations, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Current reporting adds detail rather than a new direction: Amazon drought in late 2023 lowered Rio Negro to 120-year low, isolating 500,000 people (Brazil INPE, Reuters). Set beside the record, the episode indicates direction without fixing the rate. For civil society and local associations, the item is useful mainly as a date stamp on a trend.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. For civil society and local associations, the question is which of these decisions can be taken before the next event rather than during it.

18.3 Community Leaders and Inclusive Participation

Susan L. Cutter (2008) puts it plainly: resilience is place-based — community traits explain recovery far better than hazard magnitude alone.

Cutter's place-based work shows that community characteristics explain recovery better than hazard size does. Her social-vulnerability mapping combines age, income, tenure and access to services into an index that local planners can use. The method is a screening tool, not a verdict. The implication for community leaders and inclusive participation is a question of sequence: what has to be in place before the event, not after it.

The record on community leaders and inclusive participation is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is

consistent across methods (IPCC, 2023). The headline figure is blunt: ~100 million people are affected by climate disasters annually (IFRC, 2022). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how civil society and local associations should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within civil society and local associations, the test is whether the arrangement survives contact with a bad week.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Not all statements about community leaders and inclusive participation carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~7 million air-pollution deaths occur annually (WHO, 2023). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. In the work of civil society and local associations, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

How a problem is framed decides what counts as a solution, and community leaders and inclusive participation is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~100 million people are affected by climate disasters annually (IFRC, 2022). For civil society and local associations, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

A recent marker points the same way: Libya's Derna dam collapse after Storm Daniel (Sept 2023) killed >4,300 and displaced 40,000 (UN OCHA, Libyan Red Crescent). Read against the longer record, it is an illustration rather than a proof. Within civil society and local associations, the report is a reminder that the response is being designed against a target that keeps moving.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they

are also the easiest to check. Within civil society and local associations, the operational reading is to record which limit applies to each decision before the decision is taken.

18.4 Principles of Mutual Aid and Starting a Group

As Fikret Berkes (2013) argues, linking Indigenous and scientific knowledge creates legitimate, effective adaptation that neither can achieve alone.

Berkes documents how indigenous and local knowledge systems support adaptation through fine-grained observation over generations. Linking that knowledge with scientific monitoring produces options neither can generate alone. The combination requires reciprocity, not extraction. In the terms of principles of mutual aid and starting a group, the claim shifts attention from prediction to preparation.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). As far as civil society and local associations is concerned, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

The strength of the claim about principles of mutual aid and starting a group varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~7 million air-pollution deaths occur annually (WHO, 2023). A central estimate without its range is not more precise, only less honest. In the work of civil society and local associations, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question principles of mutual aid and starting a group is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: ~100 million people are affected by climate disasters annually (IFRC, 2022). For civil society and local associations, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on principles of mutual aid and starting a group rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within civil society and local associations, that means planning against a range

rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people (WMO, UN OCHA). The value of such reports is that they date the change rather than explain it. As far as civil society and local associations is concerned, what matters is not the record but the lead time it implies.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. As far as civil society and local associations is concerned, the question is which of these decisions can be taken before the next event rather than during it.

18.5 Case Study: Social Capital, Mutual Aid and Governance

The case that anchors this chapter is documented and recent: Typhoon Doksuri flooded Beijing in July 2023 with heaviest rain in 140 years, 1 million displaced (China CMA, Reuters). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. As far as civil society and local associations is concerned, the record supplies both the warning and the timetable.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence civil society and local associations can measure in its own records.

The test of any lesson drawn from the case is whether it changes a budget line, a standard or a protocol. If it changes none of the three, it remains a story.

18.6 Future Scenarios and Predictions: Social Capital, Mutual Aid and Governance

For social capital and mutual aid, the near term is relational: by 2040 neighborhoods with strong mutual aid recover 30–40% faster and report 20% fewer losses, while fragmented areas wait one to two weeks longer for help.

By 2050, trust becomes a form of infrastructure as repeated disasters exhaust formal systems, and by 2100 a high-emissions path widens the gap between connected and isolated communities (Aldrich & Meyer, 2015; IPCC, 2022).

Block captains, tool libraries, community funds, and joint decisions on water and evacuation turn neighbors into a first-response network.

Table 18.1 Future scenarios and predictions for social capital, mutual aid and community governance (2030-2100)

Horizon	What changes for this chapter
2030	recovery 30–40% faster with strong ties
2050	trust as infrastructure; repeated shocks
2100	connected vs isolated gap widens

Watch how many households know their block captain, how quickly a tool library responds, and whether decisions are shared; social cohesion is measurable through simple surveys and drills.

CHAPTER 19

RENEWABLE ENERGY AND APPROPRIATE TECHNOLOGIES

Theme: renewable energy and appropriate technologies for climate adaptation

This chapter deals with renewable energy and appropriate technologies for climate adaptation. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

19.1 Renewable Energy Systems and Solar/Battery

As Saleemul Huq (2020) argues, it is adapt or die — adaptation is survival, and loss-and-damage must be permanent in the Paris Agreement for vulnerable nations.

Huq's statement was delivered repeatedly at COPs from 2015 to 2023 and reflects his leadership at ICCCAD in Dhaka and IPCC authorship from AR3 to AR5. 'Adapt or die' was his stark shorthand that for low-lying Bangladesh and small islands, adaptation is not optional but survival. 'Loss-and-damage must be permanent in the Paris Agreement' refers to his advocacy for a third pillar beyond mitigation and adaptation to address unavoidable harms. The phrase captures that vulnerable nations face impacts they did not cause, raising justice and finance obligations. Huq linked local experience — cyclones, floods, salinity — to global negotiations, insisting that affected communities have voice. His work bridges research on community-based adaptation with policy for climate finance and compensation. For renewable energy within renewable energy and appropriate technologies for climate adaptation, his perspective centers equity, local leadership, and the limits of adaptation when warming exceeds coping capacity. Applied to renewable energy, the argument sets a test: does the measure still work when capacity is stretched?

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). None of those signals is a perfect measure, but together they indicate direction. As far as energy planners and regulators is concerned, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: ~100 million people are affected by climate disasters annually (IFRC, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. In the work of energy planners and regulators, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For energy planners and regulators, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within energy planners and regulators, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The news cycle supplies a concrete marker: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). The event is best read as a measurement, not as a verdict. For energy planners and regulators, the item is useful mainly as a date stamp on a trend.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. As far as energy planners and regulators is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

19.2 Water Technologies and Purification

Fatih Birol (2023) puts it plainly: the energy crisis sped the transition, but fossil demand must fall far faster for 1.5°C.

Birol's World Energy Outlook tracks the gap between stated goals and investment flows. Crises have accelerated renewables while leaving fossil demand too high for 1.5°C. The series is a scenario exercise, not a forecast. For water technologies and purification, the bearing is direct: the constraint is institutional as much as physical.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. In the work of energy planners and regulators, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in water technologies and purification depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). For energy planners and regulators, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within energy planners and regulators, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Anyone deciding about water technologies and purification is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as energy planners and regulators is concerned, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Current reporting adds detail rather than a new direction: Vanuatu's twin cyclones Judy and Kevin (Mar 2023) hit within 48 hours, affecting 80% of population (UNICEF, WMO). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. Within energy planners and regulators, the report is a reminder that the response is being designed against a target that keeps moving.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. In the work of energy planners and regulators, the question is which of these decisions can be taken before the next event rather than during it.

19.3 Agricultural Technologies and Adaptive Practices

As Francesco La Camera (2023) argues, renewables dominate new power, but deployment must triple by 2030 — finance is the choke point.

IRENA documents that renewables dominate new generating capacity but deployment must roughly triple by 2030 to meet agreed targets. Finance, grids and permits are the binding constraints. Cost is no longer the main obstacle in most markets. Read against agricultural technologies and adaptive practices, the point is that intention without capacity changes little.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). For energy planners and regulators, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within energy planners and regulators, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The relevant question for agricultural technologies and adaptive practices is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as energy planners and regulators is concerned, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). Averages describe the size of the problem, not who carries it. Decisions taken by energy planners and regulators — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. In the work of energy planners and regulators, the test is whether the arrangement survives contact with a bad week.

A recent marker points the same way: Italy’s Emilia-Romagna floods (May 2023) caused 15 deaths and €8.5B damage after months of drought (Italy Civil Protection). On its own it proves little; placed against the trend it shows the direction of travel. As far as energy planners and regulators is concerned, what matters is not the record but the lead time it implies.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. For energy planners and regulators, the operational reading is to record which limit applies to each decision before the decision is taken.

19.4 Early Warning Systems and AI for Climate

As Inger Andersen (2023) argues, adaptation finance gaps are widening, leaving the vulnerable to pay with lives.

UNEP's adaptation-gap reports compare finance needs with flows actually disbursed. The gap has widened every year the series has been published. Adaptation remains the smaller share of climate finance by a wide margin. The implication for early warning systems and AI for climate is a question of sequence: what has to be in place before the event, not after it.

The economics of early warning systems and AI for climate are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Within energy planners and regulators, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Any assessment of early warning systems and AI for climate will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. As far as energy planners and regulators is concerned, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The aggregate picture hides the distribution, and in early warning systems and AI for climate the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: atmospheric CO2 now stands at ~420 ppm in 2024 (NOAA, 2024). In the work of energy planners and regulators, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. For energy planners and regulators, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The news cycle supplies a concrete marker: US 2023 billion-dollar disasters hit record 28 events, costing $92.9B (NOAA NCEI). Taken alone the episode is weather; taken with the record it is a data point on a slope. In the work of energy planners and regulators, the item is useful mainly as a date stamp on a trend.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. Within energy planners and regulators, the question is which of these decisions can be taken before the next event rather than during it.

19.5 Case Study: Renewable Energy and Appropriate Technologies

One documented episode makes the argument concrete: Nigeria's 2022 floods killed 600 and displaced 1.4 million, worst in a decade (Nigeria NEMA, UN OCHA). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. In the work of energy planners and regulators, the sequence is worth reconstructing before the next season rather than after it.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence energy planners and regulators can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

19.6 Future Scenarios and Predictions: Renewable Energy and Appropriate Technologies

For renewable energy and appropriate technologies, the near term is economic: by 2040 solar-plus-storage is cheaper than diesel in 70% of off-grid areas, but heat cuts panel output 5–10% and storms damage 15% of exposed installations.

By 2050, mini-grids and battery cooperatives become the backbone of resilient power, and by 2100 a high-emissions path still leaves 20% of demand on fossil fuels without storage scale-up (IEA, 2023; IPCC, 2022).

Elevated solar arrays, cyclone clips, battery cooperatives, and repair training for young people keep power on when the grid fails.

Table 19.1 Future scenarios and predictions for renewable energy and appropriate technologies for climate adaptation (2030-2100)

Horizon	What changes for this chapter
2030	solar+storage cheaper in 50% off-grid
2050	mini-grids backbone; battery co-ops
2100	20% fossil demand without storage scale

Watch your system's output on the hottest day, battery autonomy hours, and the number of trained local technicians; technology costs are high-confidence, but storm resilience depends on installation quality.

CHAPTER 20

CLIMATE PSYCHOLOGY AND MENTAL HEALTH

Theme: climate anxiety, trauma and building psychological resilience

What follows concerns climate anxiety, trauma and building psychological resilience. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

20.1 Climate Anxiety and Rational Alarm

Kristie Ebi (2018) puts it plainly: if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with co-benefits for productivity and equity. For climate anxiety and rational alarm within climate anxiety, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. For climate anxiety and rational alarm, the practical consequence is to name the decision that would have to change.

Most disagreements about climate anxiety and rational alarm are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: ~100 million people are affected by climate disasters annually (IFRC, 2022). Within mental-health services and schools, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind climate anxiety and rational alarm is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~7 million air-pollution deaths occur annually (WHO, 2023). As far as mental-health services and schools is concerned, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not

evidence; it is decoration.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of mental-health services and schools, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Uncertainty in climate anxiety and rational alarm is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: ~100 million people are affected by climate disasters annually (IFRC, 2022). For mental-health services and schools, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

The news cycle supplies a concrete marker: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). Set beside the record, the episode indicates direction without fixing the rate. Within mental-health services and schools, the report is a reminder that the response is being designed against a target that keeps moving.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. Within mental-health services and schools, the operational reading is to record which limit applies to each decision before the decision is taken.

20.2 Youth Anxiety and Government Inaction

As Susan Clayton (2020) argues, climate anxiety is a rational alarm, especially for youth sensing betrayal by elders.

Clayton argues that climate anxiety is a rational response to a real threat rather than a disorder. Distress concentrates among people who trust the science and feel unheard by institutions. Clinical framing and political framing therefore point in different directions. Applied to youth anxiety and government inaction, the argument sets a test: does the measure still work when capacity is stretched?

The evidence base for youth anxiety and government inaction is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~7 million air-pollution deaths occur annually (WHO, 2023). The appropriate response to a strong signal with weak

local detail is not to wait for precision but to design for the range. As far as mental-health services and schools is concerned, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of mental-health services and schools, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

An honest account of youth anxiety and government inaction distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: ~100 million people are affected by climate disasters annually (IFRC, 2022). Numbers carry confidence intervals, and those intervals should travel with them into any decision. For mental-health services and schools, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Discussion of youth anxiety and government inaction usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: ~7 million air-pollution deaths occur annually (WHO, 2023). Read that way, the question for mental-health services and schools is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. Within mental-health services and schools, the test is whether the arrangement survives contact with a bad week.

Current reporting adds detail rather than a new direction: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). Read against the longer record, it is an illustration rather than a proof. As far as mental-health services and schools is concerned, what matters is not the record but the lead time it implies.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. As far as mental-health services and schools is concerned, the question is which of these decisions can be taken before the next event rather than during it.

20.3 Trauma, PTSD and Mental Health Impacts

As Kristie Ebi (2018) argues, if you have cancer, you need the doctor to tell you how serious it is — people are suffering and dying right now; risks rise without urgent investment in health systems.

Ebi's statement comes from her IPCC SR15 lead authorship and long-term research at the University of Washington CHanGE on heat and health. 'If you have cancer, you need the doctor to tell you how serious it is' is her analogy for clear risk communication — understating severity delays life-saving action. 'People are suffering and dying right now' counters the view that climate health impacts are future, citing documented rises in heat-related illness and mortality among outdoor workers, elders, and low-income groups. 'Risks rise without urgent investment in health systems' points to the need for heat early warning, expanded primary care, and climate-resilient facilities. Ebi stresses that health evidence is actionable and economically justified, with co-benefits for productivity and equity. For trauma within climate anxiety, her framing treats adaptation as public health preparedness, requiring surveillance, workforce training, and funding that reaches the front lines. For trauma, the bearing is direct: the constraint is institutional as much as physical.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). In the work of mental-health services and schools, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

Not all statements about trauma, PTSD and mental health impacts carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: ~100 million people are affected by climate disasters annually (IFRC, 2022). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. For mental-health services and schools, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

How a problem is framed decides what counts as a solution, and trauma, PTSD and mental health impacts is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~7 million air-pollution deaths occur annually (WHO, 2023). Within mental-health services and schools, the practical consequence is that the frame

has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

The record on trauma, PTSD and mental health impacts is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how mental-health services and schools should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. As far as mental-health services and schools is concerned, the practical question is what would have to change first.

A recent marker points the same way: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). The value of such reports is that they date the change rather than explain it. In the work of mental-health services and schools, the item is useful mainly as a date stamp on a trend.

Certainty is unevenly distributed across this literature. Detection is firm, projection is conditional, and cost estimates are contingent on choices not yet made. A reader who wants a single number will be disappointed; a reader who wants a defensible decision has what is needed. In the work of mental-health services and schools, the operational reading is to record which limit applies to each decision before the decision is taken.

20.4 Building Psychological Resilience

Susan Clayton (2020) puts it plainly: climate anxiety is a rational alarm, especially for youth sensing betrayal by elders.

Clayton argues that climate anxiety is a rational response to a real threat rather than a disorder. Distress concentrates among people who trust the science and feel unheard by institutions. Clinical framing and political framing therefore point in different directions. Read against building psychological resilience, the point is that intention without capacity changes little.

The strength of the claim about building psychological resilience varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: ~100 million people are affected by climate disasters annually (IFRC, 2022). A central estimate without its range is not more precise, only less honest. For mental-health services and schools, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question building psychological resilience is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical,

because the second framing admits trade-offs that the first hides. The scale is already visible: ~7 million air-pollution deaths occur annually (WHO, 2023). Within mental-health services and schools, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on building psychological resilience rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. As far as mental-health services and schools is concerned, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of mental-health services and schools, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The news cycle supplies a concrete marker: European summer 2022 heatwave caused over 61,000 excess deaths, the continent's deadliest heat event (Nature Medicine, 2023). It matters less as a record than as evidence that the baseline has already moved. For mental-health services and schools, the report is a reminder that the response is being designed against a target that keeps moving.

What emerges is less a prediction than a test. A measure is adequate if it still works when the upper end of the range arrives; a plan is credible if it names who acts, with what authority and with what money. Judged that way, the gap between what is known and what is done is not a knowledge gap. For mental-health services and schools, the question is which of these decisions can be taken before the next event rather than during it.

20.5 Case Study: Climate Psychology and Mental Health

A single episode illustrates the mechanism: Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses (IPCC, 2022; WMO, 2023). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. For mental-health services and schools, the episode fixes the date at which the argument stops being hypothetical.

Cases of this kind are useful because they compress several mechanisms into a short period. Exposure, capacity and institutional reach become visible at once, and the gaps between them stop being abstract (UNDRR, 2015). Within mental-health services and schools, the episode works as a readiness test rather than a weather story.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

20.6 Future Scenarios and Predictions: Climate Psychology and Mental Health

For climate anxiety and mental health, the near term is emotional: by 2040, 35–45% of teenagers report weekly climate anxiety, heat and displacement raise depression 15–25%, and post-flood trauma lasts 6–12 months without support.

By 2050, chronic stress becomes a public-health cost in hot and insecure regions, and by 2100 a high-emissions path doubles chronic stress burdens where heat and displacement persist (Watts et al., 2021; WHO, 2023).

Peer groups, green spaces, counseling in schools and clinics, and action projects that restore a sense of agency turn anxiety into engagement.

Table 20.1 Future scenarios and predictions for climate anxiety, trauma and building psychological resilience (2030-2100)

Horizon	What changes for this chapter
2030	anxiety 35–45% of teens weekly
2050	chronic stress raises health costs
2100	chronic burden ×2 in hot, insecure zones

Watch school counseling referrals, sleep loss during heatwaves, and community participation in recovery projects; mental-health trends are medium-confidence, but local service demand is visible.

CHAPTER 21

GOVERNANCE, JUSTICE AND CIVIC ENGAGEMENT

Theme: climate governance, justice, equity and civic engagement

This chapter examines climate governance, justice, equity and civic engagement. It moves from mechanism to evidence to what can be done, and each section states its main claim before the supporting material. Where the evidence is uncertain, the text says so; where it is strong, the language is correspondingly direct.

21.1 International Frameworks and Paris Agreement

As Diana Liverman (2020) argues, climate justice means accounting for who is most responsible, who is most vulnerable, and that those least responsible for emissions bear the greatest burdens.

Liverman's climate justice scholarship, including her 2019 testimony to the Arizona House and IPCC authorship, underpins her statement on responsibility and vulnerability. 'Accounting for who is most responsible, who is most vulnerable' calls for attributing emissions historically and geographically, not just current national totals. 'Those least responsible bear the greatest burdens' points to smallholder farmers, small islands, and low-income urban communities facing the earliest and harshest impacts despite low per-capita emissions. Liverman argues that responses must address land tenure, gender, and access to resources, not just technical fixes. She emphasizes that top-down prescriptions often fail when they ignore local knowledge and power imbalances. For international frameworks and paris agreement within climate governance, her lens requires assessing interventions by who benefits, who pays, and who decides, integrating equity metrics into planning and evaluation. In the terms of international frameworks and paris agreement, the claim shifts attention from prediction to preparation.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~100 million people are affected by climate disasters annually (IFRC, 2022). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. For courts, legislatures and civic organisations, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Averages describe the size of the problem, not who carries it. Decisions taken by courts, legislatures and civic organisations — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to

solve. Within courts, legislatures and civic organisations, the implication is a shorter list of decisions than the analysis suggests.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as courts, legislatures and civic organisations is concerned, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The next decade will show whether preparation or improvisation prevails. Three signals are worth tracking: how often events exceed design thresholds, how much public spending goes to prevention rather than repair, and whether the most exposed groups gain access before a disaster rather than after it (IPCC, 2023). The baseline is already shifting: ~100 million people are affected by climate disasters annually (IFRC, 2022). None of those signals is a perfect measure, but together they indicate direction. In the work of courts, legislatures and civic organisations, adaptation is not a state to be reached but a standing capacity to revise decisions as evidence changes. The test is whether the next surprise is met with a plan or with a scramble.

The news cycle supplies a concrete marker: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). Reported in isolation the figure looks dramatic; reported against the baseline it looks consistent. As far as courts, legislatures and civic organisations is concerned, what matters is not the record but the lead time it implies.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. For courts, legislatures and civic organisations, the operational reading is to record which limit applies to each decision before the decision is taken.

21.2 National Adaptation Plans and Local Governance

As Mark Howden (2020) argues, adapting Agriculture to Climate Change)

Howden's statement comes from his 2007 PNAS paper 'Adapting agriculture to climate change' and his role as IPCC WGII Vice-Chair. 'Dealing with barriers to effective adaptation will require a comprehensive and dynamic policy approach' means technical options alone fail when institutions, markets, and knowledge systems are misaligned. 'Science, too, has to adapt' calls for linking scientific, local, and Indigenous knowledge, and for research that is co-produced with farmers and communities. Barriers he identifies include fragmented governance, short-term incentives, and poor access to tailored forecasts and finance. Howden emphasizes iterative learning — piloting, monitoring, and adjusting — rather than one-off plans. For national adaptation plans and local governance within climate governance, this directs attention to enabling conditions for adoption: trusted extension, tenure security, and policy that supports

diversification and ecosystem-based practices alongside technology. For national adaptation plans and local governance, the practical consequence is to name the decision that would have to change.

The aggregate picture hides the distribution, and in national adaptation plans and local governance the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Within courts, legislatures and civic organisations, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). What the record does show is that authority, money and information have to move together; any one without the others stalls. As far as courts, legislatures and civic organisations is concerned, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

Anyone deciding about national adaptation plans and local governance is choosing under conditions that will not improve before the choice has to be made. That makes sequencing the central skill: measures that are cheap, reversible and useful under several futures go first, and irreversible commitments wait for better information (IPCC, 2022). The sequence matters because early options buy the time that later options need. The pressure is documented rather than projected: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of courts, legislatures and civic organisations, the test of a plan is not whether it is optimal for one forecast but whether it survives several. Decisions taken now will be judged by events that were not in the forecast.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. For courts, legislatures and civic organisations, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Current reporting adds detail rather than a new direction: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). On its own it proves little; placed against the trend it shows the direction of travel. In the work of courts, legislatures and

civic organisations, the item is useful mainly as a date stamp on a trend.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. Within courts, legislatures and civic organisations, the question is which of these decisions can be taken before the next event rather than during it.

21.3 Carbon Pricing and Climate Finance

Francesco La Camera (2023) puts it plainly: renewables dominate new power, but deployment must triple by 2030 — finance is the choke point.

IRENA documents that renewables dominate new generating capacity but deployment must roughly triple by 2030 to meet agreed targets. Finance, grids and permits are the binding constraints. Cost is no longer the main obstacle in most markets. Applied to carbon pricing and climate finance, the argument sets a test: does the measure still work when capacity is stretched?

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. As far as courts, legislatures and civic organisations is concerned, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The relevant question for carbon pricing and climate finance is not what will happen but what to do while the answer is still forming. That favours measures that are cheap, reversible and useful across several futures. Early warning, insurance, redundant systems and protected water sources meet that test; megaprojects tied to one scenario do not (IPCC, 2023). Pressure is building regardless: ~100 million people are affected by climate disasters annually (IFRC, 2022). In the work of courts, legislatures and civic organisations, the portfolio matters more than any single measure. Robustness, not optimality, is the right criterion under uncertainty.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. For courts, legislatures and civic organisations, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: economic losses from climate disasters reach

~$200–300 billion per year (World Bank, 2022). For that reason, average-based targets can be met while the most exposed end up worse off. Within courts, legislatures and civic organisations, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

A recent marker points the same way: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). Taken alone the episode is weather; taken with the record it is a data point on a slope. For courts, legislatures and civic organisations, the report is a reminder that the response is being designed against a target that keeps moving.

The evidence is not unanimous on pace and regional detail, and the gaps should be named rather than smoothed over. Where estimates differ, the difference usually traces back to assumptions about emissions, adaptation and measurement. That does not weaken the case for action; it defines the range within which action has to be robust. As far as courts, legislatures and civic organisations is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

21.4 Adaptation Finance Gaps and Investment

As Inger Andersen (2023) argues, adaptation finance gaps are widening, leaving the vulnerable to pay with lives.

UNEP's adaptation-gap reports compare finance needs with flows actually disbursed. The gap has widened every year the series has been published. Adaptation remains the smaller share of climate finance by a wide margin. For adaptation finance gaps and investment, the bearing is direct: the constraint is institutional as much as physical.

Any assessment of adaptation finance gaps and investment will turn less on what is promised now than on what is in place when the next event arrives. Plans that exist only on paper, thresholds that trigger no action, and funds that cannot be released quickly are the recurring failure modes (UNDRR, 2015). The pressure is not hypothetical: ~100 million people are affected by climate disasters annually (IFRC, 2022). Progress can be measured in lead times, in the share of exposed households covered by warning and insurance, and in how quickly services return. In the work of courts, legislatures and civic organisations, reviewing those measures on a fixed cycle works better than reviewing them after each disaster. Iteration is the difference between adaptation and repeated recovery.

The economics of adaptation finance gaps and investment are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For courts, legislatures and civic organisations, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against

one budget year is often cheap against ten.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Because disadvantages compound, the same event can be an inconvenience for one family and a permanent loss for another. Within courts, legislatures and civic organisations, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in adaptation finance gaps and investment depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~100 million people are affected by climate disasters annually (IFRC, 2022). As far as courts, legislatures and civic organisations is concerned, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

The news cycle supplies a concrete marker: Flooding in Pakistan in 2022 affected 33 million people, killed >1,700 and caused over $30 billion in damages/losses (World Bank PDNA 28 Oct 2022). The item is useful because it fixes a date to a trend that otherwise drifts. Within courts, legislatures and civic organisations, what matters is not the record but the lead time it implies.

The strands point in one direction: the physical signal is established, the human consequences depend on exposure and capacity, and the response depends on decisions taken before the event. That sequence is what makes the problem tractable, and what makes delay expensive. In the work of courts, legislatures and civic organisations, the question is which of these decisions can be taken before the next event rather than during it.

21.5 Case Study: Governance, Justice and Civic Engagement

The case that anchors this chapter is documented and recent: Hurricane Ian (Sept 2022) caused $113 billion in US damages and was the costliest Florida hurricane (NOAA NHC). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. Within courts, legislatures and civic organisations, it is the clearest recent test of whether the arrangements on paper survive contact with an event.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which courts, legislatures and civic organisations controls.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

21.6 Future Scenarios and Predictions: Governance, Justice and Civic Engagement

For governance, justice and civic engagement, the near term is budgetary: by 2040 climate lawsuits double while adaptation budgets meet only 15–25% of needs, and low-income wards receive 30% less protection.

By 2050, unequal exposure becomes a political fault line, and by 2100 a high-emissions path locks in injustice where decisions are made without community voice (IPCC, 2022; Schlosberg, 2007).

Participatory budgeting, equity audits, legal aid, and public transparency boards that track spending put resources where risk is highest.

Table 21.1 Future scenarios and predictions for climate governance, justice, equity and civic engagement (2030-2100)

Horizon	What changes for this chapter
2030	adaptation budgets 15–25% of needs
2050	unequal exposure becomes political
2100	injustice locked in without voice

Watch per-capita adaptation spending by ward, the number of community consultations, and lawsuit outcomes; distributional data are available and can be audited.

CHAPTER 22

CASE STUDIES: HEAT, FIRE, FLOOD AND URBAN EXTREMES

Theme: regional case studies of heat, wildfire, flood and urban climate extremes

The subject of this chapter is regional case studies of heat, wildfire, flood and urban climate extremes. The argument proceeds in three steps: what is happening, how we know, and what follows for decisions. Sources are cited as they are used, and contested points are marked as contested rather than presented as settled.

22.1 Heat in South Asia and Urban Heat in Global Cities

As Friederike Otto (2021) argues, the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that heat in south asia and urban heat in global cities extremes are not random bad luck but part of a detectable trend. For regional case studies of heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. The implication for heat in south asia and urban heat in global cities is a question of sequence: what has to be in place before the event, not after it.

Uncertainty in heat in south asia and urban heat in global cities is not a single quantity, and treating it as one is the most common analytic error in the field. Physical uncertainty, scenario uncertainty and model uncertainty behave differently: the first narrows with data, the second depends on choices, the third on structure (IPCC, 2021). Only the first is reduced by waiting; the other two are reduced by deciding. The range is wide enough to matter for planning: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). In the work of urban authorities in fire- and flood-prone regions, the correct response to the wide part of the range is a decision that performs acceptably across it, not a postponement until it narrows. Plans that require certainty before they begin are plans that begin too late.

Most disagreements about heat in south asia and urban heat in global cities are not disagreements about facts. They are disagreements about weight: which risks count, over what horizon, and for whom (IPCC, 2022). That is why the same evidence can support both alarm and complacency without either side misreading the data. The scale involved is not marginal: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). For urban authorities in fire- and flood-prone regions, clarity about the weighting matters as much as clarity about the measurement. Stating the criterion in advance is what separates analysis from advocacy.

The evidence base behind heat in south asia and urban heat in global cities is larger than its public profile suggests, and more uneven. Satellite records, station data, insurance claims and household surveys answer different questions, and they rarely agree on the same decimal place (IPCC, 2021). Where they disagree, the disagreement is usually about coverage rather than about the direction of change. Confidence is highest on aggregates and lowest on localities: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). Within urban authorities in fire- and flood-prone regions, the sensible use of the record is to establish direction and then test local plausibility rather than to import a global average wholesale. A number that cannot be traced to a source is not evidence; it is decoration.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale (UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The news cycle supplies a concrete marker: Wildfires in Canada in 2023 burned ~15 million ha, more than double the previous record (1989, 6.7 Mha), sending smoke across North America (CIFFC, Nature 2024). Read against the longer record, it is an illustration rather than a proof. In the work of urban authorities in fire- and flood-prone regions, the item is useful mainly as a date stamp on a trend.

Readers should know where this section stands on the evidence ladder. The direction of change is established; the pace is estimated; the local consequences are modelled. Statements that blur those levels are the ones most likely to mislead, and they are also the easiest to check. In the work of urban authorities in fire- and flood-prone regions, the operational reading is to record which limit applies to each decision before the decision is taken.

22.2 Wildfires in Australia and North America

Friederike Otto (2021) puts it plainly: the Pacific Northwest 2021 heatwave would have been virtually impossible without human-caused climate change.

Otto's statement comes from the 2021 World Weather Attribution rapid study of the Pacific Northwest heat dome, which found such heat virtually impossible without climate change. 'Virtually impossible' is a probabilistic attribution conclusion, not rhetoric, meaning the event's likelihood increased many tens of times due to warming. The method compares observed temperatures with climate model simulations of worlds with and without anthropogenic greenhouse gases. The heatwave — with Lytton, Canada reaching 49.6°C — illustrated how a few degrees of mean warming can shift extremes non-linearly. Otto's lab pioneered rapid attribution to link specific disasters to climate change within days, aiding public understanding. The finding underscores that

wildfires in australia and north america extremes are not random bad luck but part of a detectable trend. For regional case studies of heat, it strengthens the case for updating standards for infrastructure, health, and labor based on new extremes, not historical averages. In the terms of wildfires in australia and north america, the claim shifts attention from prediction to preparation.

Discussion of wildfires in australia and north america usually begins as a technical question, and that framing hides the decisions underneath it. The technical part is real: measurement, modelling and attribution have improved sharply over two decades (IPCC, 2023). But the contested part is rarely the measurement; it is what the numbers imply for who acts, who pays and who is protected. One figure sets the scale: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Read that way, the question for urban authorities in fire- and flood-prone regions is not whether the trend is real but which choices remain defensible across the plausible range. A framing that treats uncertainty as a reason to wait gets the logic backwards: uncertainty is what makes early, reversible decisions valuable. For urban authorities in fire- and flood-prone regions, that is the difference between a plan that can be defended and one that merely exists.

The evidence base for wildfires in australia and north america is broad but uneven, and the unevenness is informative. Where many independent studies agree, confidence is high; where a single model family supplies the estimate, it is not (IPCC, 2021). One figure illustrates the state of knowledge: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). The appropriate response to a strong signal with weak local detail is not to wait for precision but to design for the range. Within urban authorities in fire- and flood-prone regions, that means carrying uncertainty into the decision rather than resolving it beforehand. Plans that require certainty before acting are plans that act late.

Impacts do not follow directly from a change in the average; they follow from the interaction between a shifting hazard and a fixed exposure. A house built to a hundred-year flood standard meets a twenty-five-year event with alarming regularity once the distribution has moved (IPCC, 2022). Nothing about the house changed; the standard it was designed against did. The shift is quantifiable: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the lesson is that design standards have a shelf life and need a review date attached to them. An unrevised standard is a decision to accept a higher level of risk than the one originally debated.

An honest account of wildfires in australia and north america distinguishes three things: what has been measured, what has been modelled, and what has been assumed. Most public confusion comes from collapsing the three into one sentence (IPCC, 2021). The distinction is concrete: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). Numbers carry confidence intervals, and those intervals should travel with them into any decision. In the work of urban authorities in fire- and flood-prone regions, the discipline is to write the interval next to the estimate whenever the estimate is used. A disclosed range can be planned against; an undisclosed point estimate cannot.

Current reporting adds detail rather than a new direction: Spain's 2023 olive harvest halved after drought+heat, driving global olive oil prices up 112% (FAO, IOC). The value of such reports is that they date the change rather than explain it. For urban authorities in fire- and flood-prone regions, the report is a reminder that the response is being designed against a target that keeps moving.

What the section establishes is narrower than a prediction and firmer than an opinion. It establishes that the direction is known, that the costs of inaction are concentrated on those least able to bear them, and that the available measures are known well enough to be tested. For urban authorities in fire- and flood-prone regions, the question is which of these decisions can be taken before the next event rather than during it.

22.3 Floods in Southeast Asia and Urban Flooding

As Chuang Zhao (2017) argues, each degree of warming lowers staple yields by ~6%, with wheat most vulnerable to heat.

Zhao and colleagues combined field warming experiments with models to estimate yield responses per degree. Staples fall by roughly six per cent per degree of local warming, with wheat most sensitive. The estimate applies to present-day locations and practices. For floods in southeast asia and urban flooding, the practical consequence is to name the decision that would have to change.

The record on floods in southeast asia and urban flooding is now strong enough to be treated as a trend rather than an anomaly. The record is uneven — some regions and seasons move sharply, others barely — but the aggregate direction is consistent across methods (IPCC, 2023). The headline figure is blunt: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). What remains contested is attribution of individual events and the speed of future change, not whether the baseline has shifted. That distinction shapes how urban authorities in fire- and flood-prone regions should read the numbers: as a distribution to plan against, not a forecast to await. Where the evidence is thin, the honest response is a stated limitation rather than a confident sentence. Within urban authorities in fire- and flood-prone regions, the test is whether the arrangement survives contact with a bad week.

Between the physical change and the harm there is a sequence, and each step can be examined on its own. A shift in the mean moves the tail of the distribution first; the tail meets exposure; exposure meets capacity; capacity fails at thresholds rather than gradually (IPCC, 2022). That is why impacts arrive in jumps and why the same event produces different outcomes in neighbouring districts. The arithmetic of the shift is blunt: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the useful question is which threshold comes first and what happens on the day it is crossed. Thresholds are where a gradual problem becomes a sudden one.

Not all statements about floods in southeast asia and urban flooding carry the same warrant. Some rest on direct measurement, some on models, some on expert judgement, and the three should not be written in the same voice (IPCC, 2021). The numbers are often firmer than the uses to which they are put: sea level has risen ~20–25

cm since 1900 (IPCC, 2021). Misreading usually happens in translation — a global average applied to a locality, a scenario treated as a forecast. In the work of urban authorities in fire- and flood-prone regions, the safeguard is to record the source, period and assumptions behind each figure used. Traceability is not bureaucracy; it is what allows a claim to be checked later.

How a problem is framed decides what counts as a solution, and floods in southeast asia and urban flooding is a case in point. Framed as an environmental issue, it invites a ministry and a budget line; framed as a development issue, it invites land use, housing and finance at once (IPCC, 2022). The second framing is harder to administer precisely because it cannot be assigned to a single department. The scale involved rules out a narrow frame: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For urban authorities in fire- and flood-prone regions, the practical consequence is that the frame has to be chosen before the plan, not after it. Frames are cheap to change on paper and expensive to change in an institution.

A recent marker points the same way: Bangladesh's Sidr-like Cyclone Sitrang (Oct 2022) inundated 10 million coastal residents (Bangladesh BMD, UN OCHA). It matters less as a record than as evidence that the baseline has already moved. Within urban authorities in fire- and flood-prone regions, what matters is not the record but the lead time it implies.

It is worth separating what would change the conclusion from what would merely add detail. A revision of the observed trend, or evidence that adaptation is failing faster than expected, would matter; another confirmation of the same direction would not. That test keeps attention on the claims that carry weight. Within urban authorities in fire- and flood-prone regions, the operational reading is to record which limit applies to each decision before the decision is taken.

22.4 Storms and Hurricanes in the Caribbean

As Ana Vicedo-Cabrera (2021) argues, over one-third of heat deaths are attributable to anthropogenic warming — a burden we can already count.

Vicedo-Cabrera and colleagues attributed a large share of warm-season heat deaths to anthropogenic warming using multi-country mortality records. The finding converts a global average into a body count. Attribution of this kind depends on the choice of baseline period. Applied to storms and hurricanes in the caribbean, the argument sets a test: does the measure still work when capacity is stretched?

The chain of causation is short enough to follow and long enough to intervene in. Physical change alters the frequency and intensity of events; exposure determines who meets them; capacity determines what happens next (IPCC, 2022). Because the links are separable, they can be addressed separately — and the cheapest link is usually exposure. The pattern is not hypothetical: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). As far as urban authorities in fire- and flood-prone regions is concerned, the lesson is to intervene where the leverage is greatest rather than where the damage is most visible. Response is necessary; prevention is cheaper.

The strength of the claim about storms and hurricanes in the caribbean varies, and that variation should be visible in the prose. Detection and attribution are well supported; regional projections carry wider bands; estimates of future damage depend on assumptions about emissions and adaptation (IPCC, 2021). A concrete example of the difference: sea level has risen ~20–25 cm since 1900 (IPCC, 2021). A central estimate without its range is not more precise, only less honest. In the work of urban authorities in fire- and flood-prone regions, the discipline is to keep scenarios labelled as scenarios and observations labelled as observations. Where the evidence is thin, the correct response is a stated limitation rather than a confident sentence.

It helps to ask what question storms and hurricanes in the caribbean is really answering. Framed as a measurement problem, it invites better instruments; framed as a decision problem, it invites better institutions (IPCC, 2023). The difference is practical, because the second framing admits trade-offs that the first hides. The scale is already visible: ~500,000+ heat-related deaths occur annually (Watts et al., 2021). For urban authorities in fire- and flood-prone regions, the choice of framing determines which options are considered at all. A question that excludes trade-offs will produce an answer that cannot be implemented.

Evidence on storms and hurricanes in the caribbean rests on several independent lines rather than a single study. Satellites, ground stations, ocean measurements and model reanalyses now agree on direction, and disagree mainly on pace and regional detail (IPCC, 2021). One measure puts the magnitude in view: economic losses from climate disasters reach ~$200–300 billion per year (World Bank, 2022). Confidence is highest where those lines converge and lowest where one method or a short record carries the claim. Within urban authorities in fire- and flood-prone regions, that means planning against a range rather than a point. A decision built on the central estimate alone is a decision that will be surprised.

The news cycle supplies a concrete marker: Italy’s Emilia-Romagna floods (May 2023) caused 15 deaths and €8.5B damage after months of drought (Italy Civil Protection). Treated as a single observation it invites over-reading; treated as one of many it is informative. As far as urban authorities in fire- and flood-prone regions is concerned, the item is useful mainly as a date stamp on a trend.

Taken together, the mechanism, the observed record and the institutional response describe a risk that is measurable but not fixed. The practical question is not whether the trend is real, but which decisions today remain defensible across the full range of projections — and those are usually the decisions that are cheap to reverse. As far as urban authorities in fire- and flood-prone regions is concerned, the question is which of these decisions can be taken before the next event rather than during it.

22.5 Case Studies: Heat, Fire, Flood and Urban Extremes

One documented episode makes the argument concrete: Wildfires in Canada in 2023 burned ~15 million ha, more than double the previous record (1989, 6.7 Mha), sending smoke across North America (CIFFC, Nature 2024). It repays attention less as an outlier than as a chain of decisions — where the hazard was allowed to concentrate, how the response was organised, and who bore the residual loss. As far as urban

authorities in fire- and flood-prone regions is concerned, the record supplies both the warning and the timetable.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted savings — a sequence urban authorities in fire- and flood-prone regions can measure in its own records.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

22.6 Future Scenarios and Predictions: Case Studies — Heat, Fire, Flood and Urban Extremes

For cities facing heat, fire, flood and urban extremes, the near term is already documented: by 2040 Ahmedabad, Los Angeles and Athens see 30–50% more wildfire-smoke days, and low-lying urban zones face 40–60 flood days a year.

By 2050, 1-in-10-year urban floods become 1-in-5 in many deltas, and by 2100 a high-emissions path makes today's 1-in-10-year flood an annual event in the most exposed cities (IPCC, 2022; UNDRR, 2015).

Fire breaks, sponge-city parks, early evacuation drills, and city-to-city learning exchanges spread what already works.

Table 22.1 Future scenarios and predictions for regional case studies of heat, wildfire, flood and urban climate extremes (2030-2100)

Horizon	What changes for this chapter
2030	smoke days +30–40%; flood days +20
2050	1-in-10 urban flood → 1-in-5
2100	1-in-10 flood becomes annual

Watch smoke-day counts, flood days in your district, and evacuation times in the last drill; city-level records are strong, though each city's exposure differs.

CHAPTER 23

CASE STUDIES: DROUGHT, FOOD, WATER AND SECURITY

Theme: regional case studies of drought, food crisis, water stress and security

This chapter deals with regional case studies of drought, food crisis, water stress and security. Each section opens with a claim, tests it against the evidence and draws out the implication for the people who have to act on it. The aim is not completeness but a defensible line through a large literature.

23.1 Drought in the Horn of Africa and Sahel

Tim Wheeler (2013) puts it plainly: food shocks travel through prices and nutrition, striking import-dependent poor first.

Wheeler and von Braun trace how food shocks travel through prices, trade and nutrition. Import-dependent poor households absorb the first and deepest losses. Short-term food aid does not prevent the next shock. Read against drought in the horn of africa and sahel, the point is that intention without capacity changes little.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: ~2 billion people lack safe water (WHO & UNICEF, 2023). As far as rural authorities in drought-prone regions is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The distribution of harm is not a detail appended to the analysis; it is the analysis. Two households can face the same event and emerge with different futures, because what they own, owe and can claim determines recovery (IPCC, 2022). The aggregate picture obscures this: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). For that reason, average-based targets can be met while the most exposed end up worse off. In the work of rural authorities in drought-prone regions, the measures that matter are those judged by their effect on the bottom of the distribution, not the middle. That is also the test of whether a policy is serious.

Whether the problem remains manageable depends as much on institutions as on the underlying trend. Early warning, land-use rules, standards and social protection reduce harm, but only where authority and funding reach the level at which risk is felt (UNDRR, 2015). The common failure is fragmentation: overlapping mandates, unshared data, responsibility falling between agencies. Coordination is a design problem with measurable features — joint protocols, common baselines, clear triggers. For rural authorities in drought-prone regions, the test is whether its own arrangements have those features before the next event rather than after it. Where such arrangements exist, response times fall and losses follow.

The case for spending early rests on a comparison that is rarely made explicit: prevention against recovery, over decades rather than one budget cycle. Damage functions and observed recovery costs point the same way (World Bank, 2022). The order of magnitude is contested in detail but not in direction: ~2 billion people lack safe water (WHO & UNICEF, 2023). Prevention loses politically because its benefits are invisible: the disaster that did not happen has no constituency. Within rural authorities in drought-prone regions, the fix is institutional — budget rules that reserve funds for prevention, and appraisal that counts avoided losses. Otherwise the arithmetic is sound and the incentives are not.

The news cycle supplies a concrete marker: Tonga's Hunga Tonga eruption (Jan 2022) injected unprecedented water vapor into stratosphere, affecting climate short-term (NASA). On its own it proves little; placed against the trend it shows the direction of travel. For rural authorities in drought-prone regions, the report is a reminder that the response is being designed against a target that keeps moving.

Two cautions apply to everything in this section. First, most figures are aggregates, and aggregates travel badly between countries and neighbourhoods. Second, published estimates lag the climate they describe. Both cautions argue for revisiting the numbers rather than treating them as fixed. As far as rural authorities in drought-prone regions is concerned, the operational reading is to record which limit applies to each decision before the decision is taken.

23.2 Food Crisis in Sub-Saharan Africa and Price Volatility

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For food crisis in sub-saharan africa and price volatility within regional case studies of drought, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. The implication for food crisis in sub-saharan africa and price volatility is a question of sequence: what has to be in place before the event, not after it.

The burden falls first on people with the least room to manoeuvre. Those with insecure tenure, irregular income or caring responsibilities have less capacity to move, insure or wait for compensation (IPCC, 2022). The scale is not marginal: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). Because disadvantages compound, the

same event can be an inconvenience for one family and a permanent loss for another. In the work of rural authorities in drought-prone regions, equity is therefore a design criterion rather than a footnote. Measures that reach the most exposed first tend to perform better for everyone else as well.

Institutional performance in food crisis in sub-saharan africa and price volatility depends less on formal authority than on whether mandates, money and information arrive together. Mandates split across agencies produce the familiar gap between warning and response; budgets tied to calendar years produce the gap between relief and recovery (UNDRR, 2015). Neither gap requires bad faith, only an ordinary division of labour. The strain is already visible in the numbers: ~800 million people face food insecurity (FAO, 2023). For rural authorities in drought-prone regions, the design question is where to place the authority that can move money and people within the same week. Coordination is not a virtue to be requested; it is a structure to be built.

Money spent before an event buys more protection than money spent after it. That claim rests on damage functions and on the repeated observation that recovery costs exceed prevention costs by a wide margin (World Bank, 2022). The order of magnitude is not in doubt: ~2 billion people lack safe water (WHO & UNICEF, 2023). What is uncertain is how benefits are distributed and how quickly they appear, which is why appraisal methods matter. Within rural authorities in drought-prone regions, the relevant question is not whether prevention pays but who captures the return. Financing arrangements that ignore that question tend to fund visible damage and neglect invisible capacity.

Why things behave as they do is a question with a traceable answer. The chain runs from physical forcing through changing conditions to the decisions that place people and assets in harm's way (IPCC, 2021). Each link is observable, and that is what allows the pattern to be projected even where individual outcomes cannot be. The magnitudes involved are not small: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as rural authorities in drought-prone regions is concerned, there is more than one point of action, and the cheapest interventions are usually the earliest. Waiting for certainty at the end of the chain is the most expensive option available.

Current reporting adds detail rather than a new direction: US Southwest megadrought 2000-2022 was the driest 23-year period in 1,200 years (UCLA, Nature Climate Change). Taken alone the episode is weather; taken with the record it is a data point on a slope. Within rural authorities in drought-prone regions, what matters is not the record but the lead time it implies.

The argument, reduced to its shortest form, is this: the physical signal is robust, the human exposure is uneven, and the response is a choice. Each element rests on evidence of different strength, and the weakest link is not the science but the capacity to act before the event rather than after it. In the work of rural authorities in drought-prone regions, the question is which of these decisions can be taken before the next event rather than during it.

23.3 Water Stress in the Middle East and Groundwater Depletion

As Cynthia Rosenzweig (2020) argues, the potential risk of multi-breadbasket failure is increasing — food systems face escalating extremes that demand systemic adaptation.

Rosenzweig's statement reflects her leadership of AgMIP and NASA GISS research and her 2022 World Food Prize, with a 2019 New York Times interview on climate and food. 'Potential risk of multi-breadbasket failure is increasing' refers to the growing likelihood that major grain regions — North America, Europe, Asia — suffer simultaneous yield shocks in the same year. The drivers are overlapping heat, drought, and extreme rainfall made more likely by warming, combined with tightly coupled markets. 'Food systems face escalating extremes that demand systemic adaptation' means incremental farm-level tweaks alone are insufficient; supply chains, storage, trade, and diets must adapt together. Her work shows wheat, maize, and rice yields decline with each degree of warming, with larger regional variability. For water stress in the middle east and groundwater depletion within regional case studies of drought, this frames food and agriculture as systemic risk requiring diversification, resilient varieties, ecosystem-based practices, and early action on storage and distribution. In the terms of water stress in the middle east and groundwater depletion, the claim shifts attention from prediction to preparation.

Institutional capacity is what turns a warning into an evacuation and a budget line into a working shelter. It depends on mandates that are clear, data that are shared, and money that can be released without a new decision (UNDRR, 2015). Where those conditions are absent, the same hazard produces a larger disaster. The stakes are visible in the numbers: ~800 million people face food insecurity (FAO, 2023). For rural authorities in drought-prone regions, capability is better tested in a drill than in a document. A plan that has never been exercised is a hypothesis.

The economics favour early action, but the arithmetic is routinely misread. Damage functions show rising marginal losses with warming, while preventive spending delivers returns over decades (World Bank, 2022). The magnitudes are large: ~2 billion people lack safe water (WHO & UNICEF, 2023). The obstacle is less the total than the distribution of costs and benefits: those who pay are rarely those who are protected. Within rural authorities in drought-prone regions, the implication is to count avoided losses, not only capital outlays. Insurance can spread the residual risk; it cannot reduce the hazard itself.

The causal path can be traced step by step, which is why the broad direction is predictable even when individual events are not. A forcing alters conditions; conditions alter exposure; exposure meets a population with unequal capacity to absorb it (IPCC, 2022). Because the steps interact, impacts rarely arrive one at a time: heat comes with humidity, drought with fire risk, rainfall with runoff. The same pattern is visible in the numbers: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as rural authorities in drought-prone regions is concerned, the practical implication is to plan for combinations rather than single hazards. Systems designed for one stress at a time fail first when two arrive together.

The resulting risk is distributed unevenly, and the unevenness is structural rather than accidental. Tenure, income, age, health and gender determine how much a household can prepare, evacuate or recover (IPCC, 2022). The aggregate numbers conceal this: ~800 million people face food insecurity (FAO, 2023). Averages describe the size of the problem, not who carries it. Decisions taken by rural authorities in drought-prone regions — where to build, whom to insure, which services keep running — widen that gap or close it. A policy that protects the average household while leaving the most exposed unprotected has not solved the problem it claims to solve. In the work of rural authorities in drought-prone regions, the test is whether the arrangement survives contact with a bad week.

A recent marker points the same way: Rhine River fell to critical lows in Aug 2022, disrupting 80% of German inland shipping (German Federal Waterways, Reuters). The item is useful because it fixes a date to a trend that otherwise drifts. As far as rural authorities in drought-prone regions is concerned, the item is useful mainly as a date stamp on a trend.

Two limits apply. The first is scale: global figures say little about a district, and district figures are often unavailable. The second is time: the record is long enough to detect a trend but not long enough to rule out rare extremes. Good planning treats both as design constraints rather than reasons to postpone. For rural authorities in drought-prone regions, the operational reading is to record which limit applies to each decision before the decision is taken.

23.4 Migration-Conflict Links in Fragile States

Solomon Hsiang (2014) puts it plainly: climate variability is robustly linked to conflict risk, though institutions decide whether stress becomes violence.

Hsiang's work links climate variability to conflict risk across several hundred studies. The association is robust but conditional, with institutions and livelihoods deciding whether stress becomes violence. Climate is a risk multiplier, not a cause. For migration-conflict links in fragile states, the practical consequence is to name the decision that would have to change.

The economics of migration-conflict links in fragile states are usually presented as a bill, which is the least useful form in which to present them. Costs arrive early and are budgeted, while benefits arrive late and are unassigned; the asymmetry systematically biases public investment toward response and away from preparation (World Bank, 2023). Insurance and credit then transmit that bias into private decisions about where to build and what to protect. The magnitude is not marginal: ~2 billion people lack safe water (WHO & UNICEF, 2023). Within rural authorities in drought-prone regions, the useful accounting separates the cost of preparing from the cost of not preparing, and prices both over the same horizon. A measure that is expensive against one budget year is often cheap against ten.

Following the sequence from cause to harm makes the points of intervention visible, and there are more of them than the public debate suggests. Emissions determine the forcing; land use and insurance determine exposure; income, health and warning determine capacity; each link can be altered by policy at a different scale

(UNDRR, 2015). Because the links are separate, a failure at one does not excuse inaction at another. The scale at which the chain operates is large: wheat yields fall ~6% per 1°C warming (Zhao et al., 2017). As far as rural authorities in drought-prone regions is concerned, the practical value of thinking in a chain is that it shows which link is cheapest to move this year. Intervention points are political choices, not physical facts.

The aggregate picture hides the distribution, and in migration-conflict links in fragile states the distribution carries most of the policy content. Risk is concentrated by geography, tenure and income, and the concentration is produced by prior decisions about land, services and insurance (UNDRR, 2015). Because those decisions were made deliberately, they can be unmade deliberately. The unevenness is measurable: ~800 million people face food insecurity (FAO, 2023). In the work of rural authorities in drought-prone regions, averages are the wrong unit of account; the relevant figure is the burden carried by the group least able to carry it. A policy judged by its average effect will systematically miss the people it is meant to protect.

Governance is tested at the seams between organisations. Most failures reported after disasters are failures of handover: warnings that never reached the right desk, plans that assumed capacity another agency had already committed (UNDRR, 2015). The evidence here is thinner than in the physical sciences, and should be described as such. One indication of the stakes: ~2 billion people lack safe water (WHO & UNICEF, 2023). What the record does show is that authority, money and information have to move together; any one without the others stalls. For rural authorities in drought-prone regions, the test is whether a decision can be taken where the consequence is felt, with the resources to act on it.

The news cycle supplies a concrete marker: Amazon drought in late 2023 lowered Rio Negro to 120-year low, isolating 500,000 people (Brazil INPE, Reuters). What the report adds is timing, not a new mechanism. In the work of rural authorities in drought-prone regions, the report is a reminder that the response is being designed against a target that keeps moving.

None of this settles the political question of how much to spend and who should pay. It does settle a prior question: whether there is anything worth spending on. The evidence says there is, and that the returns depend on acting while the options are still open. Within rural authorities in drought-prone regions, the question is which of these decisions can be taken before the next event rather than during it.

23.5 Case Studies: Drought, Food, Water and Security

A single episode illustrates the mechanism: Tonga's Hunga Tonga eruption (Jan 2022) injected unprecedented water vapor into stratosphere, affecting climate short-term (NASA). Read closely, it is a record of thresholds crossed, capacities exhausted and responsibilities disputed — the same elements that recur, at different scales, in every chapter of this book. In the work of rural authorities in drought-prone regions, the sequence is worth reconstructing before the next season rather than after it.

Read as evidence rather than anecdote, the episode shows how a physical shock travels through housing, work, health and finance. Damage does not stop where the water recedes; it continues through lost income, interrupted schooling and depleted

savings — a sequence rural authorities in drought-prone regions can measure in its own records.

Within rural authorities in drought-prone regions, the transferable lesson is procedural: identify which decision failed first, and fix that decision rather than the symptom.

23.6 Future Scenarios and Predictions: Case Studies — Drought, Food, Water and Security

For drought, food, water and security hotspots, the near term repeats recent crises: by 2040 Horn of Africa and Sahel droughts like 2022 recur every 3–4 years, Mekong dry-season flows fall 20%, and Punjab groundwater drops 1 m a year, pushing food prices up 20–35% in bad years.

By 2050, multi-breadbasket failures become more likely, and by 2100 a high-emissions path doubles the risk of simultaneous crop losses across continents (FAO, 2023; IPCC, 2022).

Drought-tolerant crops, watershed recharge, grain reserves, and cross-border water agreements buffer the shock before it becomes a famine.

Table 23.1 Future scenarios and predictions for regional case studies of drought, food crisis, water stress and security (2030-2100)

Horizon	What changes for this chapter
2030	drought every 4–5 years; prices +15%
2050	multi-breadbasket risk rises
2100	simultaneous crop loss risk ×2

Watch the number of consecutive failed rainy seasons, regional grain stocks, and cross-border water releases; drought onset is forecastable, but famine depends on markets and conflict.

CHAPTER 24

ACTION PLANS AND CHECKLISTS FOR SURVIVAL

Theme: personal and community action plans, checklists and templates for surviving the future

What follows concerns personal and community action plans, checklists and templates for surviving the future. The chapter separates observation from projection, and projection from speculation, because the three carry different obligations for anyone making a decision. Readers who disagree with the conclusion should be able to see exactly where.

24.1 30-Day Personal Action Plan: Water, Food, Shelter

As Gilbert Cissé (2022) argues, informal urban settlements face compounded climate-health risks that require integrated planning.

Cissé's assessment shows that informal urban settlements carry compound climate-health risks: heat, water contamination, crowding and absent services. Adaptation that assumes formal tenure and piped water misses most of the affected population. Planning has to start from the settlement, not the grid. For 30-day personal action plan: water, the bearing is direct: the constraint is institutional as much as physical.

30-day personal action plan: water, food, shelter is only useful if it can be executed under stress, and stress is exactly when detail is lost. The plan that survives is the one written for a tired person at night with no power and no network.

The common failure is not ambition but vagueness. 'Prepare a kit' means nothing at 2 a.m.; 'two litres of water per person per day for three days, stored where the door cannot trap you' means something. Specificity is the difference between a document and a capability.

Review matters more than design. A plan tested once a year, with the people who will actually carry it out, will outperform a better plan that is never rehearsed. The test is not whether the folder looks complete but whether a household can act from memory.

24.2 1-Year Community Action Plan: Governance and Networks

As Peter Gleick (2020) argues, water crises are governance crises magnified by climate, not just hydrology.

Gleick frames water scarcity as a governance failure magnified by climate rather than a purely hydrological limit. Allocation rules, pricing and infrastructure decide who is short. Drought turns a distribution problem into a crisis. Read against 1-year community action plan: governance and networks, the point is that intention without capacity changes little.

1-year community action plan: governance and networks works at the scale of a street or a building, not a municipality. The unit that matters is the group that can reach each other without transport, because that is the group that acts first.

What makes such groups effective is boring: named contacts, agreed meeting points, a shared list of who needs help and who has a vehicle, and one person authorised to make a call without a committee. Trust is built by doing small things together before it is needed.

Sustainability depends on rotating responsibility. Groups that depend on one energetic organiser collapse when that person moves or burns out. Written handovers and shared tasks are not bureaucracy; they are what allows the group to outlast its founder.

24.3 Emergency Checklists: Go-Bags, Documents, Communication

Bill McKibben (2019) puts it plainly: we have built a world that can be undone by what we built — the fight is for a workable future.

McKibben argues that the fight is for a workable future rather than an unchanged one. The book catalogues movement successes and failures since 2007. Its purpose is strategic, not predictive. The implication for emergency checklists: go-bags is a question of sequence: what has to be in place before the event, not after it.

emergency checklists: go-bags, documents, communication should be short enough to use and specific enough to check. A checklist that runs to thirty items will be skimmed; one that fits on a card will be followed.

The items that matter most are the ones people forget under pressure: documents in one place, a paper copy of contacts, medication that does not expire this year, cash in small notes, and a way to charge a phone without mains power. Each is cheap; each is routinely missing.

Checklists also serve a social function. Sharing one with neighbours turns a private precaution into a common standard, and common standards are what make informal support reliable when formal services are stretched.

24.4 Household Templates: Risk Matrix and Water Audit

As Nick Obradovich (2018) argues, higher temperatures measurably increase mental-health hospitalizations and suicide risk.

Obradovich and colleagues matched temperature records with mental-health hospital admissions and suicide data. Both rise with heat, and the effect persists beyond a single day. The mechanism runs through sleep, medication and social contact. In the terms of household templates: risk matrix and water audit, the claim shifts attention from prediction to preparation.

household templates: risk matrix and water audit are tools for thinking, not forms to file. A risk matrix that has not changed a decision has failed, however neatly it is filled in.

The test of a household template is whether it produces a number someone acts on: litres stored, days of food, minutes to reach high ground, the one route that stays open. If the template produces only a feeling of preparedness, it is decoration.

Templates should be revised after every real event and every near miss. The revision is the learning; the document is only its memory.

24.5 Case Study: Action Plans and Checklists for Survival

The case that anchors this chapter is documented and recent: Tonga's Hunga Tonga eruption (Jan 2022) injected unprecedented water vapor into stratosphere, affecting climate short-term (NASA). What matters is not the event alone but the sequence it exposes — a hazard that exceeded design assumptions, a response constrained by mandate and money, and a recovery that revealed who was covered and who was not. For households, schools and local government, the episode fixes the date at which the argument stops being hypothetical.

The instructive part is the asymmetry between the event and its aftermath. Response times are measured in hours, recovery in years, and the difference between the two is usually decided before the event — by land use, insurance and local authority, all of which households, schools and local government controls.

What travels to other settings is not the specific hazard but the pattern of who was warned, who could move and who was compensated.

24.6 Future Scenarios and Predictions: Action Plans and Checklists for Survival

For personal and community action plans, the near term is about readiness: by 2040 families with a one-month plan cut losses 40–50% and recover twice as fast, while 30% without checklists forget documents and medicines in an evacuation.

By 2050, community plans covering water, food, shelter and governance decide whether a shock becomes a crisis, and by 2100 a high-emissions path overwhelms ad-hoc coping entirely (IFRC, 2022; IPCC, 2022).

A 30-day household plan (water, food, shelter), a one-year community plan (governance, networks), go-bags, risk matrices, and drills updated every year.

Table 24.1 Future scenarios and predictions for personal and community action plans, checklists and templates for surviving the future (2030-2100)

Horizon	What changes for this chapter
2030	1-month plan cuts losses 40–50%
2050	community plans decide outcomes
2100	ad-hoc coping overwhelmed (high path)

Watch your plan's last update date, the number of households with go-bags, and drill participation; preparedness is measurable and improves with repetition.

APPENDIX A: LAMPIRAN (CHECKLISTS & PANDUAN)

Master Emergency Checklist

- Item 1: Verify supplies, documents, and communication plans.
- Item 2: Verify supplies, documents, and communication plans.
- Item 3: Verify supplies, documents, and communication plans.
- Item 4: Verify supplies, documents, and communication plans.
- Item 5: Verify supplies, documents, and communication plans.
- Item 6: Verify supplies, documents, and communication plans.
- Item 7: Verify supplies, documents, and communication plans.
- Item 8: Verify supplies, documents, and communication plans.
- Item 9: Verify supplies, documents, and communication plans.
- Item 10: Verify supplies, documents, and communication plans.

Panduan Penulisan dan Debat Akademik

Buku ini dirancang untuk digunakan dalam diskusi akademik. Saat memperdebatkan atau memperluas klaimnya:

- Selalu sitasi sumber primer (paper peer-reviewed, laporan IPCC).
- Bedakan antara observasi, atribusi, dan proyeksi.
- Akui ketidakpastian dan tingkat kepercayaan.
- Pertimbangkan berbagai jalur sosioekonomi (SSP).
- Sertakan dimensi keadilan dan kesetaraan.
- Perbarui klaim seiring munculnya bukti baru.

APPENDIX B: GLOSARIUM

- Adaptation: Adjusting to actual or expected climate impacts.
- Mitigation: Reducing greenhouse gas emissions.
- Resilience: Capacity to absorb disturbance and reorganize.
- Vulnerability: Susceptibility to harm.
- Exposure: Presence of people or assets in hazard-prone areas.
- Sensitivity: Degree to which a system is affected by climate.
- Adaptive capacity: Ability to adjust to climate change.
- Tipping point: Threshold beyond which change becomes self-sustaining.
- Greenhouse effect: Trapping of heat by atmospheric gases.
- Carbon footprint: Total greenhouse gas emissions caused by an entity.
- SSP: Shared Socioeconomic Pathway (scenario framework).
- RCP: Representative Concentration Pathway.
- Committed warming: Warming locked in by past emissions.

APPENDIX C: INDEKS

- Adaptation
- Biodiversity
- Carbon footprint
- Carbon pricing
- Climate finance
- Committed warming
- Conflict
- Drought
- Early warning
- Ecosystem
- Equity
- Exposure
- Flood
- Food security
- Greenhouse effect
- Health
- Heat wave
- Indigenous knowledge
- Infrastructure
- Justice
- Mental health
- Migration
- Mitigation
- Mutual aid
- RCP
- Renewable energy
- Resilience
- SSP
- Sea level rise
- Sensitivity
- Social capital
- Tipping point
- Vulnerability
- Water security

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APPENDIX E: ABOUT THE AUTHOR

Setyo Wisodo is a professional working across AI development, product management and digital marketing, with a focus on AI-based solutions and digital products. Based in Indonesia.

APPENDIX F: EXTENDED ANALYTICAL NOTES

Analytical note: The relationship between climate drivers and social outcomes is mediated by institutional capacity, resource availability, and historical context. No single factor determines the result.

Analytical note: Adaptation strategies must be evaluated not only for effectiveness but also for equity, cost, and feasibility in specific contexts.

Analytical note: The temporal dimension of climate impacts requires long-term thinking that often conflicts with short-term political and economic incentives.

Analytical note: Knowledge production and knowledge use are distinct processes; bridging them requires deliberate effort and sustained engagement.

Analytical note: Uncertainty is inherent in climate science and should be communicated transparently rather than hidden or exaggerated.